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Implicit Time Optimization in Indirect Low-Thrust Fuel and Energy Optimal Trajectories: A Closed-Form Approach

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Abstract

Low-thrust propulsion has emerged in recent decades as a promising strategy for a diverse range of mission applications. The evaluation of optimal trajectories in the context of continuous low-thrust guidance is particularly crucial, demanding robust and reliable methods for solution evaluation and continuation.

Although various strategies exist for the continuation of time-optimal (TO), and, separately, energy-optimal (EO) and fuel-optimal (FO) solutions, there appears to exist a gap in literature regarding works relating TO solutions to either FO or EO solutions. This dissertation addresses this gap by developing a closed-form relationship connecting TO solutions to FO and EO solutions that implicitly optimize time.

Using the M-ARGO mission as a case study, a comprehensive analysis of existing formulations is conducted to identify a suitable relationship and its applicability conditions. The results show that, regardless of the problem formulation, a set of analytical requirements governing the indirect formulation, specifically the Euler-Lagrange equations and shooting function, must be met to allow for the existence, throughout the entire trajectory, of a closed-form relationship between the costate paths of solutions. Specifically, it is found that, starting from TO solutions, the optimal FO and EO costate paths of solutions that implicitly optimize time can be obtained simply by scaling the TO costate path, achieving optimality without altering the trajectory.

These analytical hypotheses are supported by numerical experiments obtained using solar electric propulsion (SEP) with variable propulsive parameters in two-body problem (2BP) dynamics, and with constant propulsive parameters in both 2BP and circular restricted three-body problem (CR3BP) dynamics.

Keywords: Low-Thrust Optimization, M-ARGO Mission, Time Optimization, Fuel Optimization, Energy Optimization, Implicit Time Optimization, Homotopy Methods, Continuation Techniques, Closed-Form Approach

Abstract in lingua italiana

La propulsione a bassa spinta è emersa negli ultimi decenni come una strategia promettente per un'ampia gamma di applicazioni di missione. La valutazione delle traiettorie ottimali nel contesto della guida continua a bassa spinta è particolarmente cruciale, richiedendo metodi robusti e affidabili per l'analisi e la continuazione delle soluzioni.
Sebbene esistano diverse strategie per la continuazione delle soluzioni che ottimizzano tempo (TO) e, separatamente, delle soluzioni che ottimizzano energia (EO) e carburante (FO), sembra esistere una lacuna nella letteratura per quanto riguarda lavori che mettono in relazione le soluzioni TO con quelle FO o EO. Questa dissertazione affronta questa lacuna sviluppando una relazione in forma chiusa che collega le soluzioni TO a soluzioni FO ed EO che ottimizzano implicitamente il tempo.
Utilizzando la missione M-ARGO come caso studio, viene condotta un'analisi completa delle formulazioni esistenti per identificare una possibile relazione e le condizioni per la sua applicabilità. I risultati ottenuti dimostrano che, indipendentemente dalla formulazione del problema, un set di requisiti analitici che governano la formulazione indiretta, nello specifico le equazioni di Eulero-Lagrange e la funzione di shooting, deve essere soddisfatto affinché esista, lungo l'intera traiettoria, una relazione in forma chiusa tra i percorsi dei costati delle soluzioni. In particolare, si dimostra che, partendo dalle soluzioni TO, i percorsi ottimali dei costati per le soluzioni FO ed EO che ottimizzano implicitamente il tempo possono essere ottenuti semplicemente scalando il percorso dei costati della soluzione TO, raggiungendo l'ottimalità senza modificare la traiettoria.
Queste ipotesi analitiche sono supportate da esperimenti numerici ottenuti utilizzano la propulsione elettrica solare (SEP) con parametri propulsivi variabili in una dinamica a due corpi (2BP), e con parametri propulsivi costanti sia nella dinamica 2BP che nella dinamica del problema dei tre corpi ristretto circolare (CR3BP).

Parole chiave: Ottimizzazione a Bassa Spinta, Missione M-ARGO, Ottimizzazione del Tempo, Ottimizzazione del Carburante, Ottimizzazione dell'Energia, Ottimizzazione del Tempo Implicita, Metodi di Omotopia, Tecniche di Continuazione, Approccio in Forma Chiusa

Contents

Abstract **i**

Abstract in lingua italiana **iii**

Contents **v**

1 Introduction **1**
1.1 M-ARGO Mission . . . 3
1.2 Continuous Guidance . . . 5
1.3 Optimal Control Problem Methods Overview . . . 5
1.4 Homotopy Functions And Continuation . . . 7
1.5 Objectives And Research Question . . . 9
1.6 Thesis Outline . . . 10

2 Indirect Optimal Control Problem Formulation **11**
2.1 Optimal Control Problem Statement . . . 11
2.2 Calculus Of Variations and PMP optimality . . . 12
2.3 Shooting Method And Analytic Derivatives . . . 15
2.4 Corners And Interior Point Constraints . . . 16

3 Homotopy And Continuation Methods **19**
3.1 Homotopy Functions . . . 19
3.2 Continuation methods . . . 21
3.2.1 Natural Parameter Continuation Methods . . . 22
3.2.2 Predictor-Corrector Methods . . . 23
3.2.3 Piecewise Linear Methods . . . 24
3.3 State Of The Art . . . 25
3.3.1 RAMSES Mission Feasibility Assessment . . . 26
3.3.2 M-ARGO Mission Feasibility Assessment . . . 27

4 Implicit Time Optimization **29**
4.1 M-ARGO Optimal Control Problem Statement . . . 29
4.2 Lagrange Time-Optimal Formulation . . . 32
4.3 Lagrange Fuel-Optimal Formulation . . . 34
4.4 Corners And Interior Point Constraints . . . 36
4.5 Alternative Formulations . . . 38
4.5.1 Mayer Time-Optimal Formulation . . . 38
4.5.2 Mayer Fuel-Optimal Formulation . . . 39
4.6 Requirements To Implicit Time Optimization . . . 40
4.6.1 Throttle Requirements . . . 41
4.6.2 Primer Vector Requirements . . . 43
4.7 Optimal Costate Path Scaling . . . 44
4.7.1 Validity Assessment . . . 45
4.8 Formalized Formulation Requirements . . . 47
4.8.1 Variable Parameters Dynamics . . . 48
4.8.2 Constant Parameters Dynamics . . . 50

5 Numerical Methodology **55**
5.1 Shooting Method . . . 55
5.1.1 Dimensional Normalization . . . 56
5.1.2 Integration Scheme . . . 57
5.1.3 Switching Detection . . . 57
5.2 Solution Space Exploration Method . . . 58
5.2.1 Time-Optimal Continuation Strategy . . . 58
5.2.2 Fuel And Energy-Optimal Continuation Strategy . . . 60

6 Results **65**
6.1 Variable Parameters Results . . . 65
6.1.1 Porkchop Plots . . . 65
6.1.2 Solution Scaling Evaluation . . . 70
6.2 Constant Parameters Results . . . 75
6.2.1 Porkchop Plots . . . 75
6.2.2 Solution Scaling Evaluation . . . 79
6.2.3 Scaling Parameter Lower-Bound Test . . . 84
6.3 Performance Comparison . . . 91

7 Conclusions And Future Development **93**
7.1 Conclusions . . . 93

7.2 Future Development 94

Bibliography **95**

A Appendix A **99**
A.1 LTO jacobian 99
A.2 LEFO jacobian 101

B Appendix B **103**

List of Figures **109**

List of Tables **113**

List of Symbols **115**

List of Acronyms **119**

Ringraziamenti **121**

1 | Introduction

Over the last few decades, advancements in technology have made access to space significantly easier. Small satellites, particularly CubeSats, have been regarded with increasing popularity as their introduction to the market has reduced the entry-level cost of Low Earth Orbit (LEO) missions by at least one order of magnitude. [15]
Consequently, electric propulsion (EP) systems have assumed a pivotal role in modern space missions: Figure 1.1 presents the example of geosynchronous (GEO) missions, which in a span of around 40 years have seen a 3 fold increase of the relative fraction of EP solutions.
Unlike liquid and solid engines, which are capable of providing high thrust-to-weight ratios but relatively low specific impulses, electric systems provide significantly higher specific impulse at the expense of thrust. They are therefore also characterized by lower fuel consumption rates. These features, together with their high reliability, render them best suitable for long missions with higher payload fraction. [14][15]

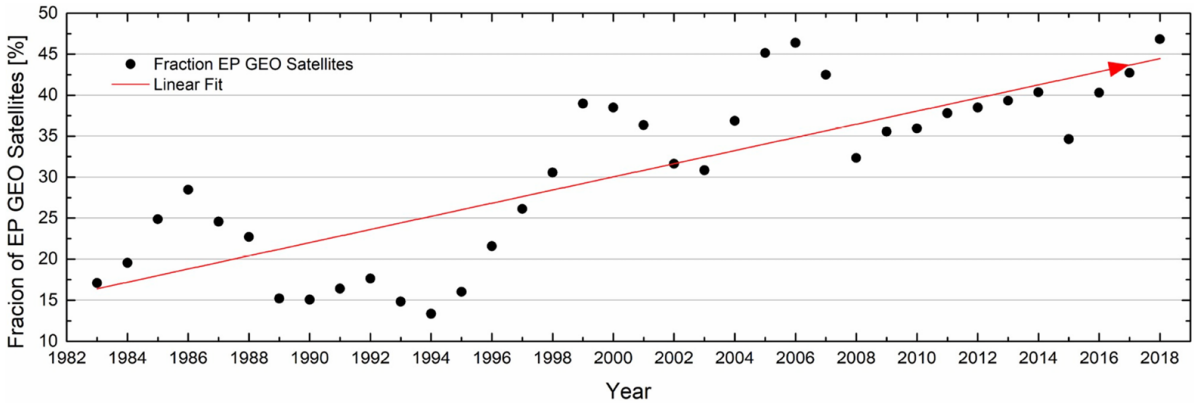


Figure 1.1: Relative fraction of launched EP-based GEO satellites per year between 1981-2018. [14]

Although most commonly employed for near-Earth (NEO) missions, as per Figure 1.2, over the past two decades there has been a growing number of effective deep-space exploration missions exploiting electric propulsion.

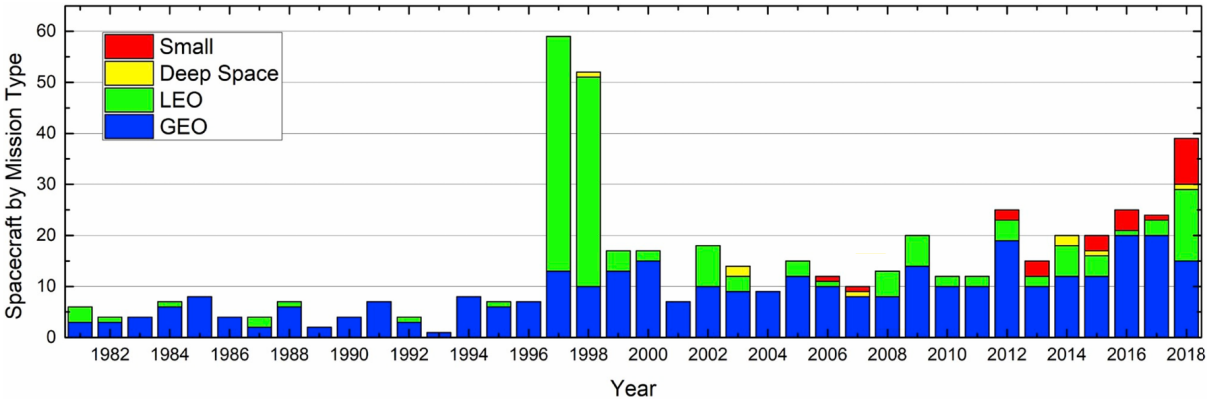


Figure 1.2: EP mission types per year between 1981-2018. [14]

Some notable examples include:

- **Deep Space 1:** Launched in October 1998, the DS1 mission saw the first use of solar-electric propulsion (SEP) on a deep-space mission. Equipped with a single xenon-ion engine, the spacecraft successfully performed the planned flyby of the asteroid Braille in 1999. [5]
- **Hayabusa 1:** Launched in 2003 by JAXA, and propelled by microwave discharge ion engines, the Hayabusa 1 explorer successfully performed a rendezvous with an asteroid in 2005 at a distance of 1.7 AU from the Sun. The robotic spacecraft collected sample materials, which were returned to Earth aboard the satellite in 2010. [13]
- **SMART-1:** Launched in 2003, it demonstrated ESA's first mission employing Hall-effect plasma thruster as the main SEP system. The mission successfully performed an Earth-Moon transfer which ended with deliberate impact on the Moon in 2006. [10]
- **Hayabusa 2:** Successor of Hayabusa 1, and using the same propulsion architecture, Hayabusa 2 was the second successful JAXA asteroid sample return mission, launched in 2014. [21]
- **BepiColombo:** Joint effort of ESA and JAXA, it is an interdisciplinary mission aimed at the exploration of Mercury and its environment. Equipped with SEP ion thrusters, it departed in 2018, planning on reaching Mercury in 2026 after several planetary flybys. [2]

The computation of low-thrust optimal trajectories can be a challenging step in the mission design phase for a spacecraft equipped with electrical propulsion.
Optimal transfers are typically computed and evaluated over large launch time windows and for wide ranges of possible transfer times. Optimality is usually evaluated in terms of transfer time and fuel consumption, and all possible maneuvers are visualized in porkchop plots. Porkchop plots are contour plots that represent the transfer solution space, characterized by the departure date t_0, the time-of-flight ToF, and typically either the expended propellant mass m_p or the imparted Δv.
The generation of a porkchop plot generally involves the solution of thousands of control problems. Therefore, robust and efficient techniques are required to keep the computational effort limited.

1.1. M-ARGO Mission

The Miniaturised Asteroid Remote Geophysical Observer (M-ARGO) is planned to be the first ESA standalone deep-space mission to rendezvous and characterize near-Earth asteroids (NEA).
M-ARGO is a 12U CubeSat planned to piggyback a larger spacecraft destined to the Sun-Earth Lagrange point L_2 (SEL2). From SEL2 orbit, it is set to depart towards a selected NEA using solar electric propulsion (SEP) ion thrusters. [30]

The M-ARGO mission objectives are reported in Table 1.1:

ID	Title	Statement
1	CubeSat Demonstration	Demonstrate the capability of CubeSat nano-spacecraft systems to independently explore deep space for the first time.
2	Scientific Investigation	Rendezvous with a near-Earth asteroid and characterize its physical properties for the presence of in situ resources.
3	Technology Advancement	Advance miniaturized technologies currently under development in Europe.
4	Autonomy Experimentation	Test autonomous guidance, navigation, and control techniques and components performance during transfer to target object.

Table 1.1: M-ARGO mission objectives. [30]

Originally, M-ARGO was set to depart SEL2 between 1 Jan 2023 and 31 Dec 2024, with the maximum transfer time set to 3 years.
The propulsion system is SEP, therefore thrust T_{max} and specific impulse I_{sp} are set as function of the engine input power P_{in}, which in turn is function of the distance from the Sun. Due to technological limits, the thruster input power is bounded by a minimum P_{min} and maximum value P_{max}. At launch, the spacecraft is characterized by total wet mass $m_0 = 22.6\,kg$, and the maximum propellant available available for interplanetary transfer is set to $m_{p,max} = 2.8\,kg$.
The assumed spacecraft data are reported in Table 1.2:

m_0	$m_{p,max}$	P_{min}	P_{max}	T_{max} at $1AU$	I_{sp} at $1AU$
$22.6\,kg$	$2.8\,kg$	$20\,W$	$120\,W$	$1.86\,mN$	$3022.591\,s$

Table 1.2: M-ARGO spacecraft data assumptions. [30]

For the purpose of target selection, all possible NEA objects are subjected to a preliminary filtering process restricting the aphelion r_a with upper bound of $1.25\,AU$, the perihelion r_p with lower bound of $0.75\,AU$ and the inclination i with upper bound of $10°$.
Objects with a number of observations lower than 10 are also filtered out to guarantee sufficient accuracy in the asteroid orbital elements.
The filtered asteroids are subjected to further pruning by evaluation of the minimum transfer time ToF_{min} computed over the selected launch window, and the associated propellant mass m_p.
Among the down-selected targets, five asteroids are chosen as preliminary targets to be used for the purpose of mission and spacecraft design.
The orbital elements of the selected asteroids are reported in Table 1.3:

Name	$a\,[AU]$	$e\,[-]$	$i\,[deg]$	$\omega\,[deg]$	$\Omega\,[deg]$
2000 SG344	0.9775	0.0669	0.1121	275.3026	191.9599
2010 UE51	1.0552	0.0597	0.6239	47.2479	32.2993
2011 MD	1.0562	0.0371	2.4455	5.9818	271.5986
2012 UV136	1.0073	0.1392	2.2134	288.6071	209.9001
2014 YD	1.0721	0.0866	1.7357	34.1161	117.6401

Table 1.3: Sample downselected M-ARGO targets. [30]

The same targets will be used as study cases in this dissertation.

1.2. Continuous Guidance

Due to their intrinsic low-thrust nature, maneuvers performed with electric propulsion systems typically take place over relatively large time spans. Therefore, for the purpose of mission design and control, the maneuvers are generally considered *continuous*.

The objective of designing an optimal continuous maneuver can be summarized as the process of determining the optimal trajectory path and control strategy, guaranteeing the satisfaction of given sets of mission constraints while minimizing certain quantities of interest. Typical quantities of interest are time of transfer, fuel consumption or energy.

In particular, an Optimal Control Problem (OCP) consists in the determination of the optimal state $\mathbf{x}(t) \in \mathbb{R}^n$ and the optimal control $\mathbf{u}(t) \in \mathbb{R}^m$ that, subjected to given constraints, minimize the performance index $J \in \mathbb{R}$. [6][17]
The performance index J can be evaluated either as a function of time or as a terminal function.

Many different methods can be employed in the solution of an OCP. Given the typical high dimension, complexity and non-linearity of systems characteristic to space OCP, numerical methods are inevitably employed.

1.3. Optimal Control Problem Methods Overview

The methods employed for the solution of OCPs can be categorized into three main approach strategies [8][18]:

- **Indirect Methods** employ the Calculus of Variations (CoV) theory to formulate the first-order necessary conditions, i.e. Euler-Lagrange equations, that the state and costate functionals must satisfy for a solution to be optimal. From the Pontryagin Minimum Principle (PMP), the optimal control is determined as function of the system Hamiltonian.
  Indirect methods turn OCPs into Two Point Boundary Value Problems (TPBVP), or Multiple Point Boundary Value Problems (MPBVP) if interior point constraints are enforced.
  The the boundary value problem is then solved as a Zero Finding Problem (ZFP). [6][17]

- **Direct Methods** reduce the infinite dimensional OCP into a finite dimensional Nonlinear Programming Problem (NLP). The discretized state and control variables are to be optimized in accordance with Karush-Kuhn-Tucker optimality conditions, while at the same time satisfying the adjusted equations of motion and given problem constraints. [6]
- **Dynamic Programming Methods** employ the Bellman's Principle of Optimality, which allows for the solution of OCP by use of the Hamilton-Jacobi-Bellman partial differential equations. [18]

In addition, direct and indirect methods typically both involve one of the following techniques: [18]

- **Single shooting:** the trajectory is integrated from initial time t_0 to final time t_f, starting from initial unknowns and imposing final boundary conditions.
- **Multiple shooting:** the time interval $[t_0, t_f]$ is divided into subintervals $[t_i, t_{i+1}]$: the trajectory is integrated over each subinterval, introducing initial unknowns over each time segment, and requiring the enforcement of continuity conditions between segments.
- **Collocation:** the state and costate variables are discretized over subintervals defined by a predefined time grid. The system-governing equations are turned into discrete defect constraints, relating the values at the beginning of each subinterval to the values at the end. The solution is then retrieved by a chosen quadrature rule, allowing for the numerical integration of the dynamics.

Indirect methods guarantee the optimality of the solution, since the equations employed are derived through analytical means. However, such equations can often be cumbersome to determine, and are tied to the specific problem formulation and constraints. Furthermore, the resulting TPBVP is extremely sensible to initial conditions, particularly in the case of single shooting.

Direct methods, on the other hand, are more straightforward to implement and offer greater flexibility in terms of problem formulation. The solution is robust and more tolerant to initial guesses. However, these methods typically require higher computational efforts, and offer no insights on possible improvements to the solution. [8]

This dissertation will focus on solution strategies inherent to indirect methods.

1.4. Homotopy Functions And Continuation

As previously stated, both direct and indirect methods require reasonable initial guess solutions in order to guarantee optimal convergence. In particular, for indirect methods, the formulation is inherently highly sensible to the initial guesses, which can often involve variables that do not bear physical interpretation, such as costate variables and Lagrange multipliers.
To mitigate these issues, several strategies have been developed to allow the formulation of reasonable guess solutions:

- **The Adjoint Control Transformation** (ACT) allows for the estimation of unknown initial costates through reasonable approximations of parameters with clear physical significance. [1][9][26][27]
- **Direct pre-optimization** involves performing a coarse direct optimization of the OCP, with the resulting solution serving as initial guess for the indirect optimizer. [19]
- **Problem convexification** refers to the process of reformulating the OCP into a simplified convex optimization problem by modifying the dynamics or relaxing the constraints. The convex formulation ensures global convergence, and its solution can be used to initialize indirect optimization. [19][35][36]

However, even such strategies may become too computationally expensive or inefficient when it is required to solve thousands of OCPs, i.e. the generation of mission porkchop plots.

To address these challenges, homotopy functions and continuation methods are widely employed in the context of trajectory optimization. In particular, homotopy functions are used to increase the radii of convergence of indirect methods for OCPs, generally exploiting auxiliary problems with simpler formulation or known solution.
In fact, a homotopy is a function in the form $\mathbf{\Gamma}(\kappa, \mathbf{y}) = \mathbf{0}$, where $\mathbf{y}$ is the solution vector, and $\kappa \in [0, 1]$ the parametrization homotopy parameter.
The parameter κ allows for a gradual transformation of the problem formulation, starting with the auxiliary problem for $\kappa = 0$, and progressing to the more complex problem at hand, for $\kappa = 1$.

Continuation methods, on the other hand, are employed to generate reasonable solution guesses by leveraging knowledge of prior solutions of problems of closely related formulation.

They are used to *path-track* the homotopy paths, which consist in the ensemble of solutions of the homotopy function in the $(\kappa, \mathbf{y}) \in \mathbb{R} \times \mathbb{R}^n$ space. More specifically, continuation methods iteratively exploit prior solutions $(\kappa_j, \mathbf{y}_j)$ for $j \leq i$ to construct a solution guess to the next iteration problem $\mathbf{\Gamma}(\kappa_{i+1}, \mathbf{y}) = \mathbf{0}$.

The definition of efficient homotopy and continuation methods is highly problem specific and often challenging. Generally, the more *regular* the homotopy path, the easier it is to track. However, the regularity of the homotopy is rarely known a priori, and many different criticalities may arise: the homotopy path may present turning points, bifurcations, it may fail to exist for some values of κ, or diverge without reaching either $\kappa = 1$ or $\kappa = 0$. [22][23][33][37]

The study of homotopy functions followed the development of continuation methods. In fact, homotopy theory provides the mathematical formalism describing numerical continuation processes. However, it was only in the late 20th century that homotopy theory began receiving significant attention, with efforts focused on fully exploiting its potential by developing a comprehensive theoretical framework.

In particular, first proposed in by Chow et al. [7] in 1978, probability-one homotopy methods emerged as a concept of particular interest. A probabily-one homotopy, in fact, guarantees the existence of finite-length homotopy path linking $\{0\} \times \mathbb{R}^n$ to $\{1\} \times \mathbb{R}^n$, and ensures close to global convergence for arbitrary starting conditions.

Additionally, in 2022 Wang et al. [33] introduced the concept of employing the Theory of Functional Connections (TFC) to generate probability-one homotopy functions. The TFC builds upon the Multivariate Theory of Connections [20], which itself expands on topological concepts proper to the theory of connections.
A TFC homotopy function is defined to automatically satisfy the boundary conditions for $\kappa = 0$ and $\kappa = 1$, and is constructed to be able to regularize the homotopy path, by detecting and recovering from improperly defined paths.

These works are of particular importance as they outline methodologies to be applied in generality to OCPs.

1.5. Objectives And Research Question

The generation of accurate porkchop plots is a critical step in the mission analysis process, and, as previously stated, this process often requires the computation of several thousands of solutions in the form of time (TO) and fuel-optimal (FO) problems.
It is evident in the literature that while fuel and energy-optimal (EO) problems are frequently treated as a single energy-to-fuel-optimal (EFO) formulation through homotopic approaches [30][31][31][32][39], time-optimal problems are generally regarded as a separate topic.
In fact, there appears to be a gap in the literature regarding studies on the possibility of linking TO problems to either EO or FO problems.

In 2018, Pan et al. [24] proposed a quadratic homotopy method to link fixed-time full-thrust problems to FO solutions. It is worth noting that the initial auxiliary problem is not strictly time-optimal, as the transfer time is fixed; however, it shares the same formulation.
The novel homotopy function is inspired by the concept of Bézier curves, which are defined by a set of control points, with the first and last serving as the endpoints of the curve. Pan sets the modified TO problem and the FO problem as the first and last endpoints, respectively, and adopts the quadratic Bézier curve formulation to continuously transition from the auxiliary problem, leveraging the EO formulation, to the FO solution.

In 2021, Li et al. [16] proposed a homotopy method to link a minimum-thrust problem to a FO solution. It's pointed out that in constant thrust TO formulation, thrust and time-of-flight can be treated as interchangeable problem unknowns. The FO solution was approached using a combination of thrust continuation and exponential smoothing of the thrust profile.

Using the M-ARGO mission [30] as case study, and employing Politecnico di Milano Low-Thrust Trajectory Optimizer LT2.0 for results validation [29], this dissertation attempts to establish a connection between TO and FO problems, as well as between TO and EO problems. In particular, this effort aims to address convergence issues characteristic of both FO and EO formulations for transfer times close to the minimum set by the TO solution.

Specifically, it is argued that TO problems are but a limit case to FO and EO problems: this dissertation seeks to establish a solution, possibly in closed-form, connecting the TO solution to those of the FO and EO problems, respectively, which implicitly minimize the transfer time.

Specifically, given $ToF^*(t_0)$ the minimum transfer time and $\boldsymbol{\lambda}^*_{TO}(t_0)$ the costate solution to the TO problem for departure date t_0, given $\boldsymbol{\lambda}^*_{FO}(t_0)$ and $\boldsymbol{\lambda}^*_{EO}(t_0)$ the costate solutions to the FO and EO problems, respectively, characterized by departure date t_0 and transfer time $ToF = ToF^*(t_0)$, the research questions this dissertation aims to answer are:

Research Questions:

I. Does $\mathbf{f}_1$ exist so that $\boldsymbol{\lambda}^*_{FO}(t_0) = \mathbf{f}_1(\boldsymbol{\lambda}^*_{TO}(t_0))$?

II. Does $\mathbf{f}_2$ exist so that $\boldsymbol{\lambda}^*_{EO}(t_0) = \mathbf{f}_2(\boldsymbol{\lambda}^*_{TO}(t_0))$?

1.6. Thesis Outline

This dissertation aims to address and answer the research question formulated in Section 1.5. In particular, in Chapter 2 the tools necessary to address an indirect low-thrust optimal control problem are provided. The theory regarding the Calculus of Variations and Pontryagin Minimum Principle are explored to allow for the definition of bounded-control problems with arbitrary boundary, internal and path constraints.
In Chapter 3 the homotopy theory and the strategies employed for numerical continuation are presented, in order to anticipate and understand the methods employed in the generation of porkchop plots. Some practical examples of porkchop plot design strategies are provided, including the one adopted for M-ARGO.
In Chapter 4 the TO, FO and EO optimal control problems are formulated with regard to the system dynamics and constraints characteristic of the M-ARGO mission. Alternative formulations to the TO and FO problems are also evaluated. A possible answer to the research question is formulated, and its validity and applicability are tested in the context of the given problem formulations. In particular, it is stated that a closed form relationship may exist, allowing to scale the TO solution into those of the FO and EO problems that implicitly optimize the time, but only in the modified case of constant propulsive parameters.
Chapter 5 presents the numerical strategies employed to the solution and the continuation of the problems presented in Chapter 4. The results associated to the porkchop plots generated are presented in Chapter 6, and employed to verify the veracity of the assumptions advanced in Chapter 4. The variable and constant parameters cases are separately tested to verify the possibility and assess the performance of the scaling of the interesting solutions. Additional proof to the hypotheses of Chapter 4 is presented in Appendix B.
The dissertation is concluded with Chapter 7, which briefly summarizes the steps of the analysis conducted and the experimental results, and presents hints for future work.

2 | Indirect Optimal Control Problem Formulation

As previously mentioned, electric engines have been regarded with increasing popularity in recent decades as effective in-space propulsion systems for both near-Earth and deep-space applications.
Electric thrusters are classified as Low-Thrust (LT) systems, with state-of-the-art engines typically operating within the thrust range of $T_{max} \approx 10^{-5 \div 1}\, N$, with specific impulse spanning $I_{sp} \approx 10^{2 \div 4}\, s$. [38]
Due to their inherent low thrust, these engines must operate for extended periods of time in order to impart significant accelerations. As a result, traditional impulsive guidance methods are not suitable for mission analysis involving electric propulsion. Instead, low-thrust trajectory optimization relies on continuous guidance, requiring the formulation and solution of optimal control problems.

This chapter introduces the fundamental concepts underlying the theory of optimal control problems in their generic formulation. These concepts are then applied to the solution of TO, FO and EO problems in Chapter 4.

2.1. Optimal Control Problem Statement

In the context of LT electric propulsion, the process of designing an optimal control for space problems involves generating the control history $\mathbf{u}(t)$ so that the trajectory $\mathbf{x}(t)$ is optimal. The optimality is evaluated in terms of the cost function $J \in \mathbb{R}$ to be minimized, which can be presented in the Bolza Form as [6][17]:

$$J = \phi\left(t_0, \mathbf{x}(t_0), t_f, \mathbf{x}(t_f)\right) + \int_{t_0}^{t_f} L\left(\mathbf{x}(t), \mathbf{u}(t), t\right) dt \tag{2.1}$$

where $\mathbf{x}(t) \in \mathbb{R}^n$ and $\mathbf{u}(t) \in \mathbb{R}^m$ are the state and control variables, respectively, while t_0 and t_f the initial and final time.

The Bolza's formulation can be split into its two components, namely:

- $\phi\left(t_0, \mathbf{x}\left(t_0\right), t_f, \mathbf{x}\left(t_f\right)\right)$ The Mayer's formulation, involving the terminal cost only
- $L\left(\mathbf{x}\left(t\right), \mathbf{u}\left(t\right), t\right)$ The Lagrange's formulation, involving the path cost only

It should be noted that the formulations are interchangeable: the choice changes the problem in its structure, but not in the solution. [17]
It should also be highlighted that both t_0 and t_f can be unknowns of the problem.

The optimization process is subjected to dynamical constraints given by the dynamics of the system [6][17]:

$$\dot{\mathbf{x}}\left(t\right) = \begin{Bmatrix} \dot{\mathbf{r}} \\ \dot{\mathbf{v}} \\ \dot{m} \end{Bmatrix} = \mathbf{f}\left(\mathbf{x}\left(t\right), \mathbf{u}\left(t\right), t\right) \tag{2.2}$$

The optimal trajectory might also be subjected to boundary constraints, both initial $\boldsymbol{\psi}_0 \in \mathbb{R}^p$ and final $\boldsymbol{\psi}_f \in \mathbb{R}^q$, and path constraints $\mathbf{C} \in \mathbb{R}^r$. In particular [6][17]:

$$\begin{cases} \boldsymbol{\psi}_0\left(t_0, \mathbf{x}\left(t_0\right)\right) = \mathbf{0} \\ \boldsymbol{\psi}_f\left(t_f, \mathbf{x}\left(t_f\right)\right) = \mathbf{0} \\ \mathbf{C}\left(\mathbf{x}\left(t\right), \mathbf{u}\left(t\right), t\right) \leq \mathbf{0}, \quad \forall t \in \left[t_0, t_f\right] \end{cases} \tag{2.3}$$

The optimal control problem is therefore formalized as the minimization of Equation (2.1), subject to Equation (2.2) and Equation (2.3).

2.2. Calculus Of Variations and PMP optimality

The Calculus of Variations (CoV) is a field of mathematical analysis involved in the computation of the analytic criteria for functional optimality. It requires the computation of the first order variation of the cost function J, to establish the necessary conditions for optimality.
The method firstly entails the definition of the augmented cost function $\bar{J}$, which accounts for the effects of the imposed constraints. To this purpose, appropriate sets of constant and functional multipliers are added to the formulation [6][17]:

$$\bar{J} = \phi\left(t_0, \mathbf{x}\left(t_0\right), t_f, \mathbf{x}\left(t_f\right)\right) + \boldsymbol{\nu}_0^\top \boldsymbol{\psi}_0\left(t_0, \mathbf{x}\left(t_0\right)\right) + \boldsymbol{\nu}_f^\top \boldsymbol{\psi}_f\left(t_f, \mathbf{x}\left(t_f\right)\right) + \\ + \int_{t_0}^{t_f} \left[L\left(\mathbf{x}, \mathbf{u}, t\right) + \boldsymbol{\lambda}^\top\left(t\right)\left(\mathbf{f}\left(\mathbf{x}, \mathbf{u}, t\right) - \dot{\mathbf{x}}\right) + \boldsymbol{\mu}^\top\left(t\right)\mathbf{C}\left(\mathbf{x}, \mathbf{u}, t\right)\right] dt \tag{2.4}$$

The time dependence on the state $\mathbf{x}$ and control variables $\mathbf{u}$ is henceforward dropped to simplify notation.

The system dynamics are incorporated through the introduction of Lagrange multipliers, or costates, $\boldsymbol{\lambda}\left(t\right) \in \mathbb{R}^n$, which are equal in number to the state variables. Additional multipliers $\boldsymbol{\nu}_0 \in \mathbb{R}^p$, $\boldsymbol{\nu}_f \in \mathbb{R}^q$ and $\boldsymbol{\mu}\left(t\right) \in \mathbb{R}^r$ are introduced to handle initial and final boundary constraints, as well as path constraints, respectively.
Specifically, $\boldsymbol{\nu}_0$ and $\boldsymbol{\nu}_f$ are constant, as they refer to boundary conditions. On the other hand [6]:

$$\begin{cases} \boldsymbol{\mu}\left(t\right) = 0 & \text{if} \quad \mathbf{C}\left(\mathbf{x}, \mathbf{u}, t\right) < \mathbf{0} \\ \boldsymbol{\mu}\left(t\right) > 0 & \text{if} \quad \mathbf{C}\left(\mathbf{x}, \mathbf{u}, t\right) = \mathbf{0} \end{cases} \tag{2.5}$$

The first-order optimality conditions are determined by ensuring stationarity of the augmented cost $\bar{J}$. This is achieved by imposing the condition $\delta\bar{J} = 0$, to negate first-order variations.
The Hamiltonian of the problem is defined as follows [6][17]:

$$H\left(\mathbf{x}, \mathbf{u}, t\right) = L\left(\mathbf{x}, \mathbf{u}, t\right) + \boldsymbol{\lambda}^\top\left(t\right)\mathbf{f}\left(\mathbf{x}, \mathbf{u}, t\right) + \boldsymbol{\mu}^\top\left(t\right)\mathbf{C}\left(\mathbf{x}, \mathbf{u}, t\right) \tag{2.6}$$

It can therefore be proven that imposing first-order optimality is equivalent to enforcing the Euler-Lagrange equations, defined as [6][17]:

$$\dot{\mathbf{x}} = \left(\frac{\partial H}{\partial \boldsymbol{\lambda}}\right)^\top = \mathbf{f}\left(\mathbf{x}, \mathbf{u}, t\right) \tag{2.7a}$$

$$\dot{\boldsymbol{\lambda}} = \left(-\frac{\partial H}{\partial \mathbf{x}}\right)^\top = -\left(\frac{\partial L}{\partial \mathbf{x}}\right)^\top - \left(\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right)^\top \boldsymbol{\lambda} - \left(\frac{\partial \mathbf{C}}{\partial \mathbf{x}}\right)^\top \boldsymbol{\mu} \tag{2.7b}$$

$$\mathbf{0} = \left(\frac{\partial H}{\partial \mathbf{u}}\right)^\top = \left(\frac{\partial L}{\partial \mathbf{u}}\right)^\top + \left(\frac{\partial \mathbf{f}}{\partial \mathbf{u}}\right)^\top \boldsymbol{\lambda} + \left(\frac{\partial \mathbf{C}}{\partial \mathbf{u}}\right)^\top \boldsymbol{\mu} \tag{2.7c}$$

To be solved in concordance with the boundary and transversality conditions [6][17]:

$$\boldsymbol{\psi}_0\left(t_0, \mathbf{x}\left(t_0\right)\right) = \mathbf{0} \tag{2.8a}$$

$$\boldsymbol{\psi}_f\left(t_f, \mathbf{x}\left(t_f\right)\right) = \mathbf{0} \tag{2.8b}$$

$$\mathbf{C}\left(\mathbf{x}\left(t\right), \mathbf{u}\left(t\right), t\right) \leq \mathbf{0}, \quad \forall t \in [t_0, t_f] \tag{2.8c}$$

$$\boldsymbol{\lambda}\left(t_0\right) = -\left(\frac{\partial \phi}{\partial \mathbf{x}} + \boldsymbol{\nu}_0^\top \frac{\partial \boldsymbol{\psi}_0}{\partial \mathbf{x}}\right)^\top \Bigg|_{t=t_0} \tag{2.8d}$$

$$\boldsymbol{\lambda}\left(t_f\right) = \left(\frac{\partial \phi}{\partial \mathbf{x}} + \boldsymbol{\nu}_f^\top \frac{\partial \boldsymbol{\psi}_f}{\partial \mathbf{x}}\right)^\top \Bigg|_{t=t_f} \tag{2.8e}$$

$$\Omega_0 = \left(\frac{\partial \phi}{\partial t} + \boldsymbol{\nu}_0^\top \frac{\partial \boldsymbol{\psi}_0}{\partial t} - H\right)\Bigg|_{t=t_0} = 0 \tag{2.8f}$$

$$\Omega_f = \left(\frac{\partial \phi}{\partial t} + \boldsymbol{\nu}_f^\top \frac{\partial \boldsymbol{\psi}_f}{\partial t} + H\right)\Bigg|_{t=t_f} = 0 \tag{2.8g}$$

The set of Differential Algebraic Equations (DAE), Equation (2.7), together with the complete set of boundary conditions, Equation (2.8), completely characterize the OCP as a TPBVP in its unknowns $\mathbf{x}\left(t\right)$, $\mathbf{u}\left(t\right)$, $\boldsymbol{\lambda}\left(t\right)$, $\boldsymbol{\mu}\left(t\right)$, $\boldsymbol{\nu}_0$, $\boldsymbol{\nu}_f$, t_0 and t_f. [6][17]
It is worth pointing out that the transversality conditions, namely Equation (2.8f) and Equation (2.8g), are dropped if either t_0 and t_f are specified, respectively.

The set of DAEs, Equation (2.7c), allow for the definition of the optimal control $\mathbf{u}\left(t\right)$ when it is unbounded $\mathbf{u}\left(t\right) \in \mathbb{R}^m$. Realistically, in most space applications it is instead bounded to the space of admissible controls $\mathcal{U}$. In such cases, Equation (2.7c) is replaced through the Pontryagin Minimum Principle (PMP) with [6][17]:

$$\mathbf{u}\left(t\right) = \arg\min_{\mathbf{u} \in \mathcal{U}} H\left(\mathbf{x}, \mathbf{u}, \boldsymbol{\lambda}, \boldsymbol{\mu}, t\right) \tag{2.9}$$

Thus, optimal control minimizes the Hamiltonian. Generally, the set $\mathcal{U}$ is formulated as:

$$\mathcal{U} = \{(u, \hat{\boldsymbol{\alpha}}): \quad \mathbf{u} = u\hat{\boldsymbol{\alpha}},\, u \in [0, 1],\, \|\hat{\boldsymbol{\alpha}}\| = 1\} \tag{2.10}$$

where the control $\mathbf{u}$ is decomposed in its magnitude, the throttle u, and direction $\hat{\boldsymbol{\alpha}}$.

Through the CoV it is also possible to generate criteria to verify the optimality of a solution. Firstly, by combination of Equation (2.6) and Equation (2.7), it is found that the optimal solution, denoted with superscript $*$, is such that:

$$\frac{dH^*}{dt} = \frac{\partial H^*}{\partial t} \tag{2.11}$$

Which entails that, if H does not explicitly depend on t, it is verified that $\dot{H}^* = 0$. [6][17]
Furthermore, the Legendre-Clebsch Condition is a second-order condition the optimal solution must satisfy to be a minimum. In particular, for optimal interior control $\mathbf{u}^*(t) \notin \partial\mathcal{U}$, it is verified that:

$$H^*_{\mathbf{u}}(t) = 0 \quad \wedge \quad H^*_{\mathbf{uu}}(t) \geq 0 \tag{2.12}$$

Which translates into the requirement for $H^*_{\mathbf{uu}}(t)$ to be positive semi-definite. [17]
Note that the subscript $\mathbf{u}$ indicates the variable of partial differentiation.

2.3. Shooting Method And Analytic Derivatives

One of the possible methodologies for solving the resulting TPBVP is that of the shooting method, either single or multiple. Specifically in the case of single shooting, since the problem involves unknown variables at both the initial t_0 and final time t_f, a guess of the initial unknown parameters has to be generated to allow time-marching integration of the system equations characterized by Equation (2.7). The solution is evaluated a posteriori by checking whether Equation (2.8) is verified.
A shooting function $\boldsymbol{\Gamma}$ is defined to relate the problem unknowns $\boldsymbol{\xi}$ to the final boundary constraints:

$$\boldsymbol{\Gamma}(\boldsymbol{\xi}) = \mathbf{0} \tag{2.13}$$

The TPBVP is therefore in turn transformed into a zero-finding problem, to be solved numerically through iterative corrections, i.e. Newton's Method.
The gradient of the shooting problem is defined as:

$$\mathbf{J}_{\boldsymbol{\Gamma}} = \frac{\partial \boldsymbol{\Gamma}}{\partial \boldsymbol{\xi}} \tag{2.14}$$

To increase the robustness and accuracy of the numerical solution, Equation (2.14) is computed analytically. It is usually function of the State Transition Matrix (STM), which maps small state variations between different time instant, in particular [27]:

$$\mathbf{\Phi}\left(t_i, t_j\right) = \frac{\delta \mathbf{y}\left(t_i\right)}{\delta \mathbf{y}\left(t_j\right)} \quad \text{with} \quad \mathbf{y}\left(t\right) = \begin{Bmatrix} \mathbf{x}\left(t\right) \\ \boldsymbol{\lambda}\left(t\right) \end{Bmatrix} \in \mathbb{R}^{2n} \tag{2.15}$$

The STM is subjected to variational equation [27]:

$$\begin{cases} \dot{\mathbf{\Phi}}\left(t, t_0\right) = \mathbf{D_y F} \Phi\left(t, t_0\right) \\ \mathbf{\Phi}\left(t_0, t_0\right) = \mathbf{I}_{2x \times 2n} \end{cases} \tag{2.16}$$

where $\mathbf{F} = \dot{\mathbf{y}} = \begin{Bmatrix} \dot{\mathbf{x}} \\ \dot{\boldsymbol{\lambda}} \end{Bmatrix}$.

2.4. Corners And Interior Point Constraints

Some problem formulations are inherently characterized by discontinuous control variables. This is the case for FO bang-bang control.
Furthermore, in the presence of path constraints, it may be found that constrained and unconstrained arcs may be patched together and, again, that the control variables are discontinuous. Points of discontinuity of the functional $\mathbf{u}\left(t\right)$ are called corners.
The Weierstrass-Erdmann Corner Conditions guarantee that for bounded control in the form of Equation (2.10), the state and costate variables, and the Hamiltonian are continuous. [17]

That is not the case for Interior Point Constraints (IPC), which act on the state variables at an instant t_i [6]:

$$\mathbf{N}\left(t_i, \mathbf{x}\left(t_i\right)\right) = \mathbf{0}, \quad \text{with} \quad t_i \in \left[t_0, t_f\right], \mathbf{N} \in \mathbb{R}^s \tag{2.17}$$

IPCs may introduce discontinuities in the costate variables as well as the Hamiltonian.

In particular, characterizing with t_i^- and t_i^+ variables prior and post discontinuity, respectively, it is found that [6]:

$$\begin{cases} \boldsymbol{\lambda}^\top \left(t_i^-\right) = \boldsymbol{\lambda}^\top \left(t_i^+\right) + \boldsymbol{\pi}^\top \dfrac{\partial N}{\partial \mathbf{x}\left(t_i\right)} \\ H\left(t_i^-\right) = H\left(t_i^+\right) - \boldsymbol{\pi}^\top \dfrac{\partial N}{\partial t_i} \end{cases} \tag{2.18}$$

where $\boldsymbol{\pi} \in \mathbb{R}^s$ is a set of constant Lagrange multipliers.

In the case of corner conditions, as well as interior point constraints, the STM encounters discontinuity. Assuming t_i to be the instant of discontinuity, a correction $\boldsymbol{\Psi}\left(t_i\right)$ is to be applied to the STM. The correction can be computed as [27][31][32][39]:

$$\boldsymbol{\Psi}\left(t_i\right) = \frac{\partial \mathbf{y}\left(t_i^+\right)}{\partial \mathbf{y}\left(t_i^-\right)} = \mathbf{I}_{2n\times 2n} + \frac{\partial \Delta \mathbf{y}}{\partial \mathbf{y}} + \left(\dot{\mathbf{y}}\left(t_i^+\right) - \dot{\mathbf{y}}\left(t_i^-\right) - \Delta \dot{\mathbf{y}}\right) \frac{1}{\dot{S}} \frac{\partial S}{\partial \mathbf{y}} \tag{2.19}$$

where $\Delta \mathbf{y} = \mathbf{y}\left(t_i^+\right) - \mathbf{y}\left(t_i^-\right)$, $\Delta \dot{\mathbf{y}} = \dfrac{\partial \Delta \mathbf{y}}{\partial \mathbf{y}} \dot{\mathbf{y}}\left(t_i^-\right)$, and S is the function determining the discontinuity.

Supposing there are N discontinuities $t_1, t_2, \ldots, t_N$, the STM is computed using the chain rule as [27][31][32][39]:

$$\boldsymbol{\Phi}\left(t_f, t_0\right) = \boldsymbol{\Phi}\left(t_f, t_N^+\right) \boldsymbol{\Psi}\left(t_N\right) \boldsymbol{\Phi}\left(t_N^-, t_{N-1}^+\right) \boldsymbol{\Phi}\left(t_{N-1}\right) \ldots \boldsymbol{\Phi}\left(t_2^-, t_1^+\right) \boldsymbol{\Psi}\left(t_1\right) \boldsymbol{\Phi}\left(t_1^-, t_0\right) \tag{2.20}$$

3 | Homotopy And Continuation Methods

Continuation methods, and later homotopy functions, have become indispensable strategies for solving complex problems. They are particularly well-suited for the solution of non-linear optimal control problems, often characterized by high initial sensitivity and small radii of convergence.
Homotopy theory investigates the possibility of analytic mapping of problems of known or simpler solutions into formulations of higher complexity.
Continuation methods, on the other hand, focus on the numerical strategies required for the navigation of these homotopic transformations, in other words the search of solutions belonging to the homotopy path given one of its extremals.

This chapter explores elements of homotopy theory, as well as common continuation strategies. Examples are provided to illustrate possible applications, particularly in solving indirect formulations and generating porkchop plots.

3.1. Homotopy Functions

A zero-finding problem (ZFP) is formulated as follows:

$$\mathbf{F}(\mathbf{y}) = \mathbf{0} \tag{3.1}$$

where $\mathbf{y} \in \mathbb{R}^n$ is the vector of unknowns and $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^n$ is a function assumed to be *sufficiently* smooth, meaning it is continuous in as many of its derivatives as required. Many numerical methods, i.e. Newton's method, can be used to solve Equation (3.1). The convergence of such methods is however greatly influenced by the provided initial conditions, and the possibility of encountering singularities during iterations. Such issues are highly likely in non-linear systems.

A homotopy function is defined as the function $\mathbf{\Gamma}(\kappa, \mathbf{y}) : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$, with $\kappa \in [0, 1]$, mapping two ZFPs $\mathbf{F}(\mathbf{y})$ and $\mathbf{G}(\mathbf{y})$ as:

$$\mathbf{\Gamma}(1, \mathbf{y}) = \mathbf{F}(\mathbf{y}), \quad \mathbf{\Gamma}(0, \mathbf{y}) = \mathbf{G}(\mathbf{y}) \tag{3.2}$$

The parameter κ is the homotopy parameter, while $\mathbf{G} : \mathbb{R}^n \to \mathbb{R}^n$ is an auxiliary function, defined to be similar to $\mathbf{F}(\mathbf{y})$ and to have solution $\mathbf{y}_0 : \mathbf{G}(\mathbf{y}_0) = \mathbf{0}$ easier to determine.

One commonly used form of homotopy function is the convex homotopy [22][23][33]:

$$\mathbf{\Gamma}(\kappa, \mathbf{y}) = \kappa \mathbf{F}(\mathbf{y}) + (1 - \kappa)\mathbf{G}(\mathbf{y}) \tag{3.3}$$

Based on the definition of $\mathbf{G}$, different types of homotopy can be defined [22][23][33]:

1. Newton homotopy $\mathbf{G}(\mathbf{y}) = \mathbf{F}(\mathbf{y}) - \mathbf{F}(\mathbf{y}_0)$
2. Fixed-point homotopy $\mathbf{G}(\mathbf{y}) = \mathbf{y} - \mathbf{y}_0$
3. Affine homotopy $\mathbf{G}(\mathbf{y}) = \mathbf{A}(\mathbf{y} - \mathbf{y}_0)$, with $\mathbf{A} \in \mathbb{R}^{n \times n}$

The Newton homotopy may give rise to multiple homotopy branches, as multiple solutions may exist for $\mathbf{F}(\mathbf{y}) = \mathbf{F}(\mathbf{y}_0)$ at $\kappa = 0$. Fixed-point and affine homotopies are free of problems regarding multiple starting roots. The fixed-point homotopy however may manifest poor scaling properties, therefore the affine homotopy adds an appropriate scaling factor $\mathbf{A}$. [22]

Under the assumption of regularity, it is possible to define the homotopy curve $\mathbf{c}(\theta) : J \subset \mathbb{R} \to \mathbb{R}^{n+1}$ as the set of possible solutions:

$$\mathbf{c}(\theta) = [\kappa(\theta), \mathbf{y}(\theta)] = \mathbf{\Gamma}^{-1}(\mathbf{0}) \tag{3.4}$$

where θ is the continuation parameter.

With reference to Figure 3.1, the possible homotopy curve types are categorized as [23][33]:

I Closed loop in $(0, 1) \times \mathbb{R}^n$

II Unbounded curve with one endpoint in $\{0\} \times \mathbb{R}^n$ or $\{1\} \times \mathbb{R}^n$

III Curve with both endpoints in $\{0\} \times \mathbb{R}^n$ or $\{1\} \times \mathbb{R}^n$

IV Curve with one endpoint in $\{0\} \times \mathbb{R}^n$ and one in $\{1\} \times \mathbb{R}^n$, which is called zero curve

There exist many different criticalities involving the regularity of a homotopy path. With reference to Figure 3.1, some of the most common are:

A *Bifurcations points:* where the homotopy path branches in continuity

B *Discontinuity points:* where two distinct values $\mathbf{y}(\theta^-) \neq \mathbf{y}(\theta^+)$ exist for the same $\kappa(\theta^-) = \kappa(\theta^+)$

C *Limit points:* where the homotopy parameter κ changes direction $d\kappa(\theta^-)d\kappa(\theta^+) < 0$

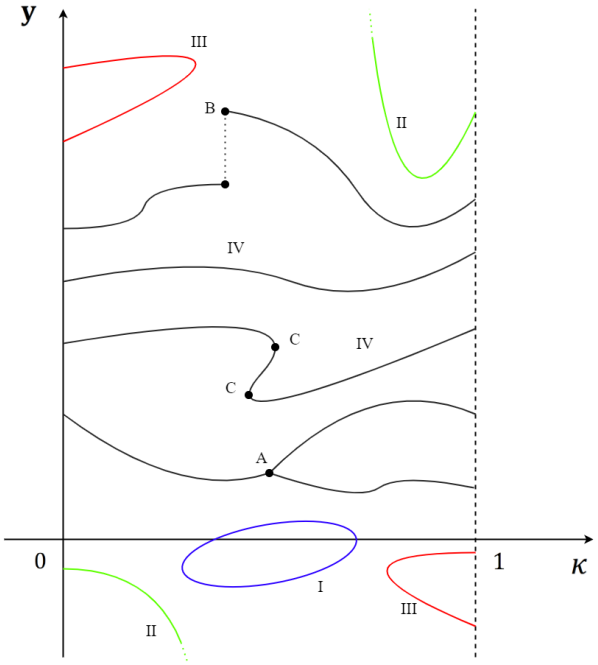


Figure 3.1: Homotopy curve types and critical points.

3.2. Continuation methods

Continuation methods are numerical methods employed to track a homotopy path starting from given solution $(0, \mathbf{y}_0)$ to $(1, \mathbf{y}^*)$, the solution of interest.
There are many available strategies to achieve zero-path following. Among the most common and reliable methods, three main classes can be identified. [11][22][33]

3.2.1. Natural Parameter Continuation Methods

Natural Parameter Continuation Methods (NPCM), also referred to as Direct Continuation Methods (DCM), represent the simplest type of continuation method. The continuation parameter θ is set to κ, and is increased monotonically during the continuation process. A guess solution to the iteration problem $\mathbf{\Gamma}(\kappa_{i+1}, \mathbf{y}) = \mathbf{0}$ is generated as function of prior solutions $(\kappa_j, \mathbf{y}_j)$ for $j \leq i$.

In particular, the 0th order NPCM (0NPCM) initiates the ZFP $\mathbf{\Gamma}(\kappa_{i+1}, \mathbf{y}) = \mathbf{0}$ using as guess the solution to the previous iteration:

$$(\kappa_{i+1}, \mathbf{y}_g): \quad \mathbf{y}_g = \mathbf{y}_i \tag{3.5}$$

Figure 3.2 shows via geometrical interpretation the iterative process of a 0NPCM:

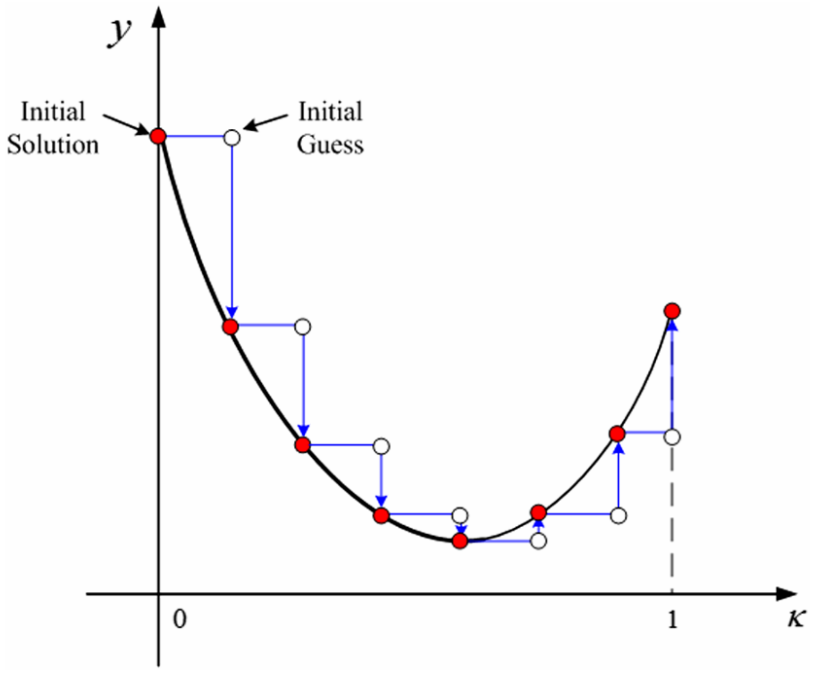


Figure 3.2: Geometric interpretation of 0NPCM. [22]

NPCMs can be constructed with arbitrarily large order exploiting polynomial extrapolation. For example, the 1st order 1NPCM:

$$(\kappa_{i+1}, \mathbf{y}_g): \quad \mathbf{y}_g = \mathbf{y}_i + \frac{\mathbf{y}_i - \mathbf{y}_{i-1}}{\kappa_i - \kappa_{i-1}}(\kappa_{i+1} - \kappa_i) \tag{3.6}$$

NPCMs of order greater than 1 are practically implemented in this work by means of MatLab `makima` function, which performs modified Akima piecewise cubic Hermite interpolation and extrapolation. Modified Akima interpolation is preferred over spline or polynomial as it tends to handle flat regions best, reducing the risk of overshoot.

The use of higher orders may be beneficial to some problems, but introduces issues of instability typical of polynomial extrapolation.
Furthermore, regardless of the order, NPCMs are bound to fail at limit points. [22][33]

3.2.2. Predictor-Corrector Methods

Predictor-Corrector (PC) methods attempt tracking the homotopy path by generating a guess solution (prediction) *sufficiently close* to the homotopy path as to guarantee local convergence via Newton method (correction).

One example is the Pseudo-Arclength Method (PAM), which uses as continuation parameter θ the path arclength s.
Given a former solution $(\kappa_i, \mathbf{y}_i) = (\kappa(s_i), \mathbf{y}(s_i))$ and the normalized tangent vector at such solution $\left(\dot{\kappa}_i, \dot{\mathbf{y}}_i\right) = \left(\dot{\kappa}(s_i), \dot{\mathbf{y}}(s_i)\right)$, the method generates a guess solution for both $\mathbf{y}$ and κ so that [22][33]:

$$(\mathbf{y}_g - \mathbf{y}_i)^\top \dot{\mathbf{y}}_i + (\kappa_g - \kappa_i)\dot{\kappa}_i - \Delta s = 0 \tag{3.7}$$

Where $\Delta s = s_{i+1} - s_i$.
Each iteration therefore involves solving the augmented system [22][33]:

$$\begin{cases} \mathbf{\Gamma}(\kappa, \mathbf{y}) = \mathbf{0} \\ (\mathbf{y} - \mathbf{y}_i)^\top \dot{\mathbf{y}}_i + (\kappa - \kappa_i)\dot{\kappa}_i - \Delta s = 0 \end{cases} \tag{3.8}$$

The augmented jacobian $\mathbf{J}_a$ of system Equation (3.8), evaluated at $(\kappa_i, \mathbf{y}_i)$, is [22][33]:

$$\mathbf{J}_a(\kappa_i, \mathbf{y}_i) = \begin{bmatrix} \dfrac{\partial \mathbf{\Gamma}(\kappa_i, \mathbf{y}_i)}{\partial \mathbf{y}} & \dfrac{\partial \mathbf{\Gamma}(\kappa_i, \mathbf{y}_i)}{\partial \kappa} \\ \dot{\mathbf{y}}_i^\top & \dot{\kappa}_i \end{bmatrix} \tag{3.9}$$

The PAM is more cumbersome in implementation over NPCM methods, but has broader convergence domain: $\mathbf{J}_a$ is in fact generally regular, and by construction the method is able to pass limit points. [4][22][33]

This feature is highlighted in Figure 3.3:

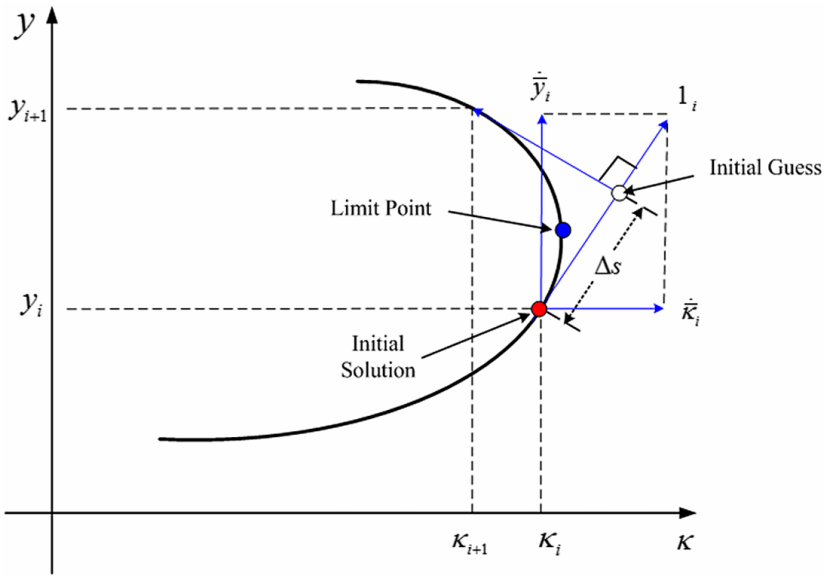


Figure 3.3: Geometric interpretation of PAM. [22]

Arguably, NPCMs can be classified as PC methods as they perform the same steps of solution prediction and correction. However, they are traditionally treated as a separate class.

Both NPCMs and PAM methods are particularly diffused in space applications. For the purpose of the dissertation only NPCMs will be further explored, as sufficient in performance for the application.

3.2.3. Piecewise Linear Methods

Piecewise-Linear (PL) methods attempt a piecewise linear approximation of the homotopy map by subdivision of the search space into cells, called triangulations or simplices. Usually slower and less efficient than the aforementioned methods, PL methods can be advantageous as they impose really low requirements on the homotopy function formulation. [11]

3.3. State Of The Art

As in the past decades electric propulsion has become an established and widely adopted propulsive solution for many different mission types, at the same time numerous strategies and solution techniques have become widespread, and the reliability of such methods continuously increases.
At the same rate, many of the challenges inherent to the indirect optimization problems have been well-defined, and, consequently, effective strategies have been developed to address them.

In the context of porkchop plots generation, the focus pivots on the solution of time-optimal (TO) and fuel-optimal (FO) trajectories.
TO solutions are characterized by constant thrust logic and are, therefore, relatively simple compared to FO solutions. Assuming the transfer time and number of revolutions associated with the transfers are limited, the search and continuation of TO solutions generally does not pose significant difficulties: NPCMs or PAM are typically sufficient for this purpose.
On the other hand, FO solution are much more complex in nature, as they are characterized by discontinuous thrust logic (Bang-Bang control). The presence of such discontinuities heavily increases the sensibility of the solution to initial conditions, and introduces challenges associated to the correct evaluation and handling of state and costate variables at discontinuity points. Consequently, convergence and continuation of FO solution often proves challenging, even when employing high reliability continuation strategies.
For both TO and FO solution, convergence may become even more challenging if additional constraints are introduced to the problem formulation.

Below, a brief overview of the most common and established continuation techniques employed in solution of OCPs:

- **Thrust continuation:** given a LT optimization problem, the minimum associated time-of-flight is tied to the maximum thrust available. In particular, a quasi-monotonic relationship is often established between T_{max}/m_0 and ToF, which entails higher ToF for lower T_{max}/m_0. [34][39]
  Since the sensitivity of TO problems increases with the transfer time, it is common to approach the problem solution starting from a shorter and simpler trajectory characterized by $T'_{max} > T_{max}$, and gradually morphing the problem with lower thrust until $T'_{max} = T_{max}$. [34][35][39]

- **Energy-to-Fuel-optimal homotopy:** Energy-optimal (EO) problems are similar in formulation to FO problems, but are characterized by continuous thrust logic. A common procedure involves approaching the FO solution via homotopic continuation, starting from the associated EO one. [30][31][32][39]
  The associated homotopy function acts as a quadratic penalty to the FO problem formulation [3], and is *embedded*, as the homotopy parameter is incorporated into the resulting system dynamics.

- **Smoothing techniques:** Discontinuities proper to a problem formulation can be approximated during continuation by use of smoothing techniques. [3] One widely used example is that of Hyperbolic Tangent Smoothing (HTS), belonging to the family of exponential smoothing functions.
  HTS has a wide range of applications, including the approximation of the Bang-Bang thrust profiles [1][16][28], or the smoothing of discontinuous thrust parameters [31][32].

Continuation and homotopy strategies, along with guess-generation techniques, can be employed individually or in combination, offering a wide range of flexible approaches for low-thrust mission analysis.
The two following sections illustrate the examples of the RAMSES and M-ARGO case studies.

3.3.1. RAMSES Mission Feasibility Assessment

The ESA Rapid Apophis Mission for SEcurity and Safety (RAMSES) is designed to rendezvous with (99942) Apophis, a potentially hazardous asteroid that will make a close approach to Earth in April 2029. With an earliest departure set for 2026, the mission involves a direct transfer using an electric engine as main propulsion, aiming to reach the asteroid before its flyby and study any changes induced by the Earth's interaction. [19]

The porkchop plots for the mission are generated via the combination of direct convex programming and indirect solution refinement.
First, the FO solution associated with earliest departure date t_0 and longest time-of-flight ToF is computed via a direct sequential convex programming algorithm. The (t_0, ToF) solution space is discretized into an uneven $M \times N$ grid. Then, in continuation from neighboring solutions, starting from the first departure date, the problems are solved for decreasing ToF until convergence halts. Subsequently, re-starting from the longest time-of-flight, a new solution is attempted for a new departure date t_0. The process iterates until the whole FO solution space is explored.

Afterwards, for each departure date, the solution of the FO problem with lowest ToF is used as the guess for the TO problem.

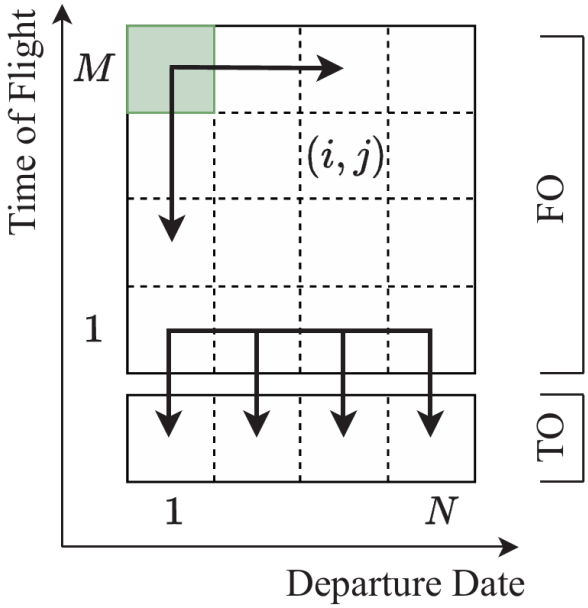


Figure 3.4: RAMSES mission porkchop plot solution space exploration strategy. [19]

Lastly, the solutions obtained with the direct convex step are used as initial conditions for an indirect algorithm, and the final porkchop plot is generated. [19]

3.3.2. M-ARGO Mission Feasibility Assessment

As stated in Section 1.1, the M-ARGO mission involves a LT transfer from SEL2 to a chosen NEA. The generation of mission porkchop plot is approached solely employing indirect optimization and continuation.
Firstly, the TO problem is solved for the earliest departure date t_0. The guess solution to the first problem is generated with the ACT. [9][27] The departure date is then increased with adaptive time step Δt_0, and new solutions are found through 0NPCM continuation, until the entire launch window has been scanned.
For each departure date t_0, the TO solution, characterized by $ToF = ToF^*(t_0)$, is used as initial guess for the FO problem associated with $(t_0, ToF^*(t_0))$. The time-of-flight of the FO solution is then gradually increased with adaptive step ΔToF through 0NPCM continuation, until the maximum ToF is reached. [30]
Figures 3.5 and 3.6 offer qualitative insight on the continuation strategies adopted.

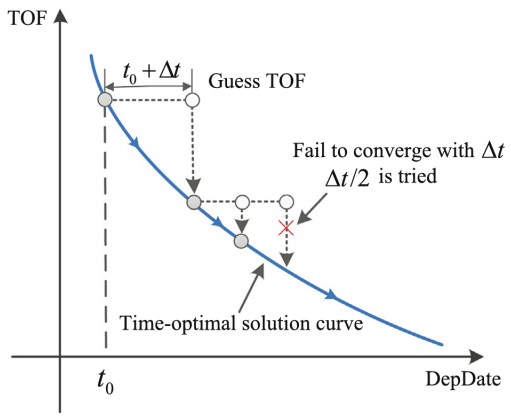


Figure 3.5: M-ARGO departure date t_0 continuation scheme for TO solutions. [30]

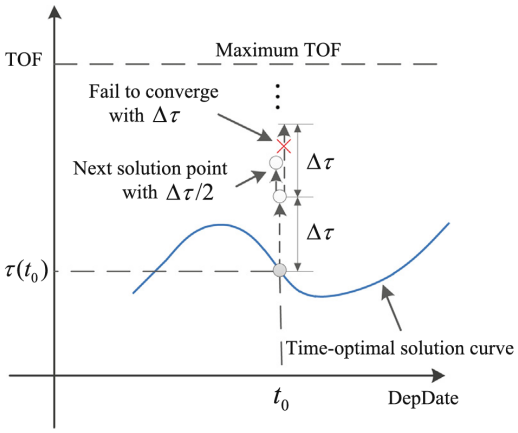


Figure 3.6: M-ARGO time-of-flight $\tau = ToF$ continuation scheme for FO solutions. [30]

4 | Implicit Time Optimization

The M-ARGO mission is taken as the case study in this dissertation. [30][31][32]
In this chapter, the CoV theory explored in Chapter 2 is applied to the derivation of the TO, FO and EO indirect formulations, in compliance with the constraints characteristic of the M-ARGO mission.
A possible answer to the research question is advanced, together with the analytical requirements regarding its applicability. In particular, a set of formal conditions is formulated to allow for the possibility of solution scaling between the optimal costates of TO and implicitly time-optimal problems.
It is then verified whether this technique is exploitable in the context of the M-ARGO mission.

4.1. M-ARGO Optimal Control Problem Statement

The heliocentric two-body problem (2BP) is used to describe the spacecraft dynamics, which in the Sun-centered ecliptic J2000 frame read [30][31][32]:

$$\dot{\mathbf{x}} = \begin{Bmatrix} \dot{\mathbf{r}} \\ \dot{\mathbf{v}} \\ \dot{m} \end{Bmatrix} = \mathbf{f}\left(\mathbf{x}, u, \hat{\boldsymbol{\alpha}}\right) = \begin{Bmatrix} \mathbf{v} \\ -\mu\dfrac{\mathbf{r}}{r^3} + u\dfrac{T_{max}}{m}\hat{\boldsymbol{\alpha}} \\ -u\dfrac{T_{max}}{I_{sp}g_0} \end{Bmatrix} \tag{4.1}$$

where μ is the Sun gravitational constant, g_0 is Earth gravitational acceleration at sea level, and u and $\hat{\boldsymbol{\alpha}}$ the throttle and thrust direction, respectively, as per Equation (2.10).

Given t_0 and t_f the departure and arrival dates, respectively, the trajectory is subjected to the following boundary conditions:

$$\boldsymbol{\psi}_0\left(\mathbf{x}\left(t_0\right)\right) = \begin{Bmatrix} \mathbf{r}\left(t_0\right) - \mathbf{r}_0 \\ \mathbf{v}\left(t_0\right) - \mathbf{v}_0 \\ m\left(t_0\right) - m_0 \end{Bmatrix} \tag{4.2a}$$

$$\boldsymbol{\psi}_f\left(\mathbf{x}\left(t_f\right)\right) = \begin{Bmatrix} \mathbf{r}\left(t_f\right) - \mathbf{r}_t\left(t_f\right) \\ \mathbf{v}\left(t_f\right) - \mathbf{v}_t\left(t_f\right) \end{Bmatrix} = \boldsymbol{\psi}\left(\mathbf{x}\left(t_f\right)\right) \tag{4.2b}$$

where $[\mathbf{x}_0, \mathbf{v}_0]$ are the SEL2 position and velocity at t_0, and m_0 is the satellite wet mass. On the other hand $[\mathbf{r}_t\left(t_f\right), \mathbf{v}_t\left(t_f\right)]$ are the known time-dependent position and velocity of the selected mission target at t_f, given by ephemerid measurements.

The engine is powered by solar radiation. The function S_p is defined as the power switching function, used to detect thruster operation logic. It is modeled as a polynomial function as [30][31][32]:

$$S_p\left(r\right) = c_0 + c_1 r + c_2 r^2 + c_3 r^3 + c_4 r^4 \tag{4.3}$$

Technological limits are imposed on the input power to the electric engine. Specifically, if S_p is greater than P_{max}, the input power saturates. On the other hand if S_p is lower than P_{min}, the engine switches off and the throttle is set to $u = 0$. [31][32]
In particular, as also shown in Figure 4.1 [30][31][32]:

$$\text{If} \quad S_p \geq P_{max} \quad \text{then} \quad P_{in} = P_{max}, \quad u \in [0, 1] \tag{4.4a}$$

$$\text{If} \quad S_p \in [P_{min}, P_{max}) \quad \text{then} \quad P_{in} = S_p, \quad u \in [0, 1] \tag{4.4b}$$

$$\text{If} \quad S_p < P_{min} \quad \text{then} \quad P_{in} = S_p, \quad u = 0 \tag{4.4c}$$

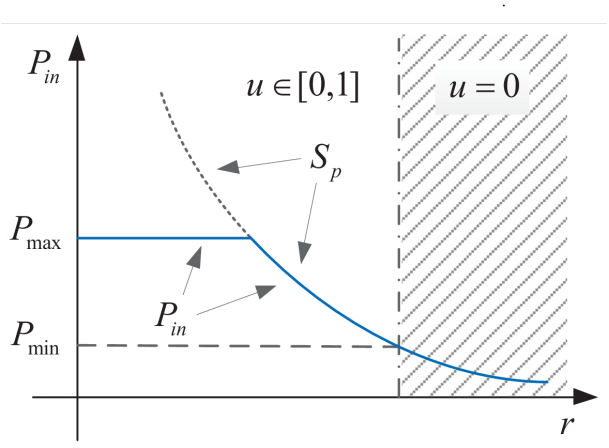


Figure 4.1: Thruster operation logic representation. [31][32]

Both the maximum thrust T_{max} and the specific impulse I_{sp} are functions of the input power P_{in}. Again, the relationships are modeled through polynomials [30][31][32]:

$$T_{max}\left(P_{in}\right) = a_0 + a_1 P_{in} + a_2 P_{in}^2 + a_3 P_{in}^3 + a_4 P_{in}^4 \tag{4.5a}$$

$$I_{sp}\left(P_{in}\right) = b_0 + b_1 P_{in} + b_2 P_{in}^2 + b_3 P_{in}^3 + b_4 P_{in}^4 \tag{4.5b}$$

The coefficients for Equation (4.3) and Equation (4.5) are collected in Table 4.1:

T_{max}	Value	Unit	I_{sp}	Value	Unit	S_p	Value	Unit
a_0	-0.7253	mN	b_0	2652	s	c_0	840.11	W
a_1	0.02481	mN/W	b_1	-18.123	s/W	c_1	-1754.3	W/AU
a_2	0	–	b_2	0.3887	s/W^2	c_2	1625.01	W/AU^2
a_3	0	–	b_3	-0.00174	s/W^3	c_3	-739.87	W/AU^3
a_4	0	–	b_4	0	–	c_4	134.45	W/AU^4

Table 4.1: Thruster model coefficients. [30][31][32]

In the scope of deriving the first-order optimality conditions, it is convenient to define the following gradients:

$$\mathbf{t}_r = \left(\frac{\partial T_{max}}{\partial \mathbf{r}}\right)^\top = \begin{cases} \dfrac{\partial T_{max}}{\partial P_{in}} \dfrac{\partial P_{in}}{\partial r} \left(\dfrac{\partial r}{\partial \mathbf{r}}\right)^\top & \text{if} \quad S_p < P_{max} \\ \mathbf{0}_{3\times 1} & \text{otherwise} \end{cases} \tag{4.6a}$$

$$\mathbf{i}_r = \left(\frac{\partial I_{sp}}{\partial \mathbf{r}}\right)^\top = \begin{cases} \dfrac{\partial I_{sp}}{\partial P_{in}} \dfrac{\partial P_{in}}{\partial r} \left(\dfrac{\partial r}{\partial \mathbf{r}}\right)^\top & \text{if} \quad S_p < P_{max} \\ \mathbf{0}_{3\times 1} & \text{otherwise} \end{cases} \tag{4.6b}$$

As the objective of the dissertation is to explore the possibility of linking the TO to FO and EO problems, particular interest is put on the available problem formulations.

4.2. Lagrange Time-Optimal Formulation

A typical formulation of the time-optimal problem involves the use of the Lagrange cost function. For given t_0 and unknown t_f [30][31][32]:

$$J_{LTO} = \int_{t_0}^{t_f} 1 \, dt \tag{4.7}$$

where the LTO subscript denotes the Lagrange TO problem.

Following the procedure indicated in Chapter 2, the system Hamiltonian is defined as [30][31][32]:

$$H_{LTO} = \boldsymbol{\lambda}^\top \mathbf{f} + 1 \tag{4.8}$$

where $\boldsymbol{\lambda} = [\boldsymbol{\lambda}_\mathbf{r}, \boldsymbol{\lambda}_\mathbf{v}, \lambda_m]^\top \in \mathbb{R}^7$ is the costate vector.

By virtue of the PMP, Equation (2.9), the primer vector $\hat{\boldsymbol{\alpha}}^*$ and optimal throttle factor u^* are defined as [30][31][32]:

$$\hat{\boldsymbol{\alpha}}^* = -\frac{\boldsymbol{\lambda}_\mathbf{v}}{\lambda_\mathbf{v}} \tag{4.9a}$$

$$u^* = \begin{cases} 0, & S_{LTO} > 0 \quad \vee \quad S_p < P_{min} \\ \in [0,1], & S_{LTO} = 0 \quad \wedge \quad S_p \geq P_{min} \\ 1, & S_{LTO} < 0 \quad \wedge \quad S_p \geq P_{min} \end{cases} \tag{4.9b}$$

where S_{LTO} is the throttle switching function, defined as [30][31][32]:

$$S_{LTO} = -\lambda_\mathbf{v} \frac{c}{m} - \lambda_m \tag{4.10}$$

with $\lambda_\mathbf{v} = \|\boldsymbol{\lambda}_\mathbf{v}\|_2$.
Through the PMP the control $\mathbf{u}$ becomes function of the problem costates, and of the distance r by dependence on S_p. Therefore $\mathbf{f}(\mathbf{x}, u, \hat{\boldsymbol{\alpha}}) = \mathbf{f}(\mathbf{x}, \boldsymbol{\lambda})$.
It ought to be noted that the technological constraint of Equation (4.4c) may conflict with PMP optimality.

Given $\mathbf{y} = [\mathbf{x},\, \boldsymbol{\lambda}]^\top$, the Euler-Lagrange equations [31]:

$$\dot{\mathbf{y}} = \mathbf{F}_{LTO}\left(\mathbf{y}\right) = \left\{ \begin{array}{c} \mathbf{v} \\ -\dfrac{\mathbf{r}}{r^3} - u\dfrac{\boldsymbol{\lambda}_\mathbf{v}}{\lambda_\mathbf{v}}\dfrac{T_{max}}{m} \\ -u\dfrac{T_{max}}{c} \\ -\dfrac{3}{r^5}\left(\mathbf{r}^\top\boldsymbol{\lambda}_\mathbf{v}\right)\mathbf{r} + \dfrac{\boldsymbol{\lambda}_\mathbf{v}}{r^3} + u\dfrac{\lambda_\mathbf{v}}{m}\mathbf{t}_r + u\dfrac{\lambda_m}{c}\left(\mathbf{t}_r - \dfrac{T_{max}}{I_{sp}}\mathbf{i}_r\right) \\ -\boldsymbol{\lambda}_\mathbf{r} \\ -u\lambda_\mathbf{v}\dfrac{T_{max}}{m^2} \end{array} \right\} \tag{4.11}$$

with $c = I_{sp}g_0$ indicating the engine exhaust velocity, and $\mathbf{t}_r$, $\mathbf{i}_r$ defined as per Equation (4.6).

The problem is characterized by unknowns $\boldsymbol{\xi}_{LTO} = \left[\boldsymbol{\lambda}^*_{LTO}\left(t_0\right), ToF^*\left(t_0\right)\right]^\top \in \mathbb{R}^8$. The final boundary constraints and shooting function, computed through Equation (2.8) [30]:

$$\boldsymbol{\Gamma}_{LTO}\left(\boldsymbol{\xi}_{LTO}\right) = \left\{ \begin{array}{c} \boldsymbol{\psi}\left(\mathbf{x}\left(t_f\right)\right) \\ \lambda_m\left(t_f\right) \\ \Omega_f \end{array} \right\} \tag{4.12}$$

with Ω_f the transversality condition at terminal time:

$$\Omega_f = H_{LTO}\left(t_f\right) - \boldsymbol{\lambda}^\top_\mathbf{r}\left(t_f\right)\mathbf{v}_t\left(t_f\right) - \boldsymbol{\lambda}^\top_\mathbf{v}\left(t_f\right)\mathbf{a}_t\left(t_f\right) \tag{4.13}$$

The quantity $\mathbf{a}_t\left(t_f\right)$ refers to the target body time-dependent acceleration at terminal time t_f, and is computed by means of the 2BP model approximation as $\mathbf{a}_t = -\mu\dfrac{\mathbf{r}_t}{r_t^3}$.

The jacobian to the TO shooting function $\mathbf{J}_{\boldsymbol{\Gamma}_{LTO}}$ is provided in the appendix Appendix A.1, Equation (A.1).

Note:

It ought to be pointed out that by construction $S_{LTO}(t) \leq 0 \quad \forall t \in [t_0, t_f]$.
In fact, by definition $\lambda_{\mathbf{v}} = \|\boldsymbol{\lambda}_{\mathbf{v}}\|_2 \geq 0$. Additionally, given the constraints of Equation (4.12) and the costate dynamics, Equation (4.11), it is found that:

$$\begin{cases} \dot{\lambda}_m(t) \leq 0 \quad \forall t \in [t_0, t_f] \\ \\ \lambda_m(t_f) = 0 \end{cases} \quad \Rightarrow \quad \lambda_m(t) \geq 0 \quad \forall t \in [t_0, t_f] \tag{4.14}$$

It can therefore be stated that the engine is always turned on with maximum throttle $u = 1$, unless arcs of insufficient power $S_p < P_{min}$ are present.
This characteristic is inherent to the TO problem, regardless of the formulation.

4.3. Lagrange Fuel-Optimal Formulation

Given fixed t_0 and t_f, the fuel-optimal cost function can be formulated in its Lagrange form [30][31][32]:

$$J_{LFO} = \int_{t_0}^{t_f} u \frac{T_{max}}{I_{sp} g_0} dt \tag{4.15}$$

where the LFO subscript denotes the Lagrange FO problem.
Since the FO thrust throttle is by nature discontinuous, the following smoothing technique is employed [3][31][32]:

$$J_{LEFO,\varepsilon} = \int_{t_0}^{t_f} \frac{T_{max}}{I_{sp} g_0} \left[u - \varepsilon u (1 - u) \right] dt \tag{4.16}$$

which yields the Lagrange energy-optimal problem for $\varepsilon = 1$, and the Lagrange fuel-optimal problem for $\varepsilon = 0$. The variable ε is the homotopy parameter used to map the solution of the EO problem into that of the FO one.
The subscript LEFO denotes the Lagrange energy-to-fuel-optimal problem.

The solution is subjected to the same boundary constraints as per the TO problem, namely Equation (4.2).

The system Hamiltonian reads [31][32]:

$$H_{LEFO,\varepsilon} = \boldsymbol{\lambda}^\top \mathbf{f} + \frac{T_{max}}{c} \left[u - \varepsilon u (1 - u) \right] \tag{4.17}$$

The optimal thrust direction $\hat{\boldsymbol{\alpha}}^*$ is defined as per Equation (4.9a), while the optimal throttle factor [31][32]:

$$u^* = \begin{cases} 0, & S_{LEFO,\varepsilon} > \varepsilon \quad \vee \quad S_p < P_{min} \\ \dfrac{\varepsilon - S_{LEFO,\varepsilon}}{2\varepsilon}, & |S_{LEFO,\varepsilon}| \leq \varepsilon \quad \wedge \quad S_p \geq P_{min} \\ 1, & S_{LEFO,\varepsilon} < -\varepsilon \quad \wedge \quad S_p \geq P_{min} \end{cases} \tag{4.18}$$

with switching function $S_{LEFO,\varepsilon}$ [31][32]:

$$S_{LEFO,\varepsilon} = -\lambda_{\mathbf{v}} \frac{c}{m} - \lambda_m + 1 \tag{4.19}$$

Once again, power constraints may conflict with the optimality of the control.
Furthermore, it is worth pointing out that the throttle logic is continuous in $S_{LEFO,\varepsilon}$ for all ε values but $\varepsilon = 0$.

The Euler-Lagrange equations for the LEFO problem are [31]:

$$\dot{\mathbf{y}} = \mathbf{F}_{LEFO,\varepsilon}(\mathbf{y}) = \\ = \begin{Bmatrix} \mathbf{v} \\ -\dfrac{\mathbf{r}}{r^3} - u\dfrac{\boldsymbol{\lambda}_{\mathbf{v}}}{\lambda_{\mathbf{v}}}\dfrac{T_{max}}{m} \\ -u\dfrac{T_{max}}{c} \\ -\dfrac{3}{r^5}\left(\mathbf{r}^\top \boldsymbol{\lambda}_{\mathbf{v}}\right)\mathbf{r} + \dfrac{\boldsymbol{\lambda}_{\mathbf{v}}}{r^3} + u\dfrac{\lambda_{\mathbf{v}}}{m}\mathbf{t}_r + \dfrac{(\lambda_m - 1 + \varepsilon)\,u - \varepsilon u^2}{c}\left(\mathbf{t}_r - \dfrac{T_{max}}{I_{sp}}\mathbf{i}_r\right) \\ -\boldsymbol{\lambda}_{\mathbf{r}} \\ -u\lambda_{\mathbf{v}}\dfrac{T_{max}}{m^2} \end{Bmatrix} \tag{4.20}$$

With respect to the LTO formulation, the LEFO problem is characterized by unknowns $\boldsymbol{\xi}_{LEFO,\varepsilon} = \boldsymbol{\lambda}^*_{LEFO,\varepsilon}(t_0) \in \mathbb{R}^7$, and shooting function [31][32]:

$$\boldsymbol{\Gamma}_{LEFO,\varepsilon}(\boldsymbol{\xi}_{LEFO,\varepsilon}) = \begin{Bmatrix} \boldsymbol{\psi}(\mathbf{x}(t_f)) \\ \lambda_m(t_f) \end{Bmatrix} \tag{4.21}$$

The jacobian to the energy-to-fuel problem shooting function $\mathbf{J}_{\mathbf{\Gamma}_{LEFO}}$ is provided in the appendix Appendix A.2, Equation (A.5).

Despite $\lambda_m(t)$ sharing the same properties as per the LTO case, nothing can be inferred on the sign of $S_{LEFO,\varepsilon}(t)$.

Note:

With reference to Equation (4.19), given an EO path $\mathbf{y}^*_{LEFO,1}(t)$ characterized by constant maximum throttle $u = 1$, it is observed that:

$$S_{LEFO,1}\left(\mathbf{y}^*_{LEFO,1}(t)\right) < -1 \quad \Rightarrow \quad S_{LEFO,0}\left(\mathbf{y}^*_{LEFO,1}(t)\right) < 0 \tag{4.22}$$

Furthermore, if $u = 1$ the Euler-Lagrange equations of the Lagrange EFO formulation described in Equation (4.20) coincide for all values of ε. In fact, using $u = 1$, the only element of Equation (4.20) influenced by ε:

$$\begin{aligned} \dot{\boldsymbol{\lambda}}_{\mathbf{r},LEFO,\varepsilon} &= \cdots + \frac{(\lambda_m - 1 + \varepsilon)\,u - \varepsilon u^2}{c}\left(\mathbf{t}_r - \frac{T_{max}}{I_{sp}}\mathbf{i}_r\right) = \\ &= \cdots + \frac{\lambda_m - 1}{c}\left(\mathbf{t}_r - \frac{T_{max}}{I_{sp}}\mathbf{i}_r\right) \qquad \forall \varepsilon \in [0,\,1] \end{aligned} \tag{4.23}$$

Therefore, an EO solution characterized by constant maximum throttle can be optimal for both the EO and FO problems, as the distinction between the two formulations is lost granted $u = 1 \quad \forall t$.

4.4. Corners And Interior Point Constraints

When $S_p < P_{min}$, the engine switches off ($u = 0$) as the power generated is insufficient. This result is not related to the minimization of the Hamiltonian and therefore might not be optimal in terms of the PMP, Equation (2.9). To ensure the necessary conditions for optimality are met, this event is treated as an interior point constraint. [6][31][32]
It is noted that the power switching function S_p, Equation (4.3), is only dependent on the state variables through the magnitude $r = \|\mathbf{r}\|$, and, in turn, $\partial S_p/\partial t = 0$.
Therefore, with reference to Equation (2.17), assuming S_p crosses P_{min} at t_c, the Hamiltonian and costate corrections read as:

$$\begin{cases} \boldsymbol{\lambda}^\top(t_c^-) = \boldsymbol{\lambda}^\top(t_c^+) + \pi\dfrac{\partial S_p}{\partial \mathbf{r}} \\ H\left(t_i^-\right) = H\left(t_i^+\right) \end{cases} \tag{4.24}$$

The Lagrange multiplier π is a scalar, specifically defined for the time-optimal π_t and the energy-to-fuel-optimal π_ε cases in their Lagrange formulations as [31][32]:

$$\pi_t = \Delta u \frac{T_{max}}{c} \frac{S_{LTO}}{\dot{S}_p} \tag{4.25}$$

$$\pi_\varepsilon = \Delta u \frac{T_{max}}{c} \frac{S_{LEFO,\varepsilon} - \varepsilon + \left(u\left(t_c^+\right) + u\left(t_c^-\right)\right)\varepsilon}{\dot{S}_p} \tag{4.26}$$

where $\Delta u = u\left(t_c^+\right) - u\left(t_c^-\right)$ and $\dot{S}_p = \dfrac{\partial S_p}{\partial \mathbf{r}}\mathbf{v}$.

Furthermore, changes in thrust logic cause STM discontinuities, to be accounted for and corrected as per Equation (2.19). Assuming that the discontinuity detected at t_c is indicated by a the switching function S crossing the threshold η, there are three possible cases [31][32]:

1. *Case 1:* $S = S_{LEFO,\varepsilon}$, $\varepsilon = 0$, $\eta = 0$; u jumps between 0 and 1
2. *Case 2:* $S = S_p$, $u \neq 0$, $\eta = P_{min}$; u jumps between 0 and a non-zero value
3. *Case 3:* $S = S_p$, $\eta = P_{max}$; u does not change, but the costate dynamics are discontinuous

Case 1 and 3 are characterized by continuous $\mathbf{y}$, but discontinuous $\dot{\mathbf{y}}$. According to Equation (2.19) [27][31][32]:

- Case 1

$$\frac{\partial S_{LEFO,\varepsilon}}{\partial \mathbf{y}} = \left[-\mathbf{i}_r^\top \frac{g_0}{m}\lambda_\mathbf{v},\ \mathbf{0}_{1\times 3},\ \frac{c\lambda_\mathbf{v}}{m^2},\ \mathbf{0}_{1\times 3},\ -\frac{c}{m}\frac{\boldsymbol{\lambda}_\mathbf{v}^\top}{\lambda_\mathbf{v}},\ -1\right] \tag{4.27a}$$

$$\dot{S}_{LEFO,\varepsilon} = \frac{c}{m\lambda_\mathbf{v}}\boldsymbol{\lambda}_\mathbf{r}^\top\boldsymbol{\lambda}_\mathbf{v} - \frac{g_0\lambda_\mathbf{v}}{m}\mathbf{i}_r^\top\mathbf{v} \tag{4.27b}$$

- Case 3

$$\frac{\partial S_p}{\partial \mathbf{y}} = \left[\frac{\partial S_p}{\partial \mathbf{r}},\ \mathbf{0}_{1\times 11}\right] \tag{4.28a}$$

$$\dot{S}_p = \frac{\partial S_p}{\partial \mathbf{r}}\mathbf{v} \tag{4.28b}$$

On the other hand, case 2 is characterized by discontinuous $\mathbf{y}$ and $\dot{\mathbf{y}}$. With reference to Equation (2.19), the increment $\Delta\mathbf{y}$ is computed as:

$$\Delta\mathbf{y} = \left[\mathbf{0}_{7\times 1},\ -\pi\left(\frac{\partial S_p}{\partial \mathbf{r}}\right)^\top,\ \mathbf{0}_{4\times 1}\right]^\top \tag{4.29}$$

4.5. Alternative Formulations

As stated in Chapter 2, depending on the form of the cost function J used, alternative equivalent formulations to the same OCP exist.
The TO and EFO problems have so far been presented in their Lagrange formulation. In this section, additional formulations are provided. In particular, the Mayer formulations of the TO and FO problems.

4.5.1. Mayer Time-Optimal Formulation

Below are summarized all the quantities of interest regarding the definition of the TPBVP associated with the Mayer time-optimal (MTO) control problem. In particular, the cost function J_{MTO}, the Hamiltonian H_{MTO}, the switching function S_{MTO}, the optimal throttle logic u^*, the Euler-Lagrange equations $\mathbf{F}_{MTO}$, and shooting function $\mathbf{\Gamma}_{MTO}$:

$$J_{MTO} = t_f \tag{4.30a}$$

$$H_{MTO} = \boldsymbol{\lambda}^\top \mathbf{f} \tag{4.30b}$$

$$S_{MTO} = -\lambda_{\mathbf{v}} \frac{c}{m} - \lambda_m \tag{4.30c}$$

$$u^* = \begin{cases} 0, & S_{MTO} > 0 \quad \vee \quad S_p < P_{min} \\ \in [0,1], & S_{MTO} = 0 \quad \wedge \quad S_p \geq P_{min} \\ 1, & S_{MTO} < 0 \quad \wedge \quad S_p \geq P_{min} \end{cases} \tag{4.30d}$$

$$\dot{\mathbf{y}} = \mathbf{F}_{MTO}(\mathbf{y}) = \begin{Bmatrix} \mathbf{v} \\ -\dfrac{\mathbf{r}}{r^3} - u\dfrac{\boldsymbol{\lambda}_{\mathbf{v}}}{\lambda_{\mathbf{v}}}\dfrac{T_{max}}{m} \\ -u\dfrac{T_{max}}{c} \\ -\dfrac{3}{r^5}\left(\mathbf{r}^\top \boldsymbol{\lambda}_{\mathbf{v}}\right)\mathbf{r} + \dfrac{\boldsymbol{\lambda}_{\mathbf{v}}}{r^3} + u\dfrac{\lambda_{\mathbf{v}}}{m}\mathbf{t}_r + u\dfrac{\lambda_m}{c}\left(\mathbf{t}_r - \dfrac{T_{max}}{I_{sp}}\mathbf{i}_r\right) \\ -\boldsymbol{\lambda}_{\mathbf{r}} \\ -u\lambda_{\mathbf{v}}\dfrac{T_{max}}{m^2} \end{Bmatrix} \tag{4.30e}$$

$$\mathbf{\Gamma}_{MTO}\left(\left[\boldsymbol{\lambda}^*_{MTO}(t_0), t_f^*\right]\right) = \begin{Bmatrix} \boldsymbol{\psi}(\mathbf{x}(t_f)) \\ \lambda_m(t_f) \\ \Omega_f \end{Bmatrix} \tag{4.30f}$$

The transversality condition Ω_f is equivalent to Equation (4.13):

$$\Omega_f = H_{MTO}\left(t_f\right) + 1 - \boldsymbol{\lambda}_{\mathbf{r}}^{\top}\left(t_f\right)\mathbf{v}_t\left(t_f\right) - \boldsymbol{\lambda}_{\mathbf{v}}^{\top}\left(t_f\right)\mathbf{a}_t\left(t_f\right) \tag{4.31}$$

The formulations of system dynamics, Equation (4.11), switching function, Equation (4.10), control logic, Equation (4.9b), and shooting function, Equation (4.12), are identical to those of the LTO case. The only difference lies in the Hamiltonian, Equation (4.8), as:

$$H_{LTO} = H_{MTO} + 1 \tag{4.32}$$

4.5.2. Mayer Fuel-Optimal Formulation

To follow, the quantities of interest of the TPBVP associated with the Mayer fuel-optimal (MFO) formulation are presented. In particular, the cost function J_{MFO}, the Hamiltonian H_{MFO}, the switching function S_{MFO}, the optimal throttle logic u^*, the Euler-Lagrange equations $\mathbf{F}_{MFO}$, and shooting function $\boldsymbol{\Gamma}_{MFO}$:

$$J_{MFO} = -m\left(t_f\right) \tag{4.33a}$$

$$H_{MFO} = \boldsymbol{\lambda}^{\top}\mathbf{f} \tag{4.33b}$$

$$S_{MFO} = -\lambda_{\mathbf{v}}\frac{c}{m} - \lambda_m \tag{4.33c}$$

$$u^* = \begin{cases} 0, & S_{MFO} > 0 \quad \vee \quad S_p < P_{min} \\ \in [0,1], & S_{MFO} = 0 \quad \wedge \quad S_p \geq P_{min} \\ 1, & S_{MFO} < 0 \quad \wedge \quad S_p \geq P_{min} \end{cases} \tag{4.33d}$$

$$\dot{\mathbf{y}} = \mathbf{F}_{MFO}\left(\mathbf{y}\right) = \begin{Bmatrix} \mathbf{v} \\ -\dfrac{\mathbf{r}}{r^3} - u\dfrac{\boldsymbol{\lambda}_{\mathbf{v}}}{\lambda_{\mathbf{v}}}\dfrac{T_{max}}{m} \\ -u\dfrac{T_{max}}{c} \\ -\dfrac{3}{r^5}\left(\mathbf{r}^{\top}\boldsymbol{\lambda}_{\mathbf{v}}\right)\mathbf{r} + \dfrac{\boldsymbol{\lambda}_{\mathbf{v}}}{r^3} + u\dfrac{\lambda_{\mathbf{v}}}{m}\mathbf{t}_r + u\dfrac{\lambda_m}{c}\left(\mathbf{t}_r - \dfrac{T_{max}}{I_{sp}}\mathbf{i}_r\right) \\ -\boldsymbol{\lambda}_{\mathbf{r}} \\ -u\lambda_{\mathbf{v}}\dfrac{T_{max}}{m^2} \end{Bmatrix} \tag{4.33e}$$

$$\boldsymbol{\Gamma}_{MFO}\left(\boldsymbol{\lambda}^{*}_{MFO}\left(t_0\right)\right) = \left\{ \begin{array}{c} \boldsymbol{\psi}\left(\mathbf{x}\left(t_f\right)\right) \\ \lambda_m\left(t_f\right) + 1 \end{array} \right\} \tag{4.33f}$$

Despite the MFO sharing the same control logic, Equation (4.18), the formulation differs from that of the LFO in its Hamiltonian, Equation (4.17), switching function, Equation (4.19), dynamics, Equation (4.20), and shooting function, Equation (4.21).

Note:

It can be proved that it is possible to relate the two formulations by performing a change of variable. In particular:

$$\lambda_{m,FTO} = \lambda_{m,MFO} + 1 \tag{4.34}$$

It must be pointed out that such operation only relates the problem formulations, but nothing is inferred on the solution paths.

4.6. Requirements To Implicit Time Optimization

Given a set OCP statement, every optimal trajectory can be uniquely identified by the departure date t_0, and either the arrival time t_f or the time-of-flight $ToF = t_f - t_0$. While for TO solutions the time-of-flight is function of the departure date itself $ToF = ToF^{*}\left(t_0\right)$, in the case of FO and EO solutions both t_0 and ToF are free variables. Obviously, in the latter cases it must be that $ToF \geq ToF^{*}(t_0)$.

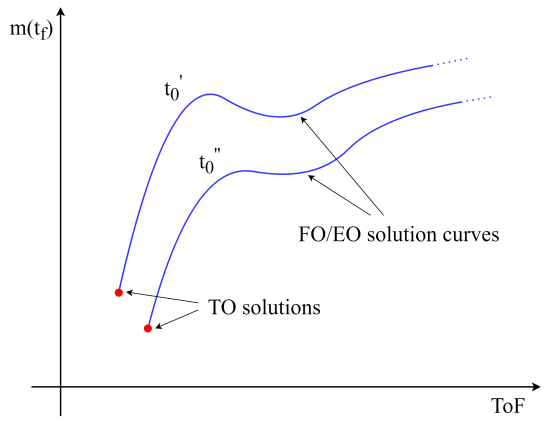


Figure 4.2: Qualitative representation of FO or EO solution curves in the $(ToF, m\left(t_f\right))$ space.

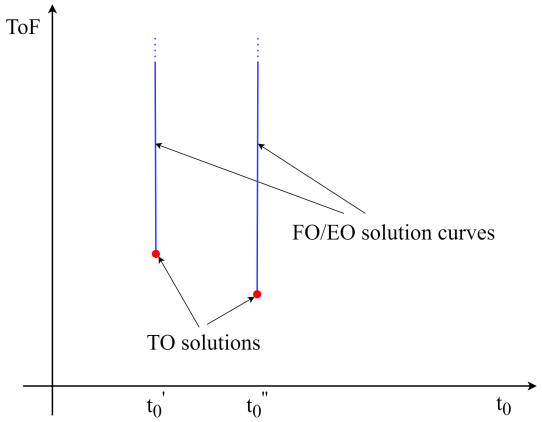


Figure 4.3: Qualitative representation of FO or EO solution curves in the (t_0, ToF) space.

It can be stated that for a given departure date t_0, the TO solution represents a limit-case to both FO and EO locii of solutions. In particular, Figures 4.2 and 4.3 show qualitative representations of possible families of solutions.
In other words, either FO or EO solutions characterized by $(t_0, ToF^*(t_0))$ implicitly minimize the time, and therefore must share the same trajectory as that of the TO solution. The objective of this dissertation is exactly to explore the relationship between TO solutions, and either FO and EO solutions that are implicitly time-optimal.

The relationship between TO and FO solutions is explored first.
Regardless of the formulation, for the TO solution to represent a limit case of the FO solution family, it is required that the trajectories obtained via the two different optimization processes must match over time, at least in their state variables:

$$\mathbf{x}^*_{TO}(t) = \mathbf{x}^*_{FO}(t) \quad \forall t \in [t_0, t_0 + ToF^*(t_0)] \tag{4.35}$$

In generality, it can be stated:

$$\mathbf{x}^*(t) = \mathbf{x}(t_0) + \int_{t_0}^{t} \mathbf{f}(\mathbf{x}^*, u^*, \hat{\boldsymbol{\alpha}}^*)\, dt \tag{4.36}$$

Since the initial conditions $\mathbf{x}(t_0)$ are the same, as per Equation (4.2a), for Equation (4.35) to be satisfied, it is therefore necessary for both solutions to share the same thrust logic throughout the entire trajectory, in terms of both throttle u^* and thrust direction $\hat{\boldsymbol{\alpha}}^*$.

4.6.1. Throttle Requirements

The simplified case characterized by constant thrust T_{max} and specific impulse I_{sp} is first considered.
Recalling that, regardless of the formulation, the TO solution is characterized by $u^*_{TO}(t) = 1 \quad \forall t$, and assuming $\mathbf{t}_r = \mathbf{i}_r = \mathbf{0}$, by virtue of Equation (4.35) and Equation (4.1), then:

$$\begin{aligned} m^*_{TO}(t_f) - m_0 &= \frac{T_{max}}{c} \int_{t_0}^{t_0+ToF^*(t_0)} u^*_{TO}(t)\, dt = \\ &= \frac{T_{max}}{c} \int_{t_0}^{t_0+ToF^*(t_0)} u^*_{FO}(t)\, dt = m^*_{FO}(t_f) - m_0 \end{aligned} \tag{4.37}$$

which holds if and only if $u^*_{FO}(t) = u^*_{TO}(t) = 1 \quad \forall t \in [t_0, t_0 + ToF^*(t_0)]$.

Taking into account variable parameters and the power constraints previously described, Equation (4.35) implies that the two trajectories must share the same T_{max} and I_{sp} profiles and possible arcs where power is insufficient, as they depend on the distance r only. Assuming there are N maximum throttle ($u = 1$) arcs, namely $\tau_i = [t_{i,on},\, t_{i,off}]$, with the throttle jumping from $u = 0$ to $u = 1$ at $t_{i,on}$ and vice versa at $t_{i,off}$, the set $\mathcal{T}$ is defined as:

$$\mathcal{T} = \bigcup_{i=1}^{N} \tau_i = \bigcup_{i=1}^{N} [t_{i,on},\, t_{i,off}] \tag{4.38}$$

Equation (4.37) can then be written as:

$$\begin{aligned} m^*_{TO}(t_f) - m_0 &= \int_{t_0}^{t_0+ToF^*(t_0)} \frac{T_{max}}{c} u^*_{TO}(t)\, dt = \\ &= \int_{\mathcal{T}_{TO}} \frac{T_{max}}{c} u^*_{TO}(t)\, dt = \\ &= \int_{\mathcal{T}_{FO}} \frac{T_{max}}{c} u^*_{FO}(t)\, dt = \\ &= \int_{t_0}^{t_0+ToF^*(t_0)} \frac{T_{max}}{c} u^*_{FO}(t)\, dt = m^*_{FO}(t_f) - m_0 \end{aligned} \tag{4.39}$$

By construction, the coasting arcs ($u = 0$) associated with the TO solutions are due to insufficient power. However, given the aforementioned observations, it can be stated that such arcs must be in common with the FO solution. Since $u^*_{TO}(t) = 1$ everywhere but those arcs, for Equations (4.35) and (4.39) to be verified, it must be:

$$\begin{cases} \tau_i^{TO} = \left[t_{i,on}^{TO}, t_{i,off}^{TO}\right] = \left[t_{i,on}^{FO}, t_{i,off}^{FO}\right] = \tau_i^{FO} \quad \forall i = 1, \dots N \\ u^*_{TO}(t) = u^*_{FO}(t) = 1 \quad \forall t \in \tau_i \quad \forall i = 1, \dots N \end{cases} \tag{4.40}$$

This translates into the requirement for the FO solution throttle to be $u_{FO}(t) = 1$ but for arcs of insufficient power.

By proof of contradiction, setting $\mathcal{T}_{TO} \neq \mathcal{T}_{FO}$ can only imply that the FO solution has additional coasting arcs. Therefore, for Equation (4.39) to hold, the two solutions would have to follow different trajectories, or not share the same T_{max} and I_{sp} profiles, in any way violating Equation (4.35).

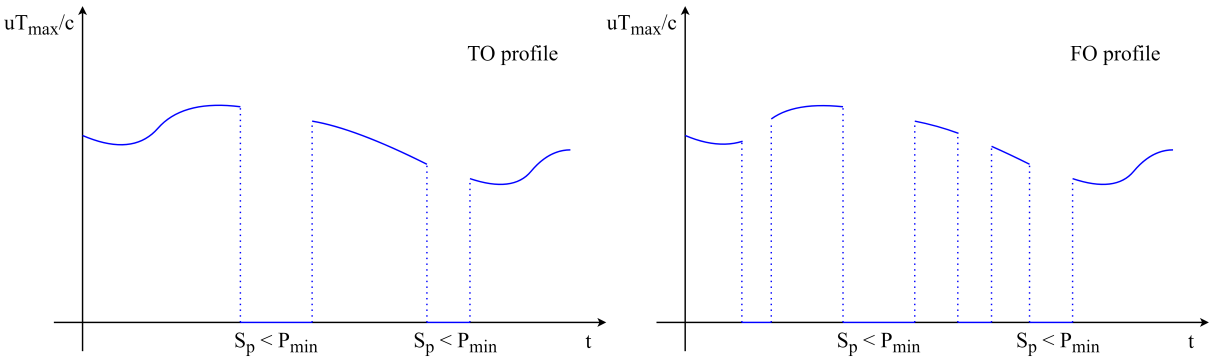


Figure 4.4: Qualitative thrust parameters profile for the proof by contradiction.

Figure 4.4 shows that, for the subtended area of the two curves to coincide, it must be that:

$$\begin{array}{ccc} \mathcal{T}_{TO} \neq \mathcal{T}_{FO} & \vee & u\dfrac{T_{max}}{c}\Big|^{*}_{TO}(t) \neq u\dfrac{T_{max}}{c}\Big|^{*}_{FO}(t) \\ \Downarrow & & \Downarrow \\ u^{*}_{TO}(t) \neq u^{*}_{FO}(t) & & \mathbf{x}^{*}_{TO}(t) \neq \mathbf{x}^{*}_{FO}(t) \end{array} \tag{4.41}$$

In both cases Equation (4.35) is violated, therefore Equation (4.40) must be true.
The sufficient condition for it to be verified is:

$$S_{FO}(t) \leq 0 \quad \forall t \in [t_0, t_0 + ToF^{*}(t_0)] \tag{4.42}$$

With respect to the presented LEFO formulation, then:

$$S_{LEFO,\varepsilon}(t) \leq -\varepsilon \quad \forall t \in [t_0, t_0 + ToF^{*}(t_0)] \tag{4.43}$$

4.6.2. Primer Vector Requirements

The secondary requirement for the TO and FO solutions to coincide is for the two trajectories to share the same thrust direction $\hat{\boldsymbol{\alpha}}^{*}$.
As stated in the previous sections, it is highlighted that regardless of the problem formulation, among the analyzed ones, it is found that:

$$\hat{\boldsymbol{\alpha}}^{*} = -\frac{\boldsymbol{\lambda}_{\mathbf{v}}}{\lambda_{\mathbf{v}}} = -\hat{\boldsymbol{\lambda}}_{\mathbf{v}} \tag{4.44}$$

The primer vector is only dependent on the unit vector $\hat{\boldsymbol{\lambda}}_{\mathbf{v}}$, regardless of its magnitude. Therefore, for Equation (4.35) to be satisfied:

$$\hat{\boldsymbol{\alpha}}^*_{TO}(t) = \hat{\boldsymbol{\alpha}}^*_{FO}(t) \quad \Leftrightarrow \quad \hat{\boldsymbol{\lambda}}^*_{\mathbf{v},TO}(t) = \hat{\boldsymbol{\lambda}}^*_{\mathbf{v},FO}(t) \quad \forall t \in [t_0,\, t_0 + ToF^*(t_0)] \tag{4.45}$$

4.7. Optimal Costate Path Scaling

The analysis of the requirement for the primer vector $\hat{\boldsymbol{\alpha}}^*$ highlights how, aside from the throttle u, the system dynamics $\mathbf{f}$ after application of the PMP are independent from the magnitude of the costate vector $\boldsymbol{\lambda}$.
In particular, regardless of the chosen formulation, the state dynamics described in Equation (4.1) are function of the costate only through the unit vector $\hat{\boldsymbol{\lambda}}_{\mathbf{v}}$ in $\hat{\boldsymbol{\alpha}}^*$. The throttle u is instead determined by the throttle switching function, which indeed depends on the magnitudes $\lambda_{\mathbf{v}}$ and λ_m. However, granting no change is made on the throttle profile u, a costate solution path $\boldsymbol{\lambda}(t)$ could be arbitrarily scaled without influencing the state trajectory $\mathbf{x}(t)$.

This observation sparks interest in further analysis of the rest of the elements pertaining to the indirect formulation.
In particular, it is first observed that, except for the transversality condition $\Omega_f = 0$, the formulation of the shooting functions associated with the Lagrange TO problem, Equation (4.12), and EFO problem, Equation (4.21), coincide. In fact, assuming
$ToF = ToF^*(t_0)$, the transversality condition to the TO solution can be neglected, as it is implicitly satisfied. The shooting functions then share the same formulations in $\boldsymbol{\lambda}$, in particular:

$$\lambda_{m,LTO}(t_f) = 0 \qquad \lambda_{m,LEFO,\varepsilon}(t_f) = 0 \tag{4.46}$$

It is also observed that the throttle switching functions, Equation (4.10) and Equation (4.19), both depend on $\boldsymbol{\lambda}$ only through the magnitudes $\lambda_{\mathbf{v}}$ and λ_m, and differ in their formulation but for a constant.

Therefore, with the aim of satisfying Equation (4.42), the possibility of introducing a scaling parameter between the TO and FO solutions is introduced.

In particular, for set departure date t_0, given $\boldsymbol{\lambda}^*_{TO}(t_0)$ the costate solution to the TO problem and $\boldsymbol{\lambda}^*_{FO}(t_0)$ the costate solution to the FO problem that implicitly optimizes time:

- Does γ exist such that $\boldsymbol{\lambda}^*_{FO}(t_0) = \gamma \boldsymbol{\lambda}^*_{TO}(t_0)$?

Assuming such a parameter existed, the resulting costate path $\boldsymbol{\lambda}^*_{FO}(t)$ for the FO problem would be required to satisfy the condition to optimality $\lambda_{m,FO}(t_f) = 0$, and the aforementioned conditions on throttle u and thrust direction $\hat{\boldsymbol{\alpha}}^*$.
In particular, if a parameter γ relating the solution paths $\boldsymbol{\lambda}^*_{FO}(t) = \gamma \boldsymbol{\lambda}^*_{TO}(t)$ existed, then it would be that $\lambda_{m,FO}(t_f) = 0$ and $\hat{\boldsymbol{\alpha}}^*_{FO}(t) = \hat{\boldsymbol{\alpha}}^*_{TO}(t)$ regardless of γ, which could perhaps be picked to satisfy the condition imposed by Equation (4.42) on S_{FO}.

Exploiting the energy-to-fuel formulation, the question can be extended to include the EO formulation. In particular:

- Does γ_ε exist such that $\boldsymbol{\lambda}^*_{EFO,\varepsilon}(t_0) = \gamma_\varepsilon \boldsymbol{\lambda}^*_{TO}(t_0)$?

More broadly:

- Does γ_ε exist such that $\boldsymbol{\lambda}^*_{EFO,\varepsilon}(t) = \gamma_\varepsilon \boldsymbol{\lambda}^*_{TO}(t)$?

Where $\boldsymbol{\lambda}^*_{EFO,\varepsilon}(t)$ represents the optimal costate path for the energy-to-fuel problem of arbitrary ε.

If such a constant γ_ε existed, then would be possible to find the solutions to either the FO or EO problems that implicitly optimize time simply by solving the TO problem of associated departure date t_0: the optimal trajectory $\mathbf{x}^*(t)$ would be the same, while the optimal costate path could be obtained simply by scaling the time-optimal costate path $\boldsymbol{\lambda}^*_{TO}(t)$.

4.7.1. Validity Assessment

Before addressing the search for γ_ε, the validity of the assumption is verified by analytical means. In particular, it is verified whether the assumption of solution path scaling would be able to generate solutions that are in turn optimal, without conflicting with existing problem constraints.

Shooting Function

Assuming the solution path $\boldsymbol{\lambda}^*_{TO}(t)$ is optimal for the Lagrange TO formulation, it satisfies the shooting function Equation (4.12), in particular $\lambda^*_{m,TO}(t_f) = 0$.
Then the path $\boldsymbol{\lambda}_{EFO,\varepsilon}(t) = \gamma\boldsymbol{\lambda}^*_{TO}(t)$ also satisfies $\lambda_{m,EFO,\varepsilon}(t_f) = 0, \quad \forall\gamma \in \mathbb{R}$.
The Lagrange EFO boundary constraints would therefore be satisfied.

State Dynamics

As previously stated, aside from the throttle u the state dynamics do not depend on the costate magnitude, and are therefore not affected by possible scaling of the solution.
More specifically, using $\hat{\boldsymbol{\alpha}}^* = \hat{\boldsymbol{\alpha}}^*(\boldsymbol{\lambda}) = -\hat{\boldsymbol{\lambda}}_{\mathbf{v}}$, it can be stated that:

$$\mathbf{f}(\mathbf{x}, u, \boldsymbol{\lambda}) = \mathbf{f}(\mathbf{x}, u, \gamma\boldsymbol{\lambda}) \quad \forall\gamma \in \mathbb{R} \tag{4.47}$$

Therefore, under the assumption of equal throttle profiles $u_{TO}(t) = u_{EFO,\varepsilon}(t)$, the TO solution path $\boldsymbol{\lambda}^*_{TO}(t)$ and the path $\boldsymbol{\lambda}_{EFO,\varepsilon}(t) = \gamma\boldsymbol{\lambda}^*_{TO}(t)$ would produce the same $\dot{\mathbf{x}}(t)$ for all $\gamma \in \mathbb{R}$.

Costate Dynamics

The hypothesis of scaling the solution must not be in contradiction to the costate dynamic constraints. Given $\boldsymbol{\lambda}_{EFO,\varepsilon}(t)$, $\boldsymbol{\lambda}_{TO}(t)$ two paths of the Euler-Lagrange equations of the EFO and TO problems, respectively, assuming $\boldsymbol{\lambda}_{EFO,\varepsilon}(t) = \gamma\boldsymbol{\lambda}_{TO}(t)$, by taking the time derivative it must be verified that:

$$\dot{\boldsymbol{\lambda}}_{EFO,\varepsilon}(t) = \gamma\dot{\boldsymbol{\lambda}}_{TO}(t) \tag{4.48}$$

Defining as $\dot{\boldsymbol{\lambda}}_{EFO,\varepsilon} = \bar{\mathbf{f}}_{EFO,\varepsilon}(\mathbf{x}, \boldsymbol{\lambda}_{EFO,\varepsilon})$ and $\dot{\boldsymbol{\lambda}}_{TO} = \bar{\mathbf{f}}_{TO}(\mathbf{x}, \boldsymbol{\lambda}_{TO})$ the functions regulating the costate dynamics in the EFO and TO formulations, respectively, it can be stated:

$$\dot{\boldsymbol{\lambda}}_{EFO,\varepsilon} = \bar{\mathbf{f}}_{EFO,\varepsilon}(\mathbf{x}, \boldsymbol{\lambda}_{EFO,\varepsilon}) = \bar{\mathbf{f}}_{EFO,\varepsilon}(\mathbf{x}, \gamma\boldsymbol{\lambda}_{TO}) = \gamma\dot{\boldsymbol{\lambda}}_{TO} \tag{4.49}$$

The function $\bar{\mathbf{f}}_{EFO,\varepsilon}$ must therefore be homogeneous of degree one with respect to the costate magnitude.

It is noted that a function $\mathbf{g} : V \to W$ is said to be homogeneous of degree n with respect to $\mathbf{w}$ if:

$$\mathbf{g}(s\mathbf{w}) = s^n \mathbf{g}(\mathbf{w}) \tag{4.50}$$

With $s \in \mathbb{R}$.

With reference to Equation (4.49), without loss of generality, using $\gamma = 1$ then:

$$\dot{\boldsymbol{\lambda}}_{TO} = \bar{\mathbf{f}}_{EFO,\varepsilon}(\mathbf{x}, \boldsymbol{\lambda}_{TO}) = \bar{\mathbf{f}}_{TO}(\mathbf{x}, \boldsymbol{\lambda}_{TO}) \tag{4.51}$$

Therefore, for $\boldsymbol{\lambda}_{EFO,\varepsilon}(t) = \gamma \boldsymbol{\lambda}_{TO}(t)$ to be verified, the TO and EFO costate dynamics must coincide and be homogeneous of order one with respect to the costate magnitude.

4.8. Formalized Formulation Requirements

It is now attempted to formalize in generality the requirements described in the previous sections.

Given $\boldsymbol{\lambda}^*_{TO}(t)$, $ToF^*(t_0)$ the solution to the TO optimal control problem characterized by departure date t_0, Euler-Lagrange equations $\mathbf{F}_{TO}$ and shooting function $\boldsymbol{\Gamma}_{TO}$:

$$\dot{\mathbf{y}} = \mathbf{F}_{TO}(\mathbf{y}) = \begin{pmatrix} \mathbf{f}(\mathbf{x}, \boldsymbol{\lambda}) \\ \bar{\mathbf{f}}_{TO}(\mathbf{x}, \boldsymbol{\lambda}) \end{pmatrix} = \begin{pmatrix} \dot{\mathbf{x}} \\ \dot{\boldsymbol{\lambda}} \end{pmatrix} \tag{4.52a}$$

$$\boldsymbol{\Gamma}_{TO}([\boldsymbol{\lambda}^*_{TO}(t_0), ToF^*(t_0)]) = \mathbf{0} \tag{4.52b}$$

Then the costate path $\boldsymbol{\lambda}^*_{EFO,\varepsilon}(t) = \gamma_\varepsilon \boldsymbol{\lambda}^*_{TO}(t) \quad \forall t \in [t_0, t_0 + ToF^*(t_0)]$, with $\gamma \in \mathbb{R}$, is an optimal solution path to the EFO optimal control problem characterized by departure date t_0, Euler-Lagrange equations $\mathbf{F}_{EFO,\varepsilon}$ and shooting function $\boldsymbol{\Gamma}_{EFO,\varepsilon}$:

$$\dot{\mathbf{y}} = \mathbf{F}_{EFO,\varepsilon}(\mathbf{y}) = \begin{pmatrix} \mathbf{f}(\mathbf{x}, \boldsymbol{\lambda}) \\ \bar{\mathbf{f}}_{EFO,\varepsilon}(\mathbf{x}, \boldsymbol{\lambda}) \end{pmatrix} = \begin{pmatrix} \dot{\mathbf{x}} \\ \dot{\boldsymbol{\lambda}} \end{pmatrix} \tag{4.53a}$$

$$\boldsymbol{\Gamma}_{EFO,\varepsilon}\left(\boldsymbol{\lambda}^*_{EFO,\varepsilon}(t_0)\right) = \mathbf{0} \tag{4.53b}$$

Under the conditions that:

1. The state dynamics $\mathbf{f}$, modified by application of the PMP, are not affected by scaling of the costate vector magnitude λ, granted the throttle path u is the same. Using $\hat{\boldsymbol{\alpha}}^* = \hat{\boldsymbol{\alpha}}^*(\boldsymbol{\lambda})$:

$$\mathbf{f}(\mathbf{x}, u, \boldsymbol{\lambda}) = \mathbf{f}(\mathbf{x}, u, \gamma\boldsymbol{\lambda}) \tag{4.54}$$

2. The costate dynamics $\bar{\mathbf{f}}_{TO}$ of the TO and $\bar{\mathbf{f}}_{EFO,\varepsilon}$ of the EFO formulations are coincident, and homogeneous of degree one with respect to the costate vector magnitude:

$$\begin{cases} \bar{\mathbf{f}}_{TO}(\mathbf{x}, \boldsymbol{\lambda}) = \bar{\mathbf{f}}_{EFO,\varepsilon}(\mathbf{x}, \boldsymbol{\lambda}) = \bar{\mathbf{f}}(\mathbf{x}, \boldsymbol{\lambda}) \\ \bar{\mathbf{f}}(\mathbf{x}, \gamma\boldsymbol{\lambda}) = \gamma\bar{\mathbf{f}}(\mathbf{x}, \boldsymbol{\lambda}) \end{cases} \tag{4.55}$$

3. The solution $\boldsymbol{\lambda}^*_{EFO,\varepsilon}(t) = \gamma_\varepsilon \boldsymbol{\lambda}^*_{TO}(t)$ is optimal for the EFO problem, satisfying the boundary conditions imposed by the shooting function $\boldsymbol{\Gamma}_{EFO,\varepsilon}(\boldsymbol{\xi}_{EFO,\varepsilon})$:

$$\begin{aligned} &\boldsymbol{\lambda}^*_{TO}(t): && \boldsymbol{\Gamma}_{TO}([\boldsymbol{\lambda}^*_{TO}(t_0), ToF^*(t_0)]) = \mathbf{0} \\ &\boldsymbol{\lambda}^*_{EFO,\varepsilon}(t): && \boldsymbol{\Gamma}_{EFO,\varepsilon}\left(\boldsymbol{\lambda}^*_{EFO,\varepsilon}(t_0)\right) = \mathbf{0} \end{aligned} \tag{4.56}$$

If conditions 4.54, 4.55 and 4.56 are met, and the scaling parameter γ_ε is chosen to ensure $u^*_{EFO,\varepsilon}(t) = 1 \quad \forall t \in [t_0,\, t_0 + ToF^*(t_0)]$, then the energy-to-fuel-optimal trajectory $\mathbf{x}^*_{EFO,\varepsilon}(t)$ and the time-optimal trajectory $\mathbf{x}^*_{TO}(t)$ coincide, in accordance with Equation (4.35).

4.8.1. Variable Parameters Dynamics

The possibility of employing solution scaling in the modalities explained in the previous sections is first explored for system dynamics characterized by variable parameters, proper of the M-ARGO mission design and described by Equations (4.3) to (4.5).

First, it is noted that, given $\hat{\boldsymbol{\alpha}}^* = -\hat{\boldsymbol{\lambda}}_{\mathbf{v}}$, as in the case of Lagrange time, fuel, or energy-optimal problems, the state dynamics fulfill the requirement imposed by Equation (4.54). It is also highlighted that, with respect to the TO problem, the Euler-Lagrange equations and the shooting function are equivalent for the Lagrange or Mayer formulations, as shown by Equations (4.11), (4.12), (4.30e) and (4.30f).

In particular, the TO dynamics satisfy the homogeneity requirement.
According to Equation (4.55), it is necessary for the EFO dynamics, Equation (4.20), to be equal to that of the TO problem. That is not the case for the Lagrange formulation: a difference is found in $\dot{\boldsymbol{\lambda}}_{\mathbf{r}}$ regardless of the homotopy parameter ε. In particular, neglecting the components shared by both formulations:

$$\begin{aligned}\dot{\boldsymbol{\lambda}}_{\mathbf{r},LTO} &= \cdots + u\frac{\lambda_m}{c}\left(\mathbf{t}_r - \frac{T_{max}}{I_{sp}}\mathbf{i}_r\right)\\ \dot{\boldsymbol{\lambda}}_{\mathbf{r},LEFO,\varepsilon} &= \cdots + \frac{(\lambda_m - 1 + \varepsilon)\,u - \varepsilon u^2}{c}\left(\mathbf{t}_r - \frac{T_{max}}{I_{sp}}\mathbf{i}_r\right)\end{aligned} \tag{4.57}$$

Furthermore, the additional terms introduce non-homogeneity: the Lagrange formulation, for either FO or EO problems, is therefore deemed not suitable for solution scaling in this example of variable parameters. In particular, for departure date t_0, given $\boldsymbol{\lambda}^*_{LTO}(t)$ optimal path to the Lagrange TO problem, and $\boldsymbol{\lambda}^*_{LEFO,\varepsilon}(t)$ optimal path to the Lagrange EFO problem implicitly optimizing time:

$$\nexists \gamma \in \mathbb{R}: \quad \boldsymbol{\lambda}^*_{LEFO,\varepsilon}(t) = \gamma_\varepsilon \boldsymbol{\lambda}^*_{LTO}(t) \quad \forall t \in [t_0,\, t_0 + ToF^*(t_0)] \tag{4.58}$$

This assumption is verified in terms of Hamiltonian optimality, Equation (2.11). In particular, given $\boldsymbol{\lambda}^*_{LTO}(t)$ optimal for the LTO formulation:

$$\dot{H}^*_{LTO} = \frac{\partial}{\partial t}\left(\boldsymbol{\lambda}^{*\top}_{LTO}\mathbf{f}^* + 1\right) = \boldsymbol{\lambda}^{*\top}_{LTO}\frac{\partial \mathbf{f}^*}{\partial t} = 0 \tag{4.59}$$

For $\boldsymbol{\lambda}^*_{LEFO,\varepsilon}(t) = \gamma_\varepsilon \boldsymbol{\lambda}^*_{LTO}(t)$ to be optimal for the LEFO problem, it must also be $\dot{H}^*_{LEFO} = 0$. However:

$$\begin{aligned}\dot{H}^*_{LEFO} &= \frac{\partial}{\partial t}\left(\boldsymbol{\lambda}^{*\top}_{LEFO,\varepsilon}\mathbf{f}^* + \frac{T_{max}}{c}\left[u - \varepsilon u\,(1-u)\right]\right) =\\ &= \gamma_\varepsilon \boldsymbol{\lambda}^{*\top}_{LTO}\frac{\partial \mathbf{f}^*}{\partial t} + \frac{\partial}{\partial t}\left(\frac{T_{max}}{c}\left[u - \varepsilon u\,(1-u)\right]\right) =\\ &= \frac{\partial}{\partial t}\left(\frac{T_{max}}{c}\left[u - \varepsilon u\,(1-u)\right]\right) \neq 0\end{aligned} \tag{4.60}$$

Analyzing different possible problem formulations, it is found that the Mayer FO dynamics, Equation (4.33e), are equivalent to that of the LTO (or MTO) formulation, and homogeneous with respect to the magnitude λ.

However, the MFO formulation does not allow Equation (4.56) to be satisfied, as the shooting function Equation (4.30f) requires:

$$\lambda_{m,MFO}(t_f) + 1 = 0 \tag{4.61}$$

Which implies:

$$\lambda_{m,MFO}(t_f) + 1 = \gamma_\varepsilon \lambda_{m,LTO}(t_f) + 1 = 1 \neq 0 \tag{4.62}$$

Therefore, it is stated that the variable parameters problem described in the previous sections is not suitable for solution scaling, regardless of the formulation employed.

4.8.2. Constant Parameters Dynamics

It can be observed that the elements that introduce non-homogeneity in the LEFO dynamics are present due to the variable nature of thrust and specific impulse.
It is now considered the simplified case of constant parameters, characterized by constant input power P_{in}, and consequently T_{max} and I_{sp}.
Therefore, introducing $\mathbf{t}_r = \mathbf{i}_r = \mathbf{0}$, it can be shown that the LTO and LEFO dynamics Equation (4.11) and Equation (4.20), respectively, reduce to the common formulation:

$$\mathbf{F}_{LTO'}(\mathbf{y}) = \mathbf{F}_{LEFO'}(\mathbf{y}) = \begin{Bmatrix} \mathbf{f}' \\ \overline{\mathbf{f}}' \end{Bmatrix} = \begin{Bmatrix} \mathbf{v} \\ -\dfrac{\mathbf{r}}{r^3} - u\dfrac{\boldsymbol{\lambda}_\mathbf{v}}{\lambda_\mathbf{v}}\dfrac{T_{max}}{m} \\ -u\dfrac{T_{max}}{c} \\ -\dfrac{3}{r^5}\left(\mathbf{r}^\top \boldsymbol{\lambda}_\mathbf{v}\right)\mathbf{r} + \dfrac{\boldsymbol{\lambda}_\mathbf{v}}{r^3} \\ -\boldsymbol{\lambda}_\mathbf{r} \\ -u\lambda_\mathbf{v}\dfrac{T_{max}}{m^2} \end{Bmatrix} \tag{4.63}$$

The superscript $'$ is used to identify the constant parameter formulation, which does not change from the variable parameters counterpart except for the Euler-Lagrange equations Equation (4.63), and the throttle control logic, which is qualitatively equivalent to the variable parameters case, Equation (4.18), but disregarding the dependency on S_p.

The state dynamics, $\mathbf{f}'$ are equivalent to those descrived in Equation (4.1), and satisfy the condition posed by Equation (4.54). The costate dynamics $\overline{\mathbf{f}}'$ are such that Equation (4.55) is also satisfied. The shooting functions are not changed and satisfy Equation (4.56). Therefore, the LTO$'$ and LEFO$'$ formulations are suitable for solution scaling.
In particular, for set departure date t_0, given $\boldsymbol{\lambda}^*_{LTO'}(t)$ the optimal path to the Lagrange TO constant parameters problem, then there exists γ_ε so that:

$$\boldsymbol{\lambda}^*_{LEFO',\varepsilon}(t) = \gamma_\varepsilon \boldsymbol{\lambda}^*_{LTO'}(t) \quad \forall t \in [t_0,\, t_0 + ToF^*(t_0)] \tag{4.64}$$

is optimal for the Lagrange EFO constant parameters problem of arbitrary ε.
Given that the the TO path $\boldsymbol{\lambda}^*_{TO}(t)$ is such that:

$$\boldsymbol{\lambda}^*_{LTO'}(t): \quad \dot{H}_{LTO'} = \frac{\partial}{\partial t}\left(\boldsymbol{\lambda}^*_{LTO'}\mathbf{f}'^* + 1\right) = 0 \tag{4.65}$$

Then, assuming constant maximum throttle $u = 1$, it is indeed verified:

$$\begin{aligned} \boldsymbol{\lambda}^*_{LEFO',\varepsilon}(t): \quad \dot{H}_{LEFO',\varepsilon} &= \frac{\partial}{\partial t}\left(\gamma_\varepsilon \boldsymbol{\lambda}^*_{LTO'}\mathbf{f}'^* + \frac{T_{max}}{c}\left[u - \varepsilon u\left(1-u\right)\right]\right) = \\ &= \frac{T_{max}}{c}\frac{\partial}{\partial t}\left(\left[u - \varepsilon u\left(1-u\right)\right]\right) = 0 \end{aligned} \tag{4.66}$$

Note:

As for the case of variable parameters, the optimal path $\mathbf{y}^*_{LEFO',1}(t)$ for the EO problem that implicitly optimizes time, characterized by $u = 1$, ensures constant maximum throttle ($u = 1$) for the FO problem as well:

$$S_{LEFO',1}\left(\mathbf{y}^*_{LEFO',1}(t)\right) < -1 \quad \Rightarrow \quad S_{LEFO',0}\left(\mathbf{y}^*_{LEFO',1}(t)\right) < 0 \tag{4.67}$$

Furthermore, in the case of variable parameters, the Euler-Lagrange equations presented in Equation (4.63) do not depend on ε. Once again, given same departure date t_0 and time-of-flight $ToF^*(t_0)$, the path $\boldsymbol{\lambda}^*_{LEFO',1}(t)$ is an optimal path for the EO problem as well as the FO problem, each of which implicitly optimize time.

The parameter γ_ε is studied with the aim of ensuring constant maximum throttle ($u = 1$). Therefore, as per Equation (4.43), it must be that the LEFO$'$ throttle switching function:

$$S_{LEFO',\varepsilon} = -\lambda_{\mathbf{v}}\frac{c}{m} - \lambda_m + 1 \leq -\varepsilon \quad \forall t \in [t_0,\, t_0 + ToF^*(t_0)] \tag{4.68}$$

Reminding that $S_{LTO'}$ is the LTO′ throttle switching function, once again non-positive by construction:

$$S_{LTO'} = -\lambda_{\mathbf{v}} \frac{c}{m} - \lambda_m \leq 0 \quad \forall t \tag{4.69}$$

Having proven the existence of γ_ε, it can be shown that:

$$S_{LEFO',\varepsilon}(t) = \gamma_\varepsilon S_{LTO'}(t) + 1 \tag{4.70}$$

By means of Equations (4.68) to (4.70), it can be deduced that:

$$\gamma_\varepsilon \geq -\frac{1+\varepsilon}{S_{LTO'}(t)} \quad \forall t \in [t_0,\, t_0 + ToF^*(t_0)] \tag{4.71}$$

Which is satisfied $\forall t \in [t_0,\, t_0 + ToF^*(t_0)]$ if:

$$\gamma_\varepsilon \geq -\frac{1+\varepsilon}{\max_{t\in[t_0,\, t_0+ToF^*(t_0)]} S_{LTO'}(t)} \tag{4.72}$$

Defining γ_L as the scaling parameter γ_ε lower bound:

$$\gamma_L = -\frac{1}{\max_{t\in[t_0,\, t_0+ToF^*(t_0)]} S_{LTO'}(t)} \tag{4.73}$$

Then the scaling parameter γ_ε:

$$\gamma_\varepsilon \geq (1+\varepsilon)\gamma_L \tag{4.74}$$

In particular, given $\boldsymbol{\lambda}^*_{LTO'}(t)$ the optimal costate path for the Lagrange TO constant parameter formulation, it is found that:

- The costate path $\boldsymbol{\lambda}^*_{LFO'}(t) = \gamma_{FO}\boldsymbol{\lambda}^*_{LTO'}(t)$ is optimal for the Lagrange FO constant parameters formulation for all γ_{FO} such that:

$$\gamma_{FO} = \gamma_{\varepsilon=0} \geq \gamma_L \tag{4.75}$$

- The costate path $\boldsymbol{\lambda}^*_{LEO'}(t) = \gamma_{EO}\boldsymbol{\lambda}^*_{LTO'}(t)$ is optimal for the Lagrange EO constant parameters formulation for all γ_{EO} such that:

$$\gamma_{EO} = \gamma_{\varepsilon=1} \geq 2\gamma_L \tag{4.76}$$

Note:

It ough to be highlighted that Equation (4.75) and Equation (4.76) only represent lower bounds. Therefore, granted their validity, there can exist infinite FO and EO solution which implicitly minimize the time. These solutions differ in the scaling of the costate path, but share the same optimal trajectory.

5 | Numerical Methodology

This chapter outlines the numerical strategies employed in the solution of the presented indirect OCPs. In particular, using the described Lagrange formulations, several time and energy-to-fuel-optimal control problems are solved to generate the porkchop plots for the preliminary targets listed in Table 1.3. The methodologies described are employed to the solution of the problems, for both the variable and constant parameters cases, drawing from the foundational work of Wang [31], Wang et al. [32], and Zhang et al. [39].

5.1. Shooting Method

The OCPs presented in this dissertation are solved according to the indirect single shooting method. The method is initiated with a guess $\boldsymbol{\xi}_g$ for the unknown variables $\boldsymbol{\xi}$, allowing for the integration from initial time t_0 to final time t_f of the complete state $\mathbf{y}$. The resulting trajectory is evaluated using the shooting function $\boldsymbol{\Gamma}(\boldsymbol{\xi})$, which measures the violation of the final boundary constraints $\boldsymbol{\delta\Gamma}$. The guess $\boldsymbol{\xi}_g$ is iteratively corrected with first-order updates $\boldsymbol{\delta\xi}$ computed by means of the jacobian $\mathbf{J}_{\boldsymbol{\Gamma}}$, according to:

$$\boldsymbol{\delta\xi} = \mathbf{J}_{\boldsymbol{\Gamma}}^{-1}\boldsymbol{\delta\Gamma} \tag{5.1}$$

The iterative process continues until the boundary constraints are satisfied within a specified tolerance, granting $\|\boldsymbol{\delta\Gamma}\| \approx 0$. However, relying solely on first-order corrections introduces limitations in precision, especially when dealing with non-linear or poorly conditioned problems. In order to mitigate this issue and enhance numerical stability, dimensional normalization is applied to the problem.
After adimensionalization, the tolerance of the iterative gradient method is set to 10^{-9}, in accordance with the precision requirements selected for the mission:

$$\|\mathbf{r}(t_f) - \mathbf{r}_t(t_f)\| \leq 10\,km \quad \wedge \quad \|\mathbf{v}(t_f) - \mathbf{v}_t(t_f)\| \leq 10^{-3}\,km/s \tag{5.2}$$

The final target position and velocity $\mathbf{r}_t(t_f)$, $\mathbf{v}_t(t_f)$, as well as the initial SEL2 position and velocity $\mathbf{r}_0$, $\mathbf{v}_0$ are computed using SPICE kernels from NASA JPL Horizons Systems website. [1]

5.1.1. Dimensional Normalization

The indirect problem formulations, in particular the Euler-Lagrange equations, are normalized according to quantities characteristic to the problem, presented in Table 5.1. Given the near-Earth nature of the targets, the length unit LU is set to $1\,AU$ and the time unit TU is computed as the period of the circular orbit of radius LU, as:

$$TU = \sqrt{\frac{LU^3}{\mu}} \tag{5.3}$$

The mass unit MU is set to the spacecraft wet mass m_0, while velocity and power units, VU and PU, respectively, are simply derived as:

$$VU = \frac{LU}{TU} \qquad PU = \frac{MU\,LU^2}{TU^3} \tag{5.4}$$

Physical constant	Value
Sun gravitational constant, μ	$1.3271 \cdot 10^{11}\,km^3/s^2$
Earth gravitational acceleration, g_0	$9.8066\,m/s^2$
Length Unit, LU	$1.4960 \cdot 10^8\,km$
Time Unit, TU	$5.0226 \cdot 10^6\,s$
Velocity Unit, VU	$29.7847\,km/s$
Mass Unit, MU	$22.6\,kg$
Power Unit, PU	$3.9917 \cdot 10^3\,W$

Table 5.1: Physical constants. [31][32]

This scaling also facilitates the convergence of the shooting method by reducing the magnitude disparity between the state and costate variables.

[1] `https://ssd.jpl.nasa.gov/horizons/`, Accessed: 2025-02-07

5.1.2. Integration Scheme

As anticipated in Chapter 4, the solution path may be discontinuous in $\dot{\mathbf{y}}$ and $\mathbf{y}$, given either the nature of the control or the enforcement of power constraints. The time marching algorithm therefore needs to be able to correctly detect and handle possible discontinuities.

The algorithm devised for the generation of the results reported in this document is an 8th order variable-step Runge-Kutta (RK) method. In particular, the Dormand-Prince scheme (DOPRI8(7)) is employed. [25] The DOPRI8(7) scheme is an embedded RK scheme which allows for the computation of the optimal step size h by comparing the 8th order step solution against the auxiliary 7th order one.

Given absolute and relative tolerances both set to 10^{-12}, in the absence of discontinuities the integrator advances with optimal step size h, computed based on the methodologies described in [12].

Integration affects both $\mathbf{y}$ and the first-order variation $\mathbf{\Phi}$ according to:

$$\dot{\mathbf{z}} = \begin{pmatrix} \dot{\mathbf{y}} \\ \text{vec}\left(\dot{\mathbf{\Phi}}\right) \end{pmatrix} = \boldsymbol{\mathcal{F}}\left(\mathbf{z}\right) = \begin{pmatrix} \mathbf{F}\left(\mathbf{y}\right) \\ \text{vec}\left(\mathbf{D_y F \Phi}\right) \end{pmatrix} \tag{5.5}$$

The STM $\mathbf{\Phi}$ dynamics refer to Equation (2.16), while $\mathbf{F}$ is the state and costate dynamics, computed as the Euler-Lagrange equations of the problem at hand.

5.1.3. Switching Detection

Given two consecutive states $\mathbf{y}_k$ and $\mathbf{y}_{k+1}$ linked by the RK integration step $\mathbf{y}_{k+1} = \boldsymbol{\varphi}_{RK}\left(\mathbf{y}_k, t_k, t_{k+1}\right)$ with $t_{k+1} = t_k + h_k$, and given the switching function S and the threshold η, a discontinuity is detected if it is found that $\left(S\left(\mathbf{y}_k\right) - \eta\right)\left(S\left(\mathbf{y}_{k+1}\right) - \eta\right) < 0$. This implies S crosses the threshold η at $t_c \in [t_k, t_{k+1}]$, and the discontinuity is then taken into account according to the flowchart presented in [31][32][39].

In particular, the crossing time t_c is a zero to the function f_c:

$$f_c\left(t_c\right) = S\left(t_c\right) - \eta = 0 \quad t_c \in [t_k, t_{k+1}] \tag{5.6}$$

A hybrid detection technique is employed. First, a bisection search is executed, initialized on the boundaries of the integration step t_k and t_{k+1}. Should the bisection method fail to converge, a Newton's method is initialized at the midpoint $t_k + h_k/2$ of the $[t_k, t_{k+1}]$ interval.

Both methods have stopping tolerance set to machine precision $2.2206 \cdot 10^{-16}$, and employ the analytic derivatives defined in Equation (4.27b) and Equation (4.28b) for throttle and power discontinuity detection, respectively.

Once t_c has been correctly evaluated, an integration step is taken to evaluate $\mathbf{y}_c = \boldsymbol{\varphi}_{RK}(\mathbf{y}_k, t_k, t_c)$, which is used to update the state across the discontinuity according to Equation (4.24), and to compute the STM correction $\boldsymbol{\Psi}(t_c)$ of Equation (2.19). The integration then restarts from t_c with updated control logic.

Note:

It's important to notice that, given the power bounds specified by Table 1.2, it is found that the engine shut-off scenario due to insufficient power $S_p < P_{min}$ is never triggered. Even if this event were to occur, it would not influence the outcomes relevant to the purpose of this study. Therefore, P_{min} is maintained at the selected value of $20\,W$, in accordance with the reference document [30]. This choice simplifies the model significantly, without loss of generality.

5.2. Solution Space Exploration Method

This section addresses the continuation strategies devised for the continuation of time-optimal and energy-to-fuel-optimal solutions. In particular, for both the variable and constant parameters cases, the continuation scheme employs variable continuation step and adaptive choice of continuation method to generate the porkchop plots relative to the selected sample mission targets of Table 1.3.

5.2.1. Time-Optimal Continuation Strategy

As stated in Chapter 4, for a given OCP a time-optimal solution is uniquely identified by the departure date t_0. Therefore, the TO solution space is one-dimensional and limited by the launch window $\left[\underline{t_0}, \overline{t_0}\right]$, identified as per Table 5.2.
Therefore, it is attempted to fully characterize the solution space by means of continuation of the departure date t_0, starting from the earliest possible solution at $\underline{t_0}$.

Parameter	Date	Modified Julian Date 2000
$\underline{t_0}$	2023 Jan 01 12 : 00 UTC	8401.5
$\overline{t_0}$	2024 Dec 31 12 : 00 UTC	9131.5

Table 5.2: Preliminary departure window. [30]

The unknowns to the Lagrange TO formulation consist in both the initial costates $\boldsymbol{\lambda}^*_{LTO}(t_0)$ and the time-of-flight $ToF^*(t_0)$. To facilitate the convergence of the method, a reasonable guess has to be provided for both. Regarding the initial costates, an estimate is computed by means of the ACT: a mapping $\mathcal{M}$ is devised so that $\boldsymbol{\lambda}_\mathbf{r}$, $\boldsymbol{\lambda}_\mathbf{v}$ are evaluated in function of the throttle switching function S, the in- and out-of-plane angles, α and β respectively, of the direction of thrust $\mathbf{u}'$, described in the spacecraft velocity centered frame, and their derivatives. Namely [9][27][31][32]:

$$\mathcal{M}\left(S,\, \dot{S},\, \alpha,\, \dot{\alpha},\, \beta,\, \dot{\beta}\right) \rightarrow (\boldsymbol{\lambda}_\mathbf{r},\, \boldsymbol{\lambda}_\mathbf{v}) \tag{5.7}$$

The transformation is tuned for the selected mission targets of Table 1.3 with the parameters reported in Table 5.3. [31][32]

Parameter	Value	Units
S	$[-1.5,\, -0.001]$	$-$
$\dot{S}$	$[-0.01,\, 0.01]$	$-$
α	$[-10,\, 10]$	deg
$\dot{\alpha}$	$[-5,\, 5]$	deg/TU
β	$[-1,\, 1]$	deg
$\dot{\beta}$	$[-0.1,\, 0.1]$	deg/TU
λ_m	1	$-$

Table 5.3: ACT parameters bounds. [31][32]

Regarding the time-of-flight $ToF^*(t_0)$, a monotonic search is performed: the convergence is tested for increasing time-of-flight values until a suitable solution is found.

Once the first solution has been computed, continuation is carried out throughout the departure window. A combination of NPCMs is employed with an adaptive step Δt_0. The continuation step size is halved when a continuation step fails and increased by a factor of 1.1 after a successful iteration, limited to the bounds $\Delta t_0 \in [0.125,\, 5]$ days.
The continuation algorithm starts with 0NPCM to find the second solution, but prioritizes 1NPCM afterward, due to its higher efficiency. If a sufficient number of prior solutions is available, second-order and successively third-order NPCM can be attempted when lower-order methods fail. If all available NPCMs fail, a new guess is generated using the ACT method, with parameters specified in Table 5.3. Should the ACT guess also fail to converge, the algorithm cycles back to the 1NPCM, and the process is iterated until a valid solution is found.

The flowchart diagram describing this continuation scheme is presented in Figure 5.1. Note that the rhombus with inscribed C symbolizes the test determining the convergence of the step, while i refers to the ith solution being attempted.

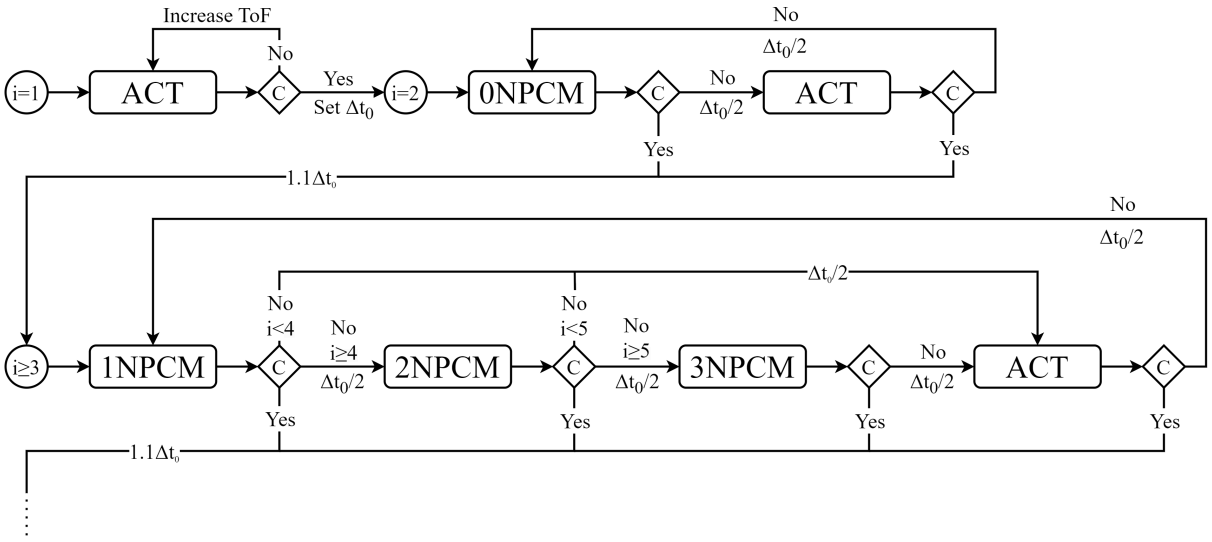


Figure 5.1: Time-optimal departure date t_0 continuation scheme.

5.2.2. Fuel And Energy-Optimal Continuation Strategy

In contrast to TO solutions, both fuel and energy-optimal trajectories are uniquely identified by the departure date t_0 and time-of-flight ToF pair, rendering the solution space two-dimensional.
Since the focus of this dissertation is on solutions characterized by the minimum possible transfer time, for each departure date t_0, continuation is carried out starting from a large time-of-flight ToF_{max}, gradually reduced to $ToF^*(t_0)$. This approach is chosen to reduce the risk of biasing the implicitly time-optimal solutions at $ToF = ToF^*(t_0)$. In fact, they are approached by exploiting other solutions belonging to the EFO family curve, rather than starting from the TO solutions themselves.

The process begins with finding the energy-optimal solution associated with the earliest departure date $\underline{t_0}$, reported in Table 5.2, and the maximum time-of-flight ToF_{max}. To reduce computational load ToF_{max} is set to $ToF_{max} = 800\,days$, down from the $3\,years$ value specified in [30]. The unknowns to the Lagrange EFO problem are the initial costates $\boldsymbol{\lambda}(t_0)$, and the guess to the first problem is generated using the ACT, with parameters provided Table 5.3.

After the first solution has been retrieved, the continuation is carried out with respect to the departure date t_0 to find all the solutions with ToF_{max}. The continuation step sizes are fixed and consistent with those used for the TO solutions. The first continuation step employs 0NPCM, while the subsequent steps favor 1NPCM. When possible, 2NPCM or 3NCPM are employed should lower-order methods fail. If none of the NPCMs succeed, the ACT is used to generate new guess solutions until convergence is achieved. The flowchart detailing the energy-optimal t_0 continuation process is reported in Figure 5.2.

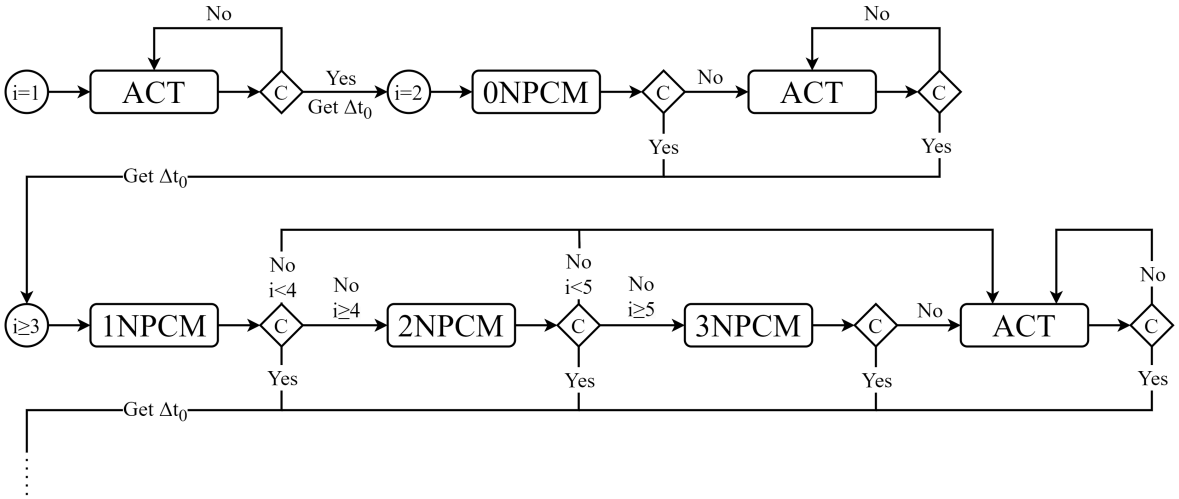


Figure 5.2: Energy-optimal departure date t_0 continuation scheme.

Once all the EO solutions with ToF_{max} have been identified, continuation is performed for each departure date t_0 to reduce ToF from ToF_{max} to $ToF^*(t_0)$. The time-of-flight continuation scheme is functionally equivalent to that employed for the departure date continuation of TO solutions. It employs an adaptive step size $\Delta ToF \in [1, 20]\ days$, halved at each failure and increased by a factor of 1.25 following each successful iteration. Despite the use of adaptive step and methods of different order, convergence becomes increasingly challenging as ToF approaches its lower bound. The continuation process is therefore set to skip to the next departure date should convergence stall within $20\,days$ of $ToF^*(t_0)$. The flowchart associated with the ToF continuation process is reported in Figure 5.3.

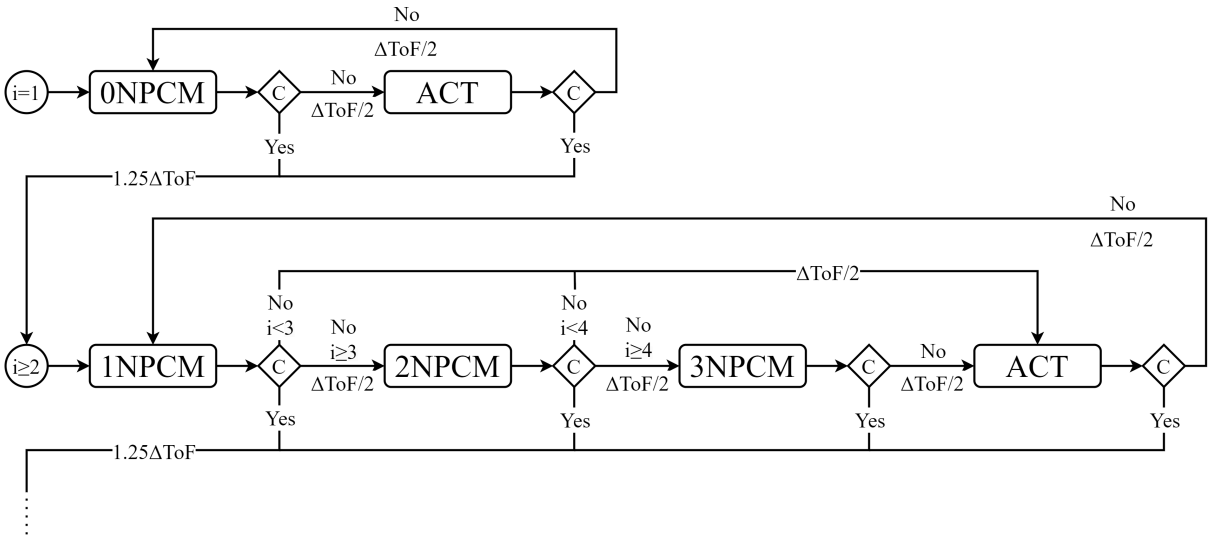


Figure 5.3: Energy-optimal time-of-flight ToF continuation scheme.

The combination of the two continuation processes allows for the complete exploration of the solution space, therefore completely characterizing the EO porkchop plot.
Figure 5.4 qualitatively summaries the solution space exploration process: (1) TO departure date t_0 continuation, (2) EO departure date t_0 continuation, (3) EO time-of-flight ToF continuation.

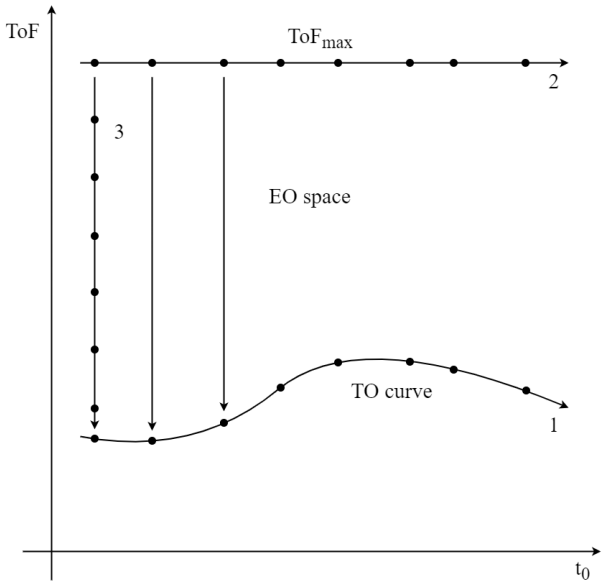


Figure 5.4: Solution space exploration strategy.

The set of energy-optimal solutions can in turn be transformed into fuel-optimal by means of the energy-to-fuel homotopy continuation, described in Section 4.3. Continuation is performed on the homotopy parameter ε for every EO solution. The ε continuation scheme is qualitatively similar to that employed for t_0 continuation of TO solutions, but in the cases of EO solutions that implicitly optimize time, $ToF = ToF^*(t_0)$, it is set to prioritize 0NPCM over 1NPCM. Having anticipated in Chapter 4 the equivalence between such energy and fuel-optimal solutions, and, in the constant parameters case, the possibility of existence of multiple optimal paths, the use of 0NPCM is justified by the need to avoid artificial alteration of the magnitude of the solution.
The continuation scheme operates with default step size $\Delta\varepsilon = 0.05$, which is halved every time the continuation step fails. The relative flowchart is presented in Figure 5.5.
Note that the rhombus with inscribed T symbolizes the test determining if ToF is minimum.

Despite the use of the smoothing technique presented in Section 4.3, the convergence of FO solutions remains challenging. Therefore, the energy-to-fuel continuation stops should either of the following conditions be met: continuation stalls or $\Delta\varepsilon < 10^{-6} \ \wedge \ \varepsilon < 10^{-5}$.

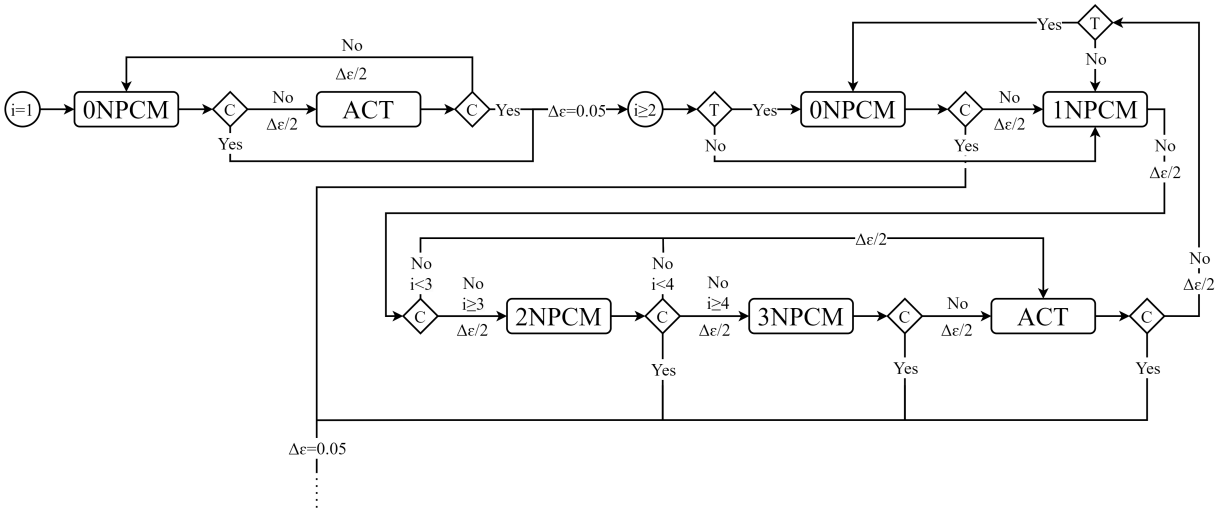


Figure 5.5: Energy-to-fuel-optimal ε continuation scheme.

6 | Results

This chapter collects the results of the indirect control problems describe in Chapter 4, solved according to the methodologies outlined in Chapter 5. In particular, the porkchop plots associated to the targets listed in Table 1.3 are presented, for both the variable and constant parameters cases. The results obtained are employed for the validation of the propositions advanced in Chapter 4.

6.1. Variable Parameters Results

This section reports the results regarding the variable parameters solutions characteristic of the M-ARGO mission, described in Chapter 4. [30] Additionally, numerical experiments are carried out to verify the assumptions of non-existence of the closed-form relation introduced in Section 4.8.1. In this section, the notation is condensed to $LTO \rightarrow TO$, $LEFO, 1 \rightarrow EO$, and $LEFO, 0 \rightarrow FO$.

6.1.1. Porkchop Plots

First, the porkchop plot associated with the rendezvous mission to $2014\,YD$ is reported in Figure 6.1 in the following page. Contour lines are plotted every $0.2\,kg$ of propellant mass, with the black dashed line representing the $2.8\,kg$ limit set in Table 1.2. The continuous red line represents the ensemble of TO solutions.

Sampled trajectories belonging to the solution space are showcased: in particular, two time-optimal and two fuel-optimal trajectories, labeled A through D, are presented with reference to Figures 6.2 to 6.5.
The trajectories are characterized by red arcs representing maximum throttle ($u = 1$) arcs and blue coasting ($u = 0$) arcs. The side plots show the evolution of parameters of interest. In particular, from top to bottom, the left column shows the throttle profile u, the throttle switching function S, and the mass profile m. The switching function S is identified by $S = S_{LTO}$, Equation (4.10), for TO solutions, and by $S = S_{LEFO,\varepsilon}$, Equation (4.19), for FO solutions.

The right column instead shows the power switching function S_p, the specific impulse I_{sp}, and the maximum thrust T_{max}.
It is noted that, with respect to the trajectories, the z-axis scale is much lower compared to the x and y-axes.

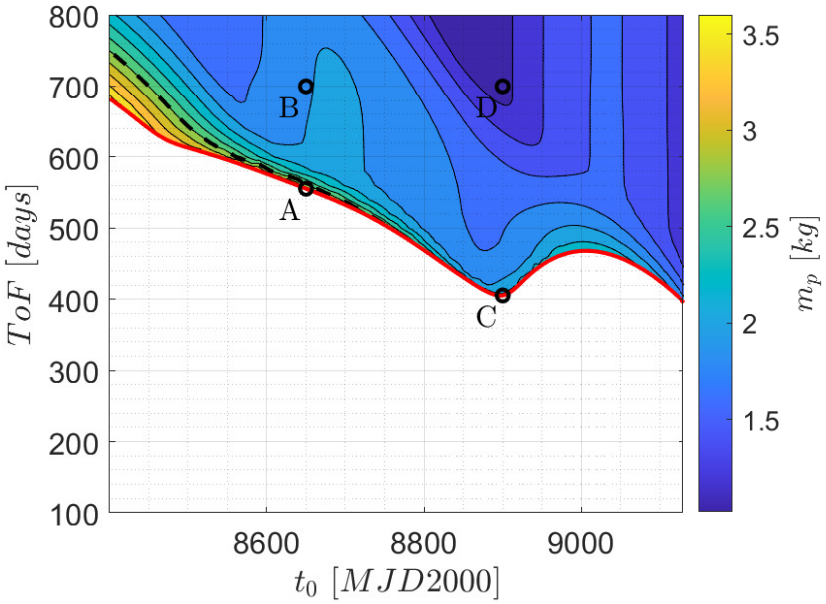


Figure 6.1: $2014\,YD$ porkchop plot.

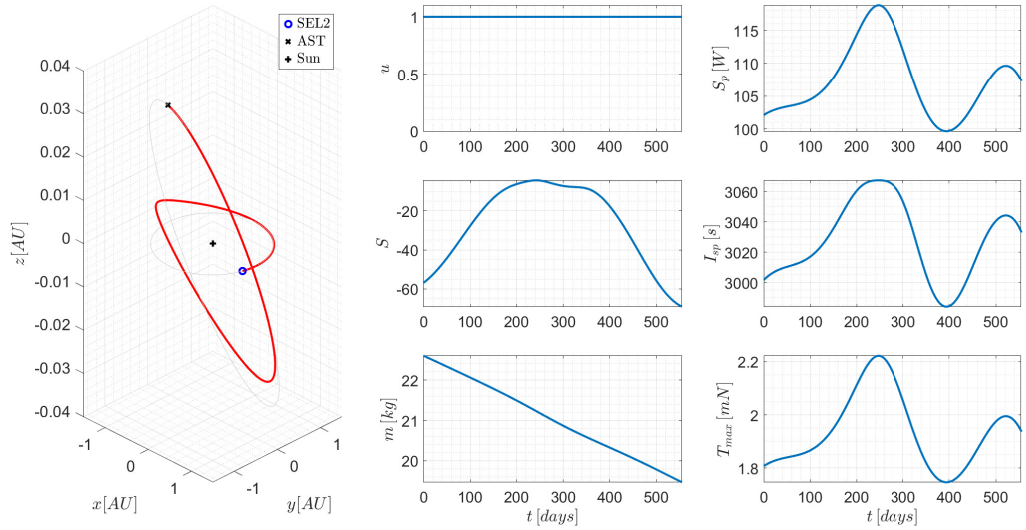


Figure 6.2: SEL2 to $2014\,YD$ (AST) transfer, sample trajectory A.

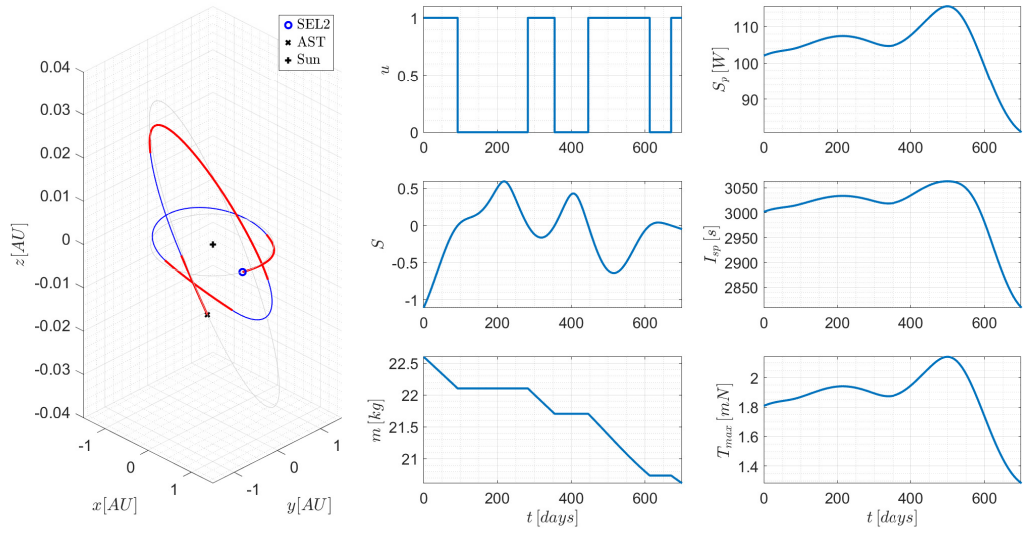


Figure 6.3: SEL2 to 2014 YD (AST) transfer, sample trajectory B.

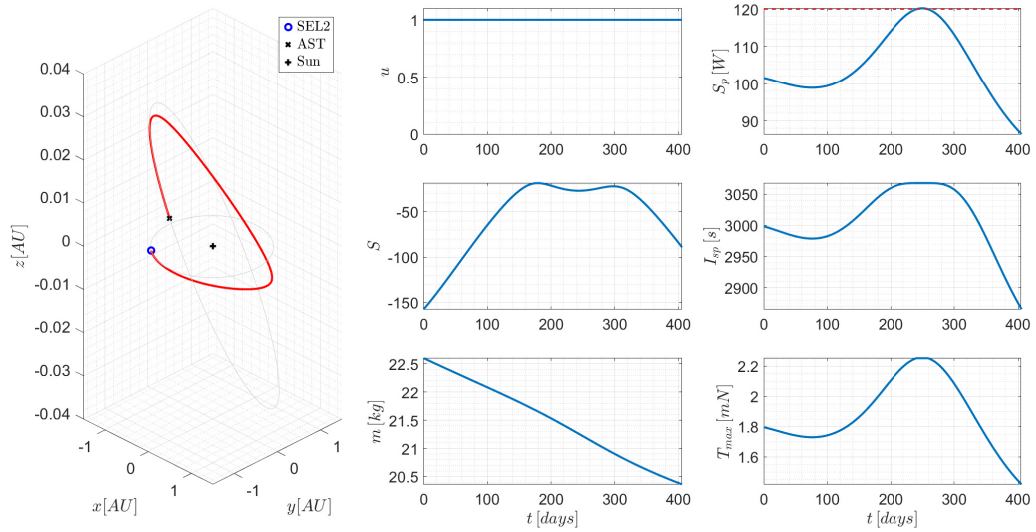


Figure 6.4: SEL2 to 2014 YD (AST) transfer, sample trajectory C.

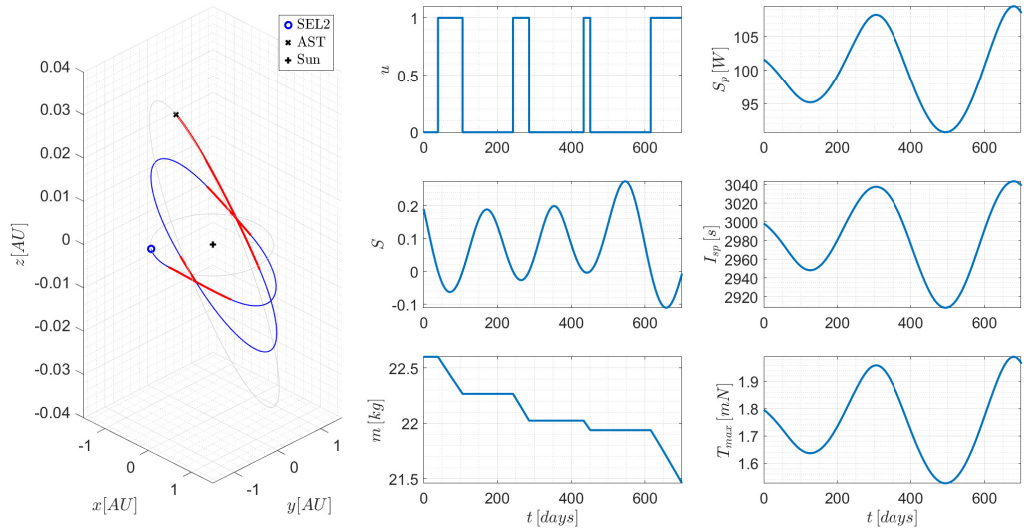


Figure 6.5: SEL2 to 2014 YD (AST) transfer, sample trajectory D.

Table 6.1 collects departure date t_0, transfer time ToF and fuel consumption m_p of the showcased trajectories, together with the final constraints violations $\|\Delta\mathbf{r}\,(t_f)\|$, $\|\Delta\mathbf{v}\,(t_f)\|$, and the Hamiltonian maximum relative variation δH, computed as:

$$\delta H = \left| \frac{\max H\,(t) - \min H\,(t)}{\max H\,(t)} \right| \tag{6.1}$$

Sample	Type	t_0 $[MJD2000]$	ToF $[days]$	m_p $[kg]$	$\|\Delta\mathbf{r}\,(t_f)\|$ $[km]$ $\|\Delta\mathbf{v}\,(t_f)\|$ $[km/s]$	δH $[-]$
A	TO	8650	555.2799	3.1312	$4.3220 \cdot 10^{-6}$ $8.0774 \cdot 10^{-12}$	$2.77367 \cdot 10^{-13}$
B	FO	8650	700.0000	1.9837	$1.2723 \cdot 10^{-5}$ $1.1056 \cdot 10^{-12}$	$7.2977 \cdot 10^{-13}$
C	TO	8900	405.3370	2.2380	$1.5073 \cdot 10^{-6}$ $2.9960 \cdot 10^{-10}$	$1.4665 \cdot 10^{-14}$
D	FO	8900	700.0000	1.1405	$8.9748 \cdot 10^{-6}$ $1.6373 \cdot 10^{-12}$	$3.4808 \cdot 10^{-12}$

Table 6.1: Summary of the results of the sample SEL2-2014 YD trajectories.

First, it can be observed that the final constraints are respected well within the specified tolerances of Equation (5.2). Furthermore, it can be noted that the Hamiltonian shows relative variation comparable or lower with respect to the integration tolerances of 10^{-12}: it can be assumed constant, proving the optimality of the solution paths, in accordance with Equation (2.11).
Additionally, the optimal trajectories are validated against the results obtained with the LT2.0 solver, and are confirmed to be accurate. [29]

Below are presented the porkchop plots associated with the remaining targets of Table 1.3.

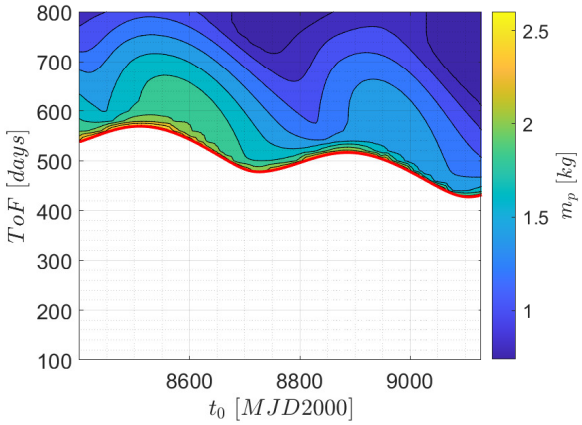


Figure 6.6: 2000 $SG344$ porkshop plot.

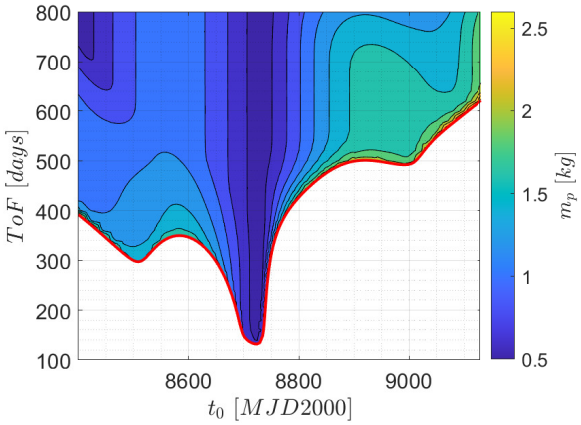


Figure 6.7: 2010 $UE51$ porkchop plot.

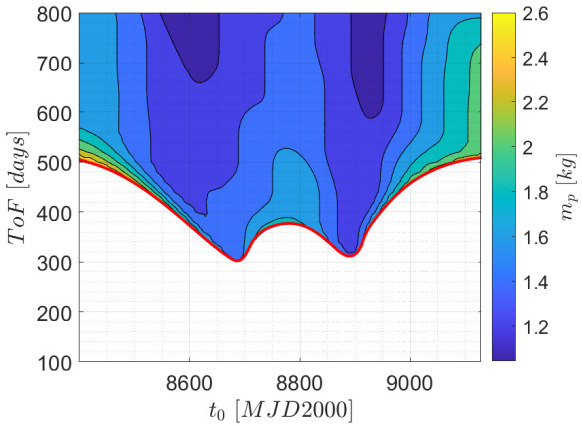


Figure 6.8: 2011 MD porkchop plot.

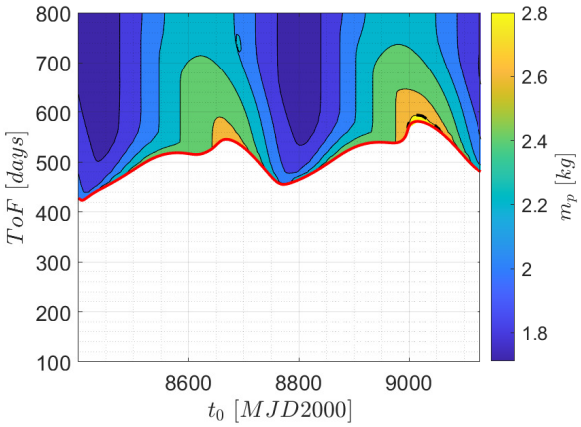


Figure 6.9: 2012 $UV136$ porkshop plot.

The following Table 6.2 collects results of interest regarding the performance of the continuation scheme. In particular, the number of indirect solutions characterizing the porkchop plots is shown. In addition, the rate of convergence of the ToF continuation scheme is presented, indicating the number of implicit time-optimal FO solutions correctly evaluated over the total number of TO solutions. It is observed that the continuation scheme performs well, as it is capable of correctly capturing all but a few implicit time-optimal solutions, $ToF = ToF^*(t_0)$. However, convergence towards the time-optimal limit remains challenging, as the continuation step often has to be greatly reduced. Lastly, the rate of convergence of the energy-to-fuel-optimal homotopy continuation is computed as the number of solutions where ε reached $\varepsilon = 0$ over the total number of solutions. It is pointed out that where $\varepsilon \approx 0$, the error in the evaluation of the propellant mass is negligible.

Target	Total TO Solutions	ToF Convergence Rate	Total FO Solutions	E-to-F Convergence Rate
2000 $SG344$	159	99.37%	2584	97.45%
2010 $UE51$	220	98.18%	4890	94.54%
2011 MD	159	100.00%	3348	99.31%
2012 $UV136$	202	100.00%	3253	97.11%
2014 YD	170	98.24%	2743	96.94%

Table 6.2: Porkchop plot continuation scheme performance.

6.1.2. Solution Scaling Evaluation

Employing the results associated with the porkchop plots presented, the FO trajectories that implicitly optimize time are compared to the TO ones. In particular, for each of such pair of trajectories, the relationship between the initial solutions $\boldsymbol{\lambda}^*_{FO}(t_0)$ and $\boldsymbol{\lambda}^*_{TO}(t_0)$ is first explored.

The following plots, Figures 6.10 to 6.14, show the mean value of the ratios of the costate elements and the associated standard deviation, as:

$$\begin{aligned} \overline{\gamma} &= \frac{1}{7}\sum_{i=1}^{7}\frac{\lambda_{i,FO}(t_0)}{\lambda_{i,TO}(t_0)} \\ \sigma_\gamma &= \sqrt{\frac{\sum_{i=1}^{7}\left(\lambda_{i,FO}(t_0)/\lambda_{i,TO}(t_0)-\overline{\gamma}\right)^2}{7}} \end{aligned} \tag{6.2}$$

It is worth noting that the plots are semi-logarithmic in the y-axis. The blue line represents the mean $\overline{\gamma}$, while the red line and the black dashed line represent the γ_L bound and the $2\gamma_L$ bound, respectively, with γ_L computed in analogy to Equation (4.73).

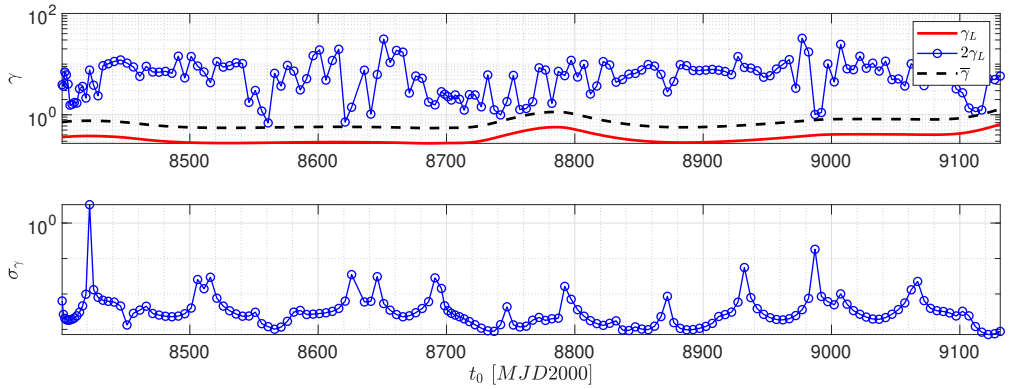


Figure 6.10: Study of 2000 $SG344$ solutions that implicitly optimize time.

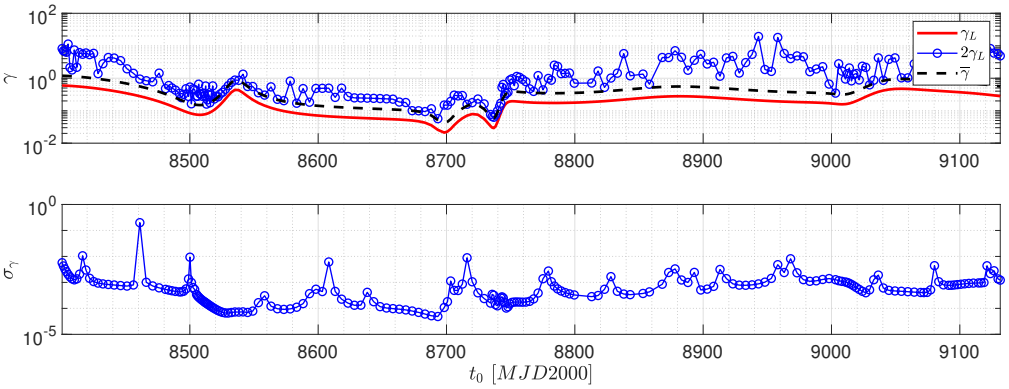


Figure 6.11: Study of 2010 $UE51$ solutions that implicitly optimize time.

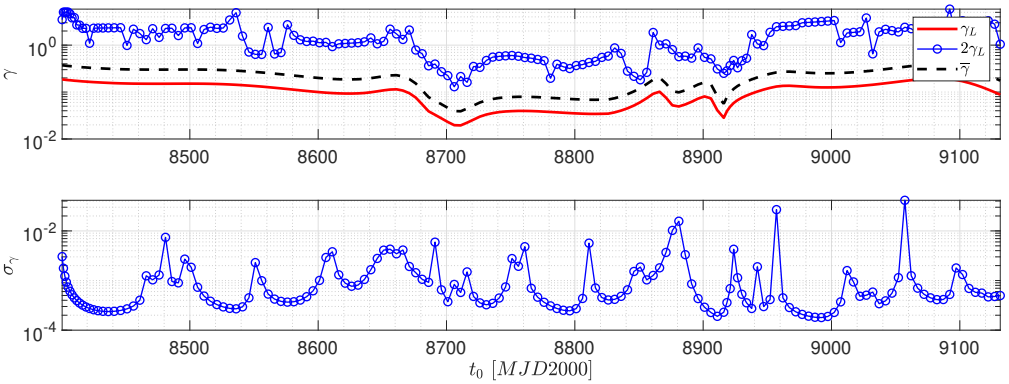


Figure 6.12: Study of 2011 MD solutions that implicitly optimize time.

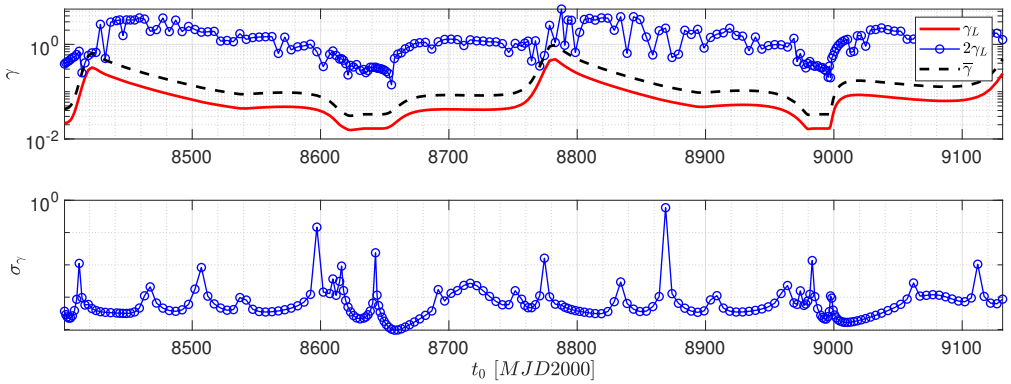


Figure 6.13: Study of 2012 $UV136$ solutions that implicitly optimize time.

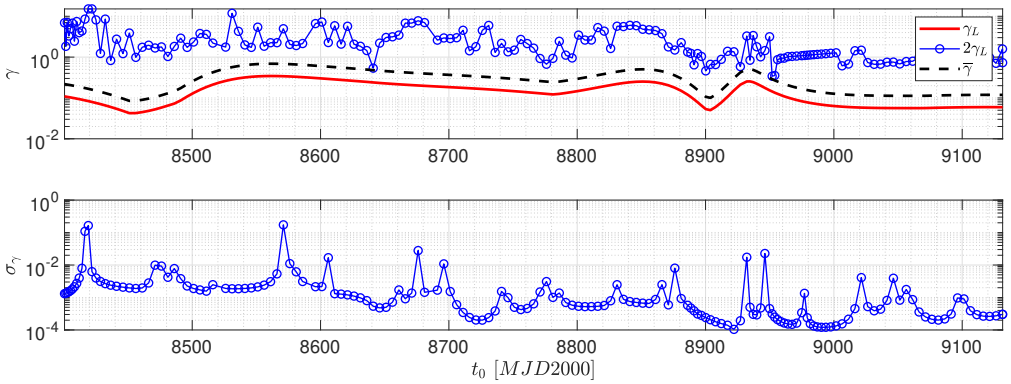


Figure 6.14: Study of 2014 YD solutions that implicitly optimize time.

It is apparent that the values of σ_γ computed are relatively high, even accounting for possible numerical noise. This observation supports the assumptions formulated in Section 4.8.1. In particular, for a set departure date t_0, given $\boldsymbol{\lambda}^*_{TO}(t_0)$ the optimal solution to the TO problem, and $\boldsymbol{\lambda}^*_{FO}(t_0)$ the optimal solution to the FO problem that implicitly optimizes time, the experimental results show that generally:

$$\nexists \gamma \in \mathbb{R}: \quad \boldsymbol{\lambda}^*_{FO}(t_0) = \gamma \boldsymbol{\lambda}^*_{TO}(t_0) \tag{6.3}$$

The equation can only hold in form of an approximation $\boldsymbol{\lambda}^*_{FO}(t_0) \approx \overline{\gamma} \boldsymbol{\lambda}^*_{TO}(t_0)$.
Furthermore, it is observed that no constant exists so that the relationship holds for different time instants: it is not possible, even in approximation, to identify a scaling parameter between the solution paths.

In fact, it can be shown that the divergence between $\boldsymbol{\lambda}^*_{FO}(t)$ and $\boldsymbol{\lambda}^*_{TO}(t)$ is sufficient to render Equation (6.3) true $\forall t$:

$$\nexists \gamma \in \mathbb{R}: \quad \boldsymbol{\lambda}^*_{FO}(t) = \gamma \boldsymbol{\lambda}^*_{TO}(t) \quad \forall t \in [t_0,\, t_0 + ToF^*(t_0)] \tag{6.4}$$

It is observed that even a small initial deviation is enough to make the possibility of scaling not exploitable. In particular, for departure date t_0, given $\boldsymbol{\lambda}^*_{TO}(t_0)$ solution to the TO problem, and the $\overline{\gamma}$ computed as per Equation (6.2) with $\boldsymbol{\lambda}^*_{FO}(t_0)$ the associated FO solution that implicitly optimizes, the approximation $\boldsymbol{\lambda}_{FO}(t_0) = \overline{\gamma}\boldsymbol{\lambda}^*_{TO}(t_0)$ gives rise to a path $\boldsymbol{\lambda}_{FO}(t)$ that is not optimal. The difference between the Euler-Lagrange equations of the TO and FO formulations, due to the introduction of non-homogeneous elements, causes the costate path to generate a significant divergence between the solution trajectories, even in the cases where σ_γ is minimum.
Figure 6.15 shows the final constraints violations of FO trajectories that implicitly optimize time for Table 1.3 targets, tested with $\boldsymbol{\lambda}_{FO}(t_0) = \overline{\gamma}\boldsymbol{\lambda}^*_{TO}(t_0)$, with $\overline{\gamma}$ computed as per Equation (6.2):

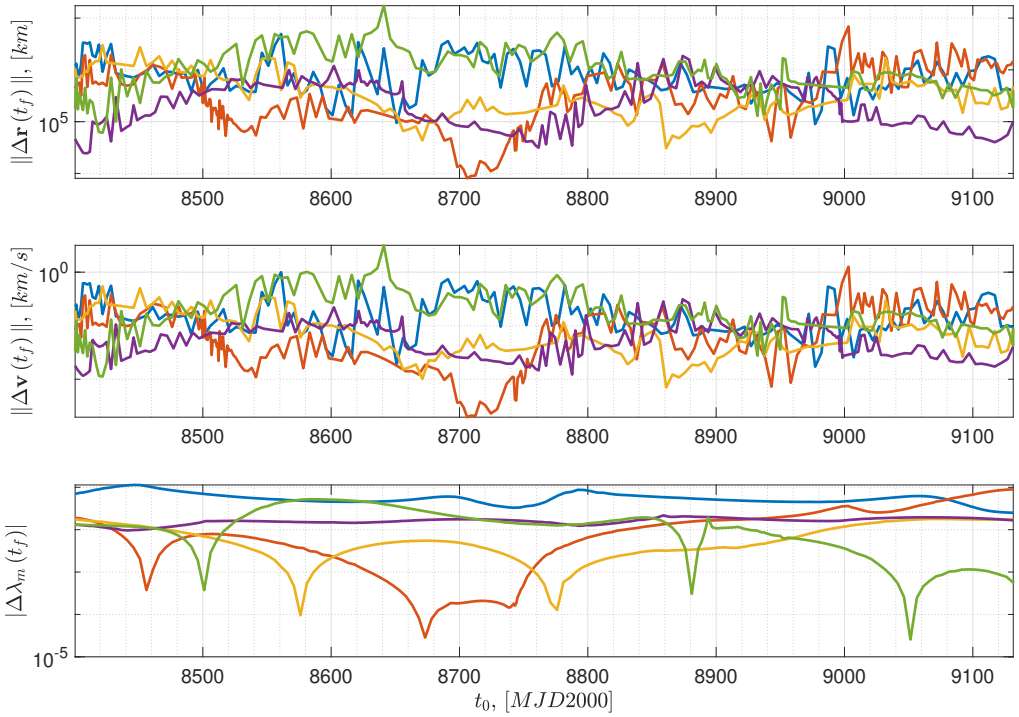


Figure 6.15: Table 1.3 targets final constraints violations using $\boldsymbol{\lambda}_{FO}(t_0) = \overline{\gamma}\boldsymbol{\lambda}^*_{TO}(t_0)$.

In particular, it is observed that, regardless of the sample target and for all departure dates t_0, the final position error $\|\Delta\mathbf{r}(t_f)\|$ is not lower than $8.0435 \cdot 10^3\, km$.

The final velocity error $\|\Delta\mathbf{v}(t_f)\|$ falls within the range of $[1.9555 \cdot 10^{-3}, 3.1723]$ km/s, while $|\lambda(t_f)|$ is not lower than $2.5588 \cdot 10^{-5}$.
Table 6.3 shows the particular cases associated with the lowest σ_γ per each target:

Target	$\min \sigma_\gamma$	$\|\Delta\mathbf{r}(t_f)\|$ $[km]$	$\|\Delta\mathbf{v}(t_f)\|$ $[km/s]$	$\lvert\lambda_m(t_f)\rvert$, $[-]$
$2000\,SG344$	$7.1630 \cdot 10^{-4}$	$4.0609 \cdot 10^{5}$	$9.4534 \cdot 10^{-2}$	$2.5913 \cdot 10^{-2}$
$2010\,UE51$	$4.8381 \cdot 10^{-4}$	$4.7580 \cdot 10^{4}$	$1.2754 \cdot 10^{-2}$	$1.8626 \cdot 10^{-4}$
$2011\,MD$	$1.7950 \cdot 10^{-4}$	$2.4329 \cdot 10^{5}$	$4.6249 \cdot 10^{-2}$	$1.0833 \cdot 10^{-2}$
$2012\,UV136$	$9.5192 \cdot 10^{-5}$	$1.6702 \cdot 10^{5}$	$4.2216 \cdot 10^{-2}$	$1.6088 \cdot 10^{-2}$
$2014\,YD$	$1.0437 \cdot 10^{-4}$	$2.9866 \cdot 10^{5}$	$6.0706 \cdot 10^{-2}$	$1.0445 \cdot 10^{-2}$

Table 6.3: Final constraints violations using $\boldsymbol{\lambda}_{FO}(t_0) = \overline{\gamma}\boldsymbol{\lambda}^*_{TO}(t_0)$ with lowest σ_γ.

On a side note, since the optimal paths of the variable parameters solutions cannot be analytically related, as per Equation (6.4), it cannot be definitively stated that γ_L serves as a boundary for the scaling parameter. However, Figures 6.10 to 6.14 show that $\overline{\gamma}$ numerically appears to satisfy $\overline{\gamma} \geq 2\gamma_L$. This observation may offer valuable insights for future research.

As anticipated in Section 4.8.1, it is observed that, for solutions that implicitly optimize time, indeed the EO path $\boldsymbol{\lambda}^*_{EO}(t)$ and the FO path $\boldsymbol{\lambda}^*_{FO}(t)$ coincide. In fact, $\boldsymbol{\lambda}^*_{FO}(t)$ is computed by 0NPCM continuation, which does not apply any modification to $\boldsymbol{\lambda}^*_{EO}(t)$.
The FO and EO solutions that implicitly optimize time are identical, both described by Figures 6.10 to 6.14. Therefore, the considerations on the FO solutions apply to the EO ones as well, and, given the optimal path $\boldsymbol{\lambda}^*_{TO}(t)$ to the TO problem for set departure date t_0:

$$\nexists\gamma \in \mathbb{R}: \quad \boldsymbol{\lambda}^*_{EO}(t) = \gamma\boldsymbol{\lambda}^*_{TO}(t) \quad \forall t \in [t_0,\, t_0 + ToF^*(t_0)] \tag{6.5}$$

where $\boldsymbol{\lambda}^*_{EO}(t)$ is the optimal path to the EO problem that implicitly optimizes time.

The possibility of employing scaling only to generate a guess to the solution, rather than as the solution itself, arises. However, it should be pointed out that this option does not prove to yield any significant improvements in convergence.
Firstly, it is in fact not possible to infer on γ without solving both the TO and EFO problem. Secondly, assuming $\overline{\gamma}$ was known from the TO solution, initializing the gradient search for the EFO problem with $\boldsymbol{\lambda}_{EFO,\varepsilon}(t_0) = \overline{\gamma}\boldsymbol{\lambda}^*_{TO}(t_0)$ leads to a solution characterized by a different approximate scaling parameter $\boldsymbol{\lambda}^*_{EFO,\varepsilon}(t_0) \approx \widetilde{\gamma}\boldsymbol{\lambda}^*_{TO}(t_0)$, with $\overline{\gamma} \neq \widetilde{\gamma}$.

6.2. Constant Parameters Results

In the following subsections the porkchop plots characterized with the constant parameters formulation described in Section 4.8.2 are reported. In particular, the results presented are obtained using the propulsive parameters of Table 6.4, which, with reference to Equation (4.3), would correspond to the variable parameters identified at heliocentric distance $r \leq 0.9279\,AU$.

Input power, P_{in}	Thrust, T_{max}	Specific impulse, I_{sp}
$120\,W$	$2.2519\,mN$	$3067.8\,s$

Table 6.4: Constant propulsive parameters.

The results are used to assess the propositions advanced in Section 4.8.2, in particular to verify the existence of the closed-form scaling solution between the costate path of FO, or EO, trajectories that implicitly optimize time, and TO trajectories. The notation in this section is condensed to $LTO' \rightarrow TO$, $LEFO', 1 \rightarrow EO$, and $LEFO', 0 \rightarrow FO$.

6.2.1. Porkchop Plots

The porkchop plot associated to the target $2014\,YD$ is presented first. In analogy to the previous section, the continuous red line represents the ensemble of TO solutions, while the dashed black line sets the $2.8\,kg$ propellant mass limit. The acronym CP is used to differentiate the results with respect to the previous section.

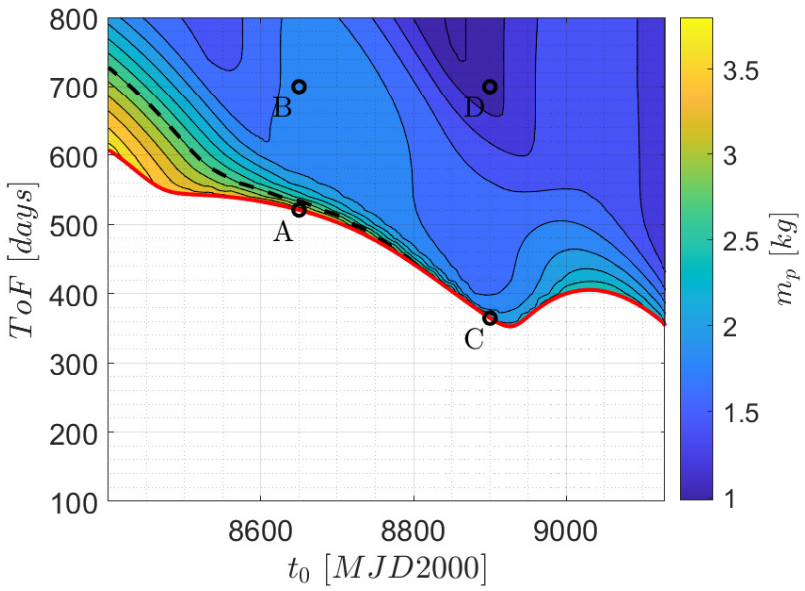


Figure 6.16: $2014\,YD$ porkchop plot (CP).

Sample constant parameters trajectories labeled A through D are presented in Figures 6.17 to 6.20. Regarding the trajectories, red and blue curves represent thrust and coasting arcs, respectively. The side plots indicate throttle u, switching function S, mass profile m, input power P_{in}, thrust T_{max} and specific impulse I_{sp}.

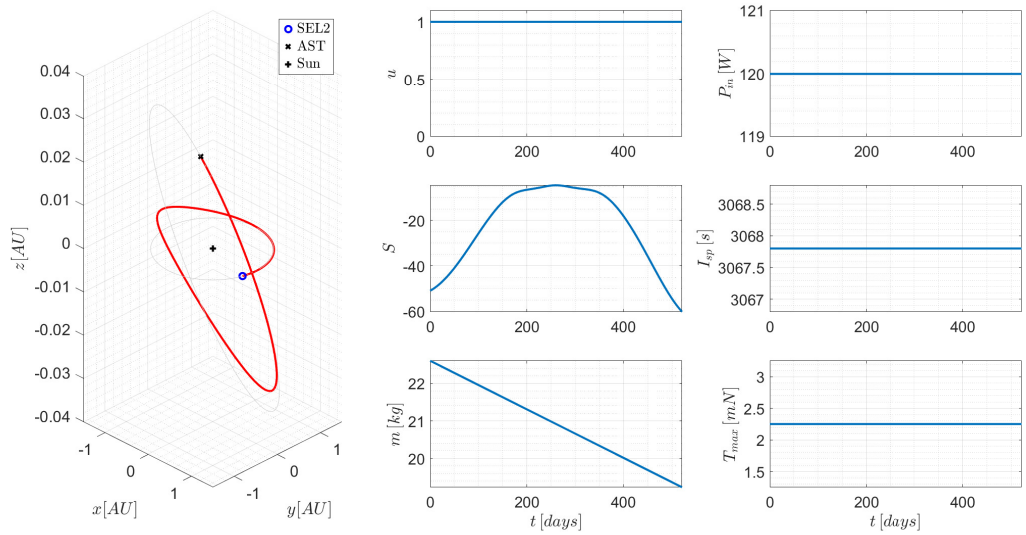


Figure 6.17: SEL2 to 2014 YD (AST) transfer, sample trajectory A (CP).

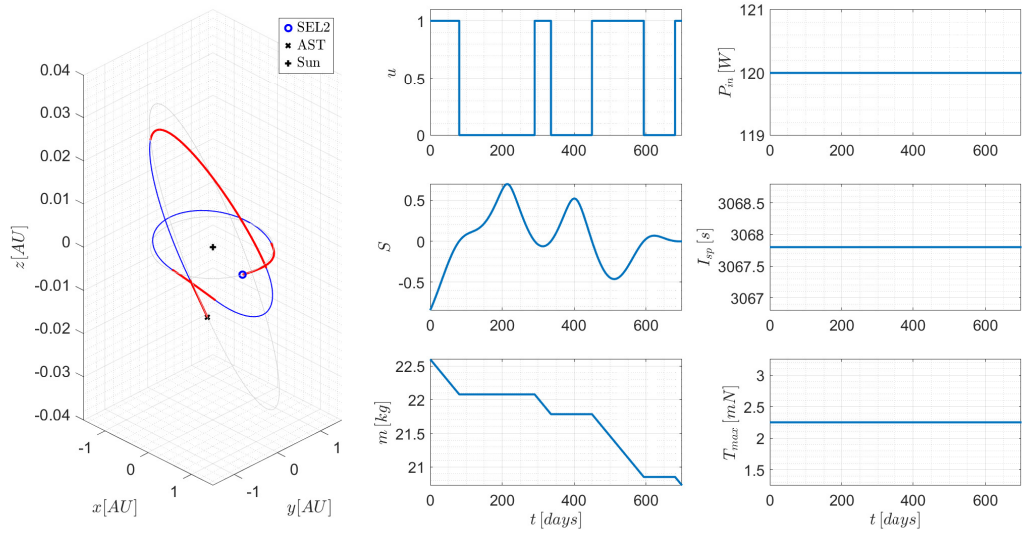


Figure 6.18: SEL2 to 2014 YD (AST) transfer, sample trajectory B (CP).

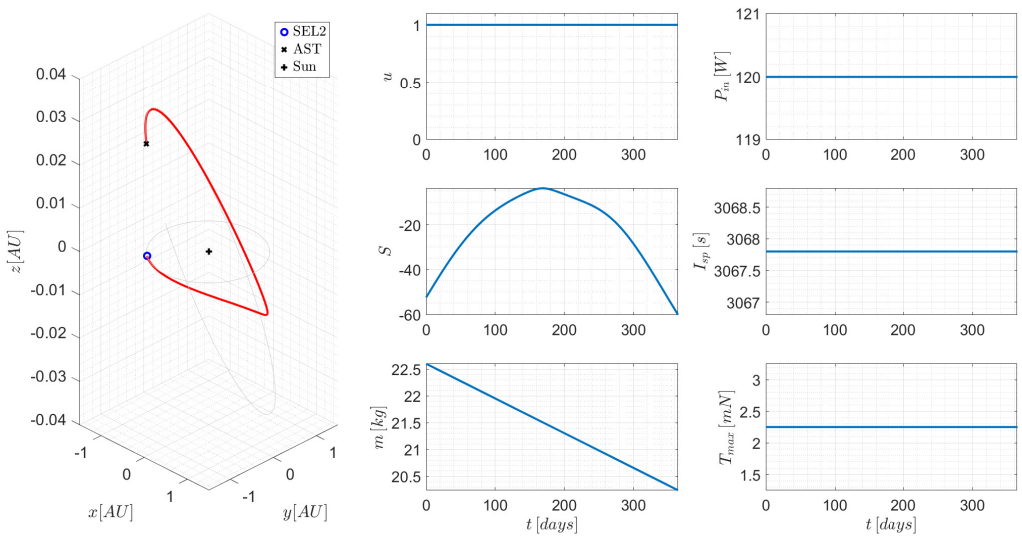


Figure 6.19: SEL2 to 2014 YD (AST) transfer, sample trajectory C (CP).

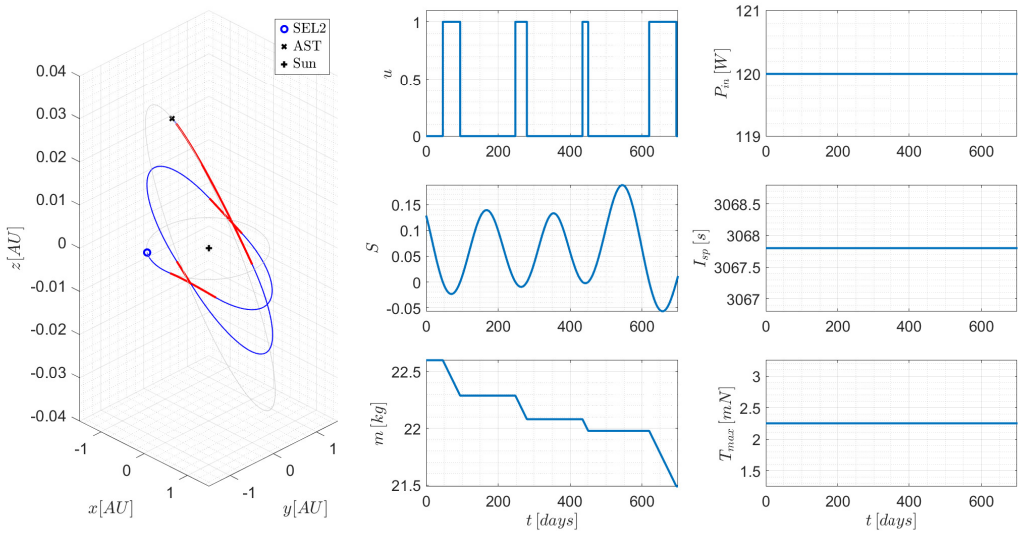


Figure 6.20: SEL2 to 2014 YD (AST) transfer, sample trajectory D (CP).

The key characteristics regarding the sample trajectories are presented in Table 6.5, collecting departure date t_0, transfer time ToF, propellant mass m_p, constraints violations $\|\Delta\mathbf{r}\,(t_f)\,\|$, $\|\Delta\mathbf{v}\,(t_f)\,\|$, and Hamiltonian deviation δH computed in accordance with Equation (6.1).

Sample	Type	t_0 $[MJD2000]$	ToF $[days]$	m_p $[kg]$	$\|\Delta\mathbf{r}(t_f)\|$ $[km]$ $\|\Delta\mathbf{v}(t_f)\|$ $[km/s]$	δH $[-]$
A	TO	8650	520.8553	3.3685	$3.6121 \cdot 10^{-6}$ $3.5678 \cdot 10^{-9}$	$1.7677 \cdot 10^{-13}$
B	FO	8650	700.0000	1.8727	$2.6653 \cdot 10^{-5}$ $2.3440 \cdot 10^{-11}$	$5.8838 \cdot 10^{-13}$
C	TO	8900	364.6330	2.3581	$2.9959 \cdot 10^{-6}$ $3.1955 \cdot 10^{-11}$	$1.1763 \cdot 10^{-14}$
D	FO	8900	700.0000	1.1112	$1.8375 \cdot 10^{-6}$ $3.4405 \cdot 10^{-13}$	$5.4677 \cdot 10^{-12}$

Table 6.5: Summary of the results of the sample SEL2-2014 YD trajectories (CP).

The constant parameters porkchop plots associated with the remaining targets of Table 1.3 are presented in Figures 6.21 to 6.24 in the following page.

Table 6.6 reports the parameters of interest describing the performance of the continuation scheme.

Target	Total TO Solutions	ToF Convergence Rate	Total FO Solutions	E-to-F Convergence Rate
2000 $SG344$	167	100.00%	3002	95.80%
2010 $UE51$	251	98.80%	6354	94.81%
2011 MD	162	98.15%	3827	99.19%
2012 $UV136$	177	100.00%	3830	94.52%
2014 YD	166	100.00%	3021	96.89%

Table 6.6: Porkchop plot continuation scheme performance (CP).

The performance of the continuation scheme is comparable to that of the variable parameters case. Again, it is observed that, although the convergence still proves challenging, the continuation scheme is effective in the correct capture of the majority of solutions belonging to the TO limit. The solutions characterized by $\varepsilon \approx 0$ again show negligible error in the evaluation of the propellant mass.

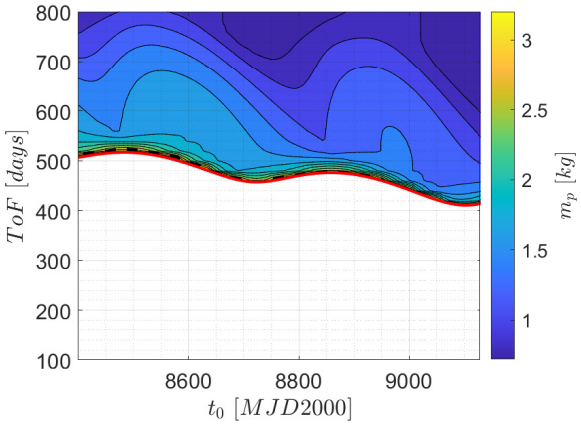


Figure 6.21: 2000 $SG344$ porkchop plot (CP).

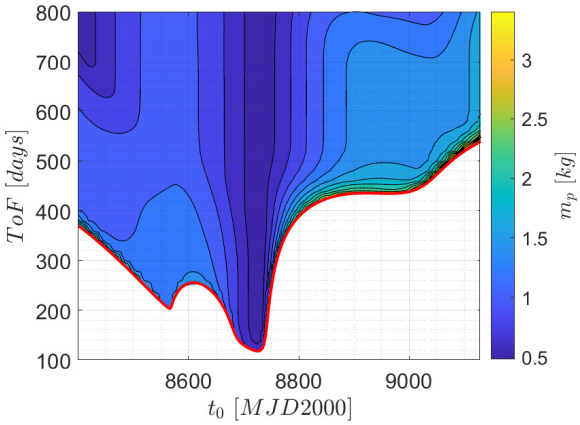


Figure 6.22: 2010 $UE51$ porkchop plot (CP).

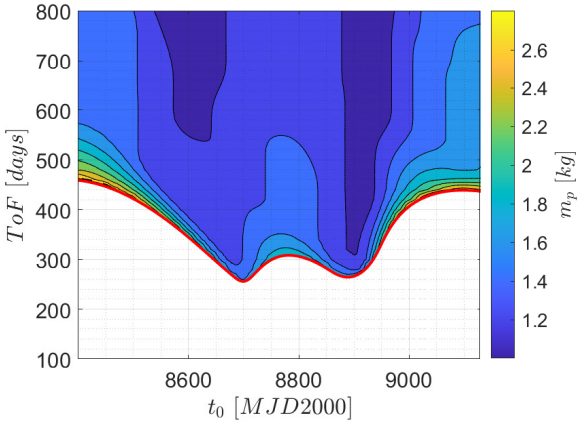


Figure 6.23: 2011 MD porkchop plot (CP).

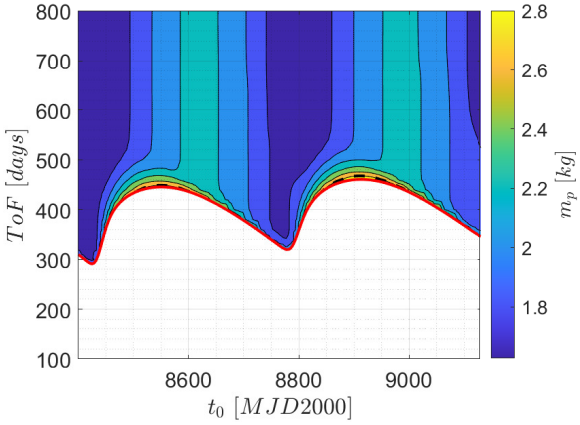


Figure 6.24: 2012 $UV136$ porkchop plot (CP).

6.2.2. Solution Scaling Evaluation

In analogy to the previous sections, the relationship between the TO solution $\boldsymbol{\lambda}^*_{TO}(t_0)$ and the solution of the FO problem that implicitly optimizes time $\boldsymbol{\lambda}^*_{FO}(t_0)$ is first explored. In particular, the $\overline{\gamma}$ values, computed as per Equation (6.2), and the bounds γ_L and $2\gamma_L$, derived from Equation (4.73), associated to the porkchop plots solutions are presented through Figures 6.25 to 6.29.

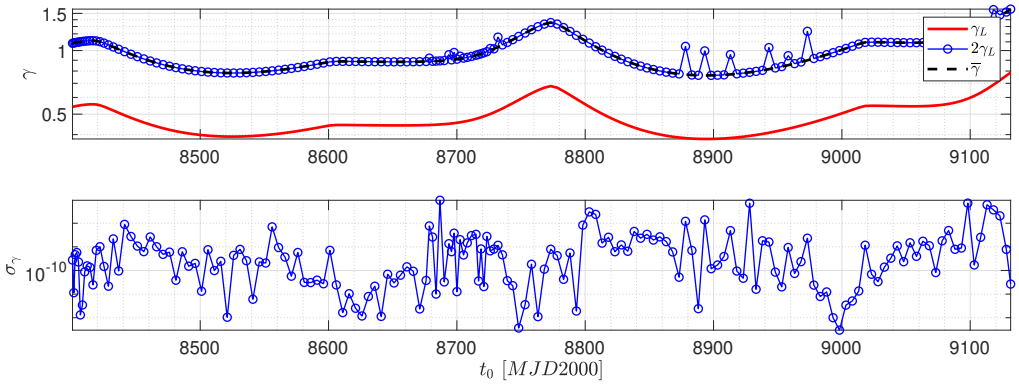


Figure 6.25: Study of 2000 $SG344$ solutions that implicitly optimize time (CP).

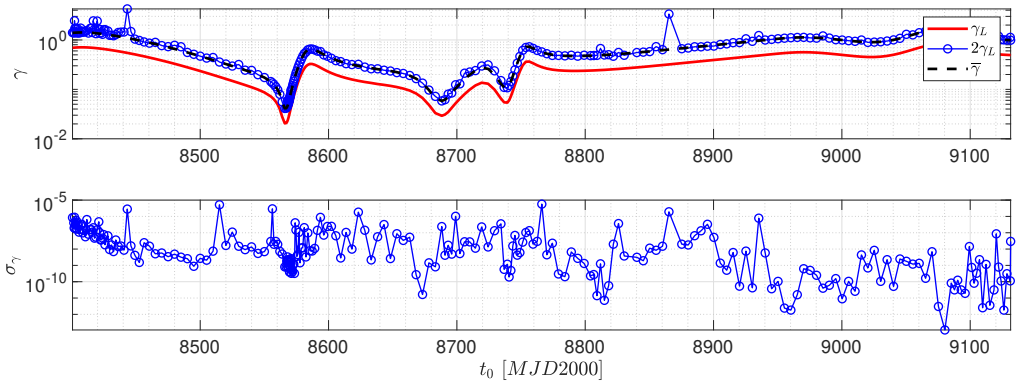


Figure 6.26: Study of 2010 $UE51$ solutions that implicitly optimize time (CP).

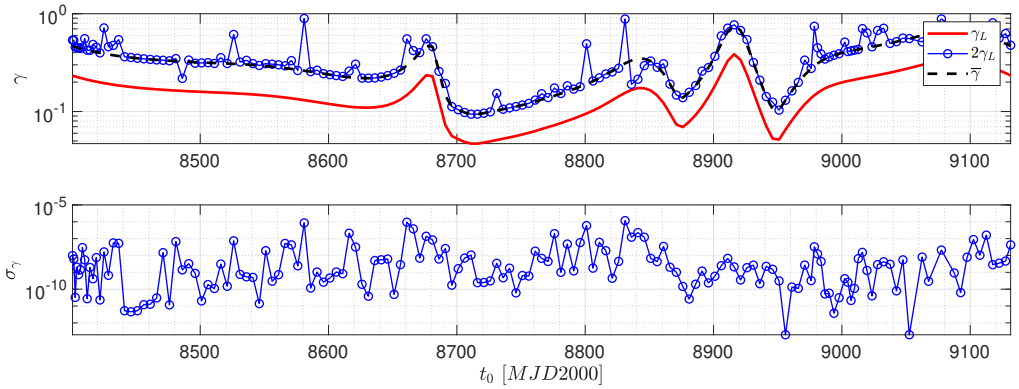


Figure 6.27: Study of 2011 MD solutions that implicitly optimize time (CP).

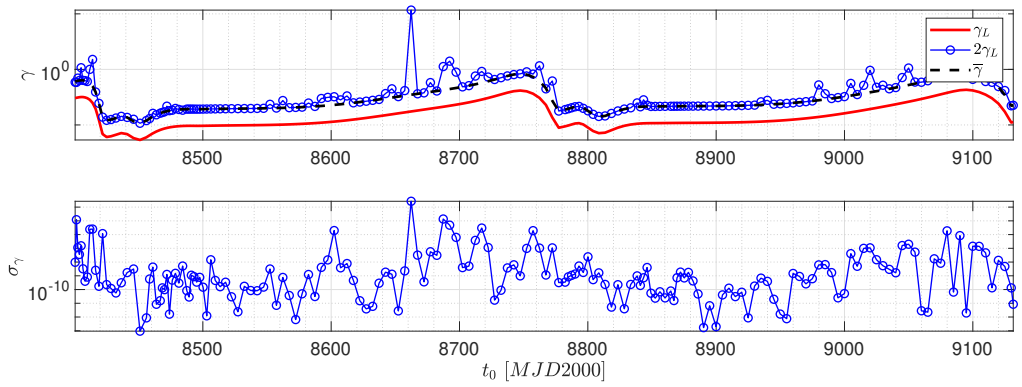


Figure 6.28: Study of 2012 $UV136$ solutions that implicitly optimize time (CP).

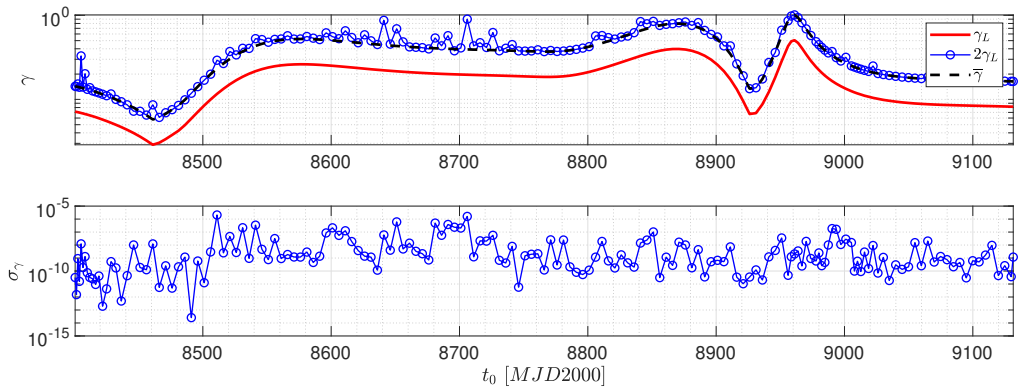


Figure 6.29: Study of 2014 YD solutions that implicitly optimize time (CP).

It is apparent that the $\overline{\gamma}$ profile is much smoother and comparable to the $2\gamma_L$ bound with respect to the variable parameters case. Furthermore, the associated values of σ_γ, computed according to Equation (6.2), are generally much lower.
Therefore, even accounting for numerical error, for a set departure date t_0, given $\boldsymbol{\lambda}^*_{TO}(t_0)$ and $\boldsymbol{\lambda}^*_{FO}(t_0)$ the optimal costate solutions to the TO problem and the FO problem that implicitly optimizes time, respectively, the experimental results confirm that:

$$\exists \gamma \in \mathbb{R}: \quad \boldsymbol{\lambda}^*_{FO}(t_0) = \gamma \boldsymbol{\lambda}^*_{TO}(t_0) \tag{6.6}$$

Furthermore, it is observed that the solution $\boldsymbol{\lambda}^*_{FO}(t_0) = \overline{\gamma}\boldsymbol{\lambda}^*_{TO}(t_0)$ flows into the path $\boldsymbol{\lambda}^*_{FO}(t) = \overline{\gamma}\boldsymbol{\lambda}^*_{TO}(t)$, where $\boldsymbol{\lambda}^*_{FO}(t)$ is indeed optimal for the FO problem.

In fact, all FO problems that implicitly optimize time are tested with $\boldsymbol{\lambda}^*_{FO}(t) = \overline{\gamma}\boldsymbol{\lambda}^*_{TO}(t)$, and the optimality of such a path is confirmed. In particular, Figure 6.30 shows the final constraints violations of FO trajectories that implicitly optimize time for Table 1.3 targets subject to such test:

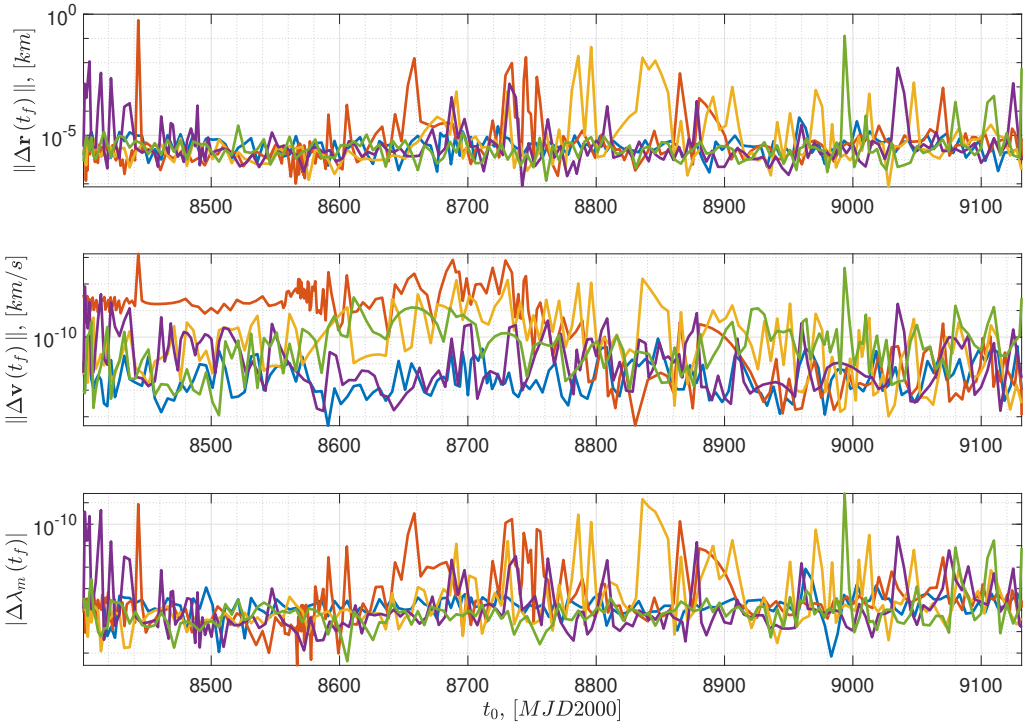


Figure 6.30: Table 1.3 targets final constraints violations using $\boldsymbol{\lambda}^*_{FO}(t) = \overline{\gamma}\boldsymbol{\lambda}^*_{TO}(t)$ (CP).

It is observed that among all the trajectories analyzed, the final position error $\|\Delta\mathbf{r}(t_f)\|$ is lower than $5.6172 \cdot 10^{-1}\, km$, while the final velocity error $\|\Delta\mathbf{v}(t_f)\|$ is lower than $1.3686 \cdot 10^{-7}\, km/s$, both below acceptable optimal tolerance. On the other hand, $|\lambda_m(t_f)|$ ranges $[2.6888 \cdot 10^{-17}, 2.6941 \cdot 10^{-9}]$. The upper bound may appear large, but it is observed to be lower than the one associated with the TO solutions, and therefore even the highest $|\lambda_m(t_f)|$ values are associated with optimal solutions.

The transfer from SEL2 to $2014\,YD$ with departure date $t_0 = 8993.6\,MJD2000$ is the one associated with the highest boundary constraints violations among all possible transfers, and is presented as example for the comparison between the paths $\boldsymbol{\lambda}^*_{FO}(t)$ and $\overline{\gamma}\boldsymbol{\lambda}^*_{TO}(t)$, presented in Figure 6.31 in the following page. In particular, the two paths are shown to be related throughout the entire trajectory by $\overline{\gamma} = 0.3416$.

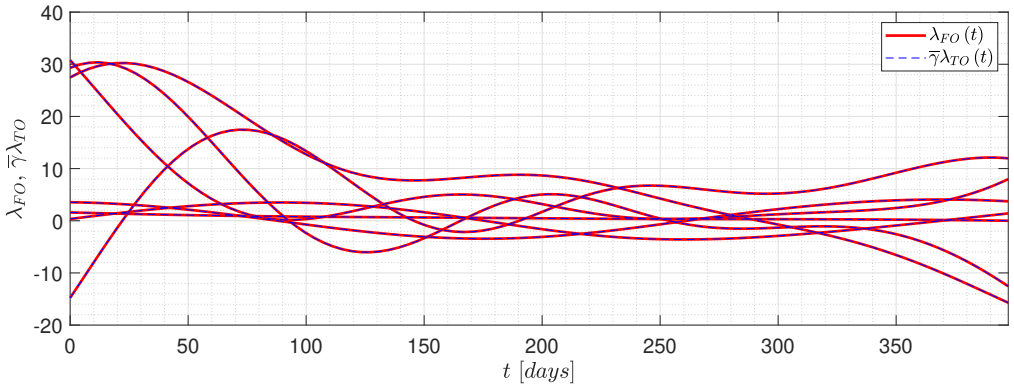


Figure 6.31: TO and implicitly time-optimal FO costate paths for SEL2-2012 to $UV136$ transfer, $t_0 = 8662.3\,MJD2000$.

Given the aforementioned experimental results, it is proved that in the case of constant parameters, the conditions defined in Section 4.8 are met, and, given $\boldsymbol{\lambda}^*_{TO}(t)$ the TO costate path for arbitrary departure date t_0:

$$\exists \gamma \in \mathbb{R}: \quad \boldsymbol{\lambda}^*_{FO}(t) = \gamma \boldsymbol{\lambda}^*_{TO}(t) \quad \forall t \in [t_0,\, t_0 + ToF^*(t_0)] \tag{6.7}$$

where $\boldsymbol{\lambda}^*_{FO}(t)$ is optimal for the FO problem that implicitly optimizes time.

It is observed that, as anticipated in Section 4.8.2, the EO and FO solutions $\boldsymbol{\lambda}^*_{EO}(t)$ and $\boldsymbol{\lambda}^*_{FO}(t)$, respectively, that implicitly optimize time indeed coincide. In fact, excluding a few numerical exceptions, $\boldsymbol{\lambda}^*_{FO}(t)$ is computed in continuation using the 0NPCM from $\boldsymbol{\lambda}^*_{EO}(t)$, without applying any change. Therefore, Figures 6.25 to 6.29 describe both the FO and EO trajectories.
In other words, the results described in this section apply equally to the EO and FO cases, and it is proven that indeed:

$$\exists \gamma \in \mathbb{R}: \quad \boldsymbol{\lambda}^*_{EO}(t) = \gamma \boldsymbol{\lambda}^*_{TO}(t) \quad \forall t \in [t_0,\, t_0 + ToF^*(t_0)] \tag{6.8}$$

where $\boldsymbol{\lambda}^*_{EO}(t)$ is optimal for the EO problem that implicitly optimizes time.

6.2.3. Scaling Parameter Lower-Bound Test

From the analysis presented in Figures 6.25 to 6.29 it can be observed that the scaling parameters identified are such that $\overline{\gamma} \approx 2\gamma_L$, for both FO and EO solutions, consistently with the lower bounds identified in Section 4.8.2. However, it is argued that the solutions approximate the bound of $2\gamma_L$ identified for EO problems by Equation (4.76), due to their derivation via continuation from EO solutions. In particular, it is argued that FO solutions may converge with $\overline{\gamma} \approx \gamma_L$, as per Equation (4.75).
Therefore, in this section, the convergence of the FO and EO problems which implicitly optimize time is examined toward the lowest admissible γ.
Specifically, for each departure date t_0, given the TO optimal path $\boldsymbol{\lambda}^*_{TO}(t_0)$, and given $\boldsymbol{\lambda}^*_{FO}(t_0)$ and $\boldsymbol{\lambda}^*_{EO}(t_0)$ optimal paths for the FO and EO problems implicitly minimizing time, respectively, the objective is to determine:

$$\begin{aligned}
\gamma_{FO} &= \min_{k\in\mathbb{R},\, k\geq 1} k\gamma_L \quad s.t. \quad \boldsymbol{\lambda}^*_{FO}(t) = \gamma_{FO}\boldsymbol{\lambda}^*_{TO}(t) \quad \forall t \in [t_0,\, t_0 + ToF^*(t_0)] \\
\gamma_{EO} &= \min_{k\in\mathbb{R},\, k\geq 1} 2k\gamma_L \quad s.t. \quad \boldsymbol{\lambda}^*_{EO}(t) = \gamma_{EO}\boldsymbol{\lambda}^*_{TO}(t) \quad \forall t \in [t_0,\, t_0 + ToF^*(t_0)]
\end{aligned} \tag{6.9}$$

Fuel-Optimal Test

The procedure is described for the FO case, but applies in analogy to the EO one.
In particular, for each departure date t_0, given the time-optimal path $\boldsymbol{\lambda}^*_{TO}(t)$, starting for an arbitrarily large $\gamma_{FO,i}$, it is iteratively tested whether the path $\gamma_{FO,i}\boldsymbol{\lambda}^*_{TO}(t)$ satisfies the boundary conditions described by the shooting function $\boldsymbol{\Gamma}_{LEFO',0}$, which is analogous to that of the variable parameters case described in Equation (4.21). If that is the case, then $\boldsymbol{\lambda}^*_{FO}(t) = \gamma_{FO,i}\boldsymbol{\lambda}^*_{TO}(t)$ is an optimal path for the FO problem and $\gamma_{FO,i}$ is stored. The scaling parameter is iteratively decreased $\gamma_{FO,i+1} < \gamma_{FO,i}$, until convergence halts. In other words, convergence stops when $\gamma_{FO} = \gamma_{FO,i}$ is found, as $\gamma_{FO,i+1}$ causes:

$$S_{LEFO',0}\left(\mathbf{x}^*(t),\, \gamma_{FO,i+1}\boldsymbol{\lambda}^*_{TO}(t)\right) = \gamma_{FO,i+1} S_{LTO'}\left(\mathbf{x}^*(t),\, \boldsymbol{\lambda}^*_{TO}(t)\right) + 1 > 0 \tag{6.10}$$

The results are presented in terms of γ_{FO}/γ_L, and compared to the lower bound described in Section 4.8.2. Additionally, the difference between $S_{LEFO',0}(t)$ and the threshold $\varepsilon = 0$ is presented as $|\max S_{LEFO',0}(t)|$.
The tests carried out on the FO problems are presented through Figures 6.32 to 6.36.

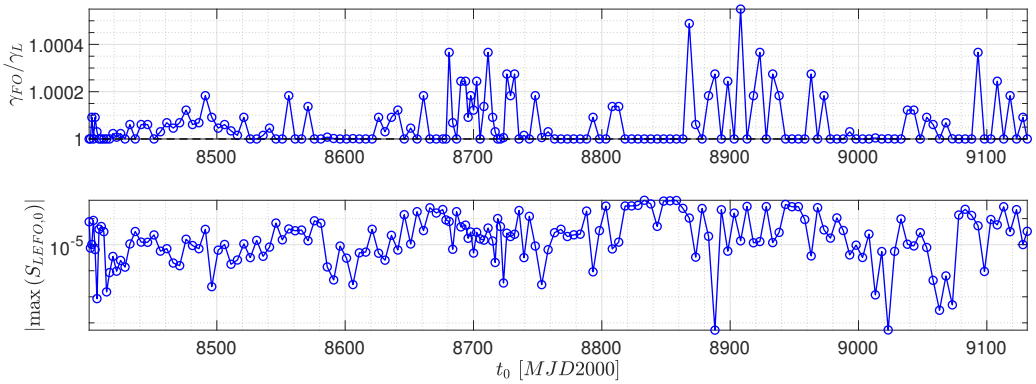


Figure 6.32: Scaling parameter γ_{FO} lower bound test, 2000 $SG344$.

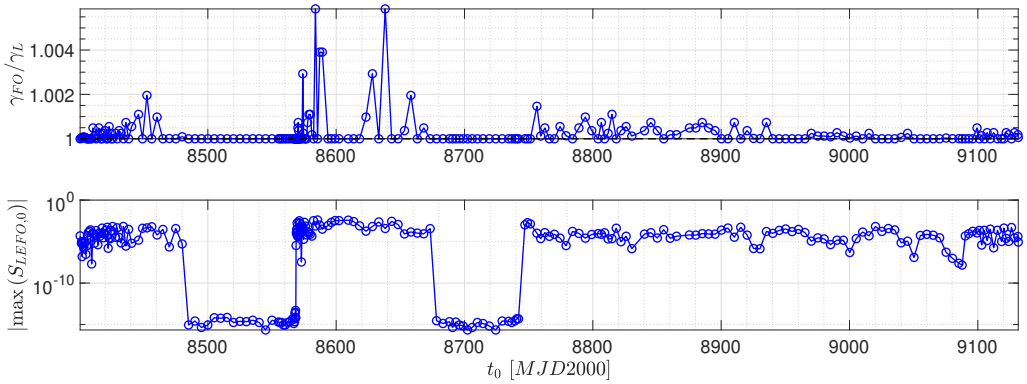


Figure 6.33: Scaling parameter γ_{FO} lower bound test, 2010 $UE51$.

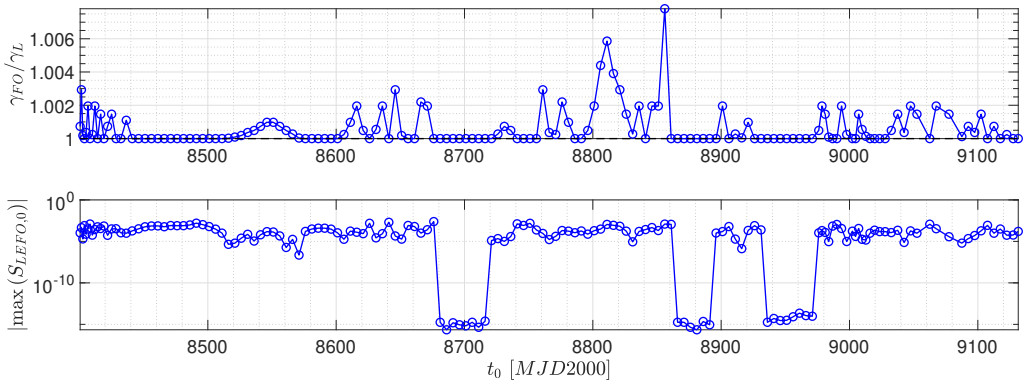


Figure 6.34: Scaling parameter γ_{FO} lower bound test, 2011 MD.

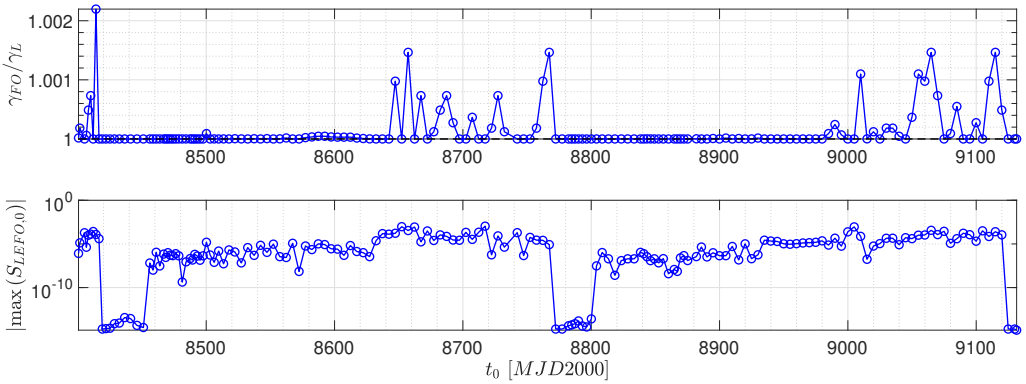


Figure 6.35: Scaling parameter γ_{FO} lower bound test, $2012\,UV136$.

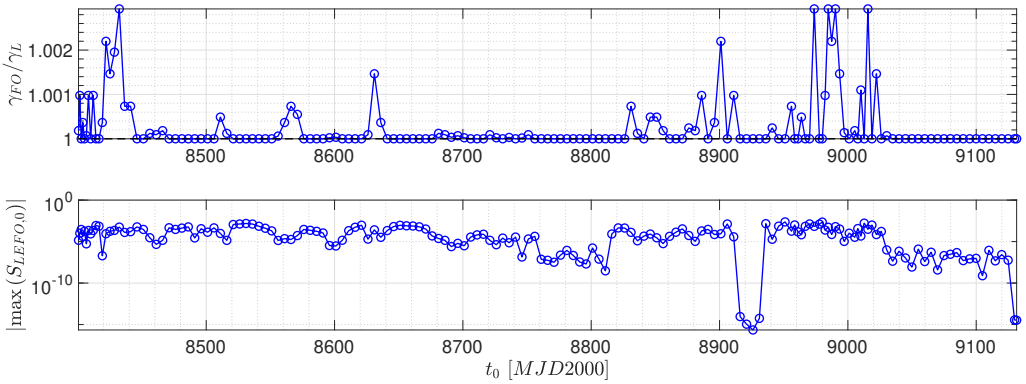


Figure 6.36: Scaling parameter γ_{FO} lower bound test, $20014\,YD$.

The experimental results show that the convergence of a FO problem that implicitly optimizes time can be ensured using the path $\boldsymbol{\lambda}^*_{FO}(t) = \gamma_{FO}\boldsymbol{\lambda}^*_{TO}(t)$, with γ_{FO} as low as $\gamma_{FO} = \gamma_L$.
Some variance can be observed in the evaluations of γ_{FO} and $|\max S_{LEFO',0}(t)|$. Specifically, γ_{FO} deviates from the analytical bound less than $1\%\gamma_L$, and $|\max S_{LEFO',0}(t)|$ at worst ranges in $10^{-2\div 3}$. These variations can be attributed to the precision of the numerical evaluation of $\max S_{LTO'}(t)$.

Therefore, the lower bound $\gamma_{FO} \geq \gamma_L$ identified with the analysis performed in Section 4.8.2 is verified. It is also highlighted that, in the case of implicitly time-optimal solutions, the experiment also confirms that the optimal costate path is not unique, as all paths $\boldsymbol{\lambda}^*_{FO}(t) = \gamma_{FO}\boldsymbol{\lambda}^*_{TO}(t)$ with $\gamma_{FO} \geq \gamma_L$ are admissible and optimal.

With reference to the transfer previously associated to greatest constraint violations, namely the SEL2-2014 YD transfer with $t_0 = 8993.6\,MJD2000$, the comparison between $S_{TO} = S_{LTO'}\left([\mathbf{x}^*(t), \boldsymbol{\lambda}^*_{TO}(t)]\right)$ and $S_{FO} = S_{LEFO',0}\left([\mathbf{x}^*(t), \gamma_{FO}\boldsymbol{\lambda}^*_{TO}(t)]\right)$ is presented in Figure 6.37, to showcase the effect of path scaling. The scaling parameter γ_{FO} is the lowest identified by the experiment with reference to Figure 6.36.

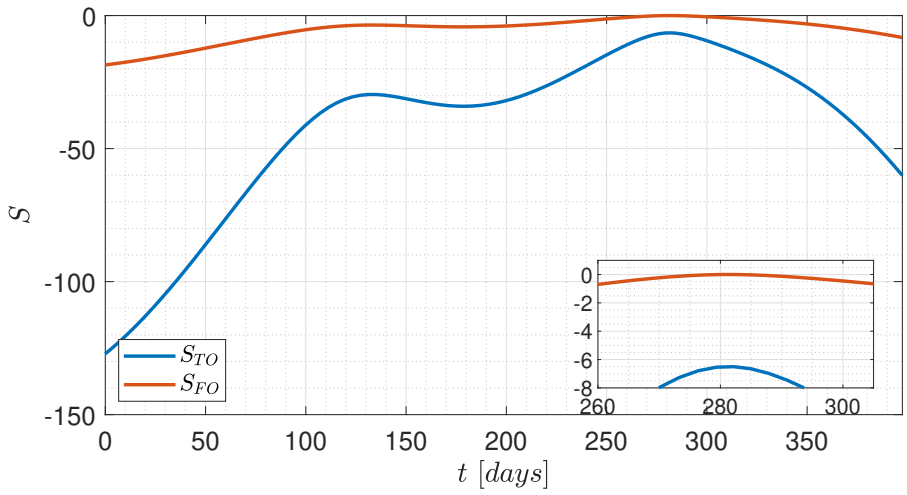


Figure 6.37: S_{TO} and S_{FO} comparison employing $\boldsymbol{\lambda}^*_{FO}(t) = \gamma_{FO}\boldsymbol{\lambda}^*_{TO}(t)$.

Energy-Optimal Test

The experiment is repeated for the EO problem. The results are presented in terms of γ_{EO}/γ_L, and $|\max S_{LEFO',1}(t) + 1|$, with reference to Figures 6.38 to 6.42.

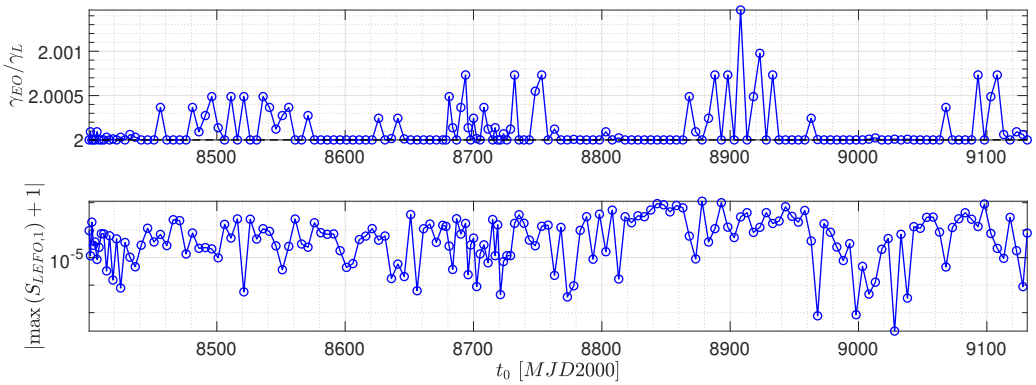


Figure 6.38: Scaling parameter γ_{EO} lower bound test, 2000 $SG344$.

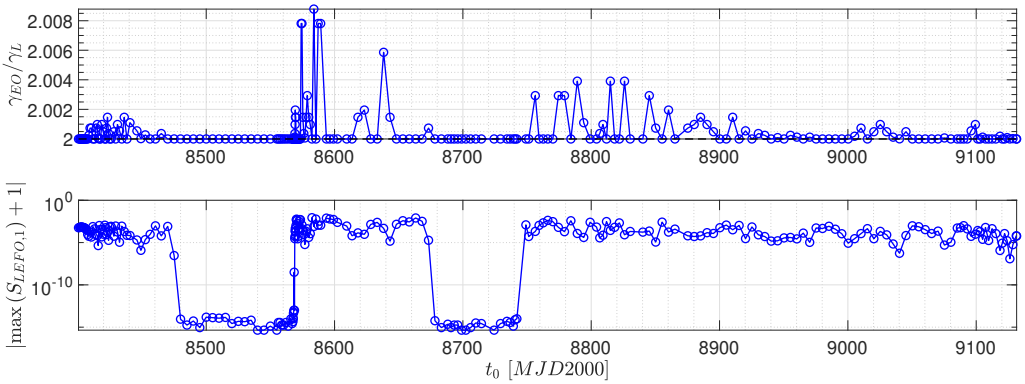


Figure 6.39: Scaling parameter γ_{EO} lower bound test, 2010 $UE51$.

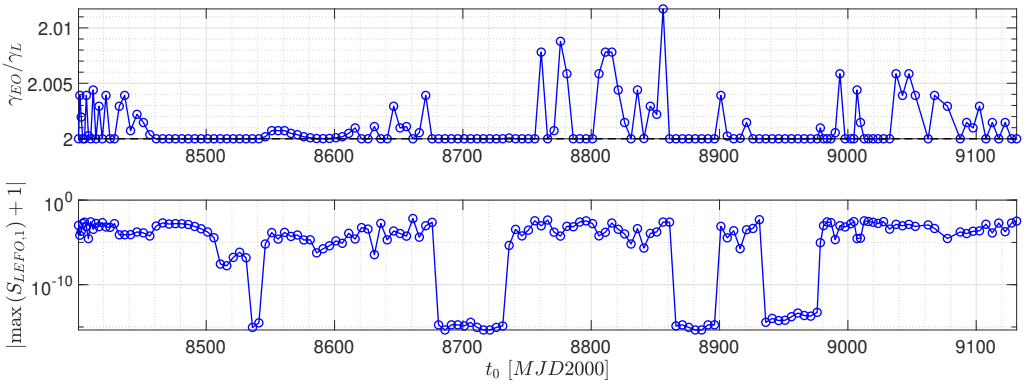


Figure 6.40: Scaling parameter γ_{EO} lower bound test, 2011 MD.

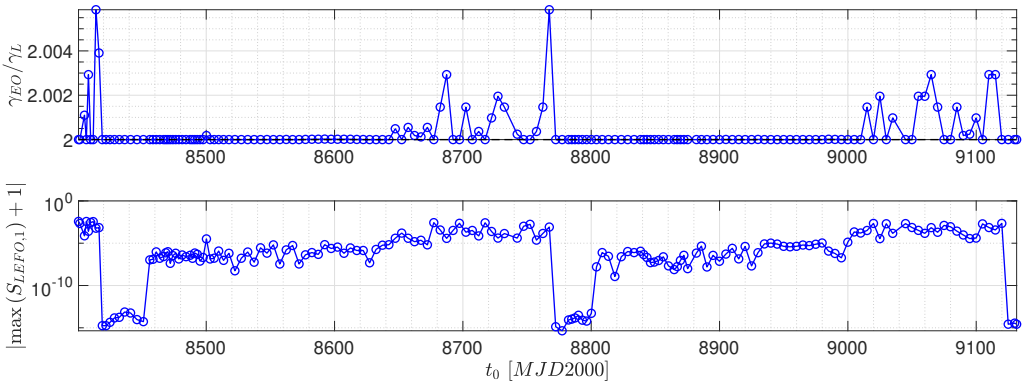


Figure 6.41: Scaling parameter γ_{EO} lower bound test, 2012 $UV136$.

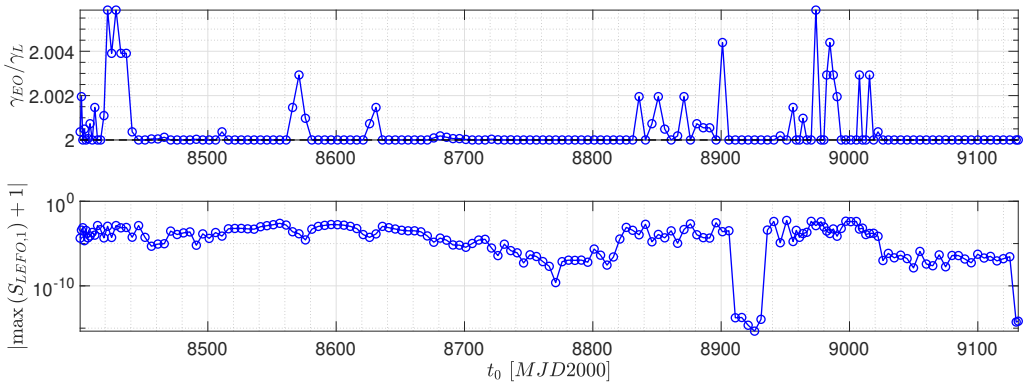


Figure 6.42: Scaling parameter γ_{EO} lower bound test, $2014\,YD$.

The experimental results show the convergence of an EO problem that implicitly optimizes time is guaranteed with path $\boldsymbol{\lambda}^*_{EO}(t) = \gamma_{EO}\boldsymbol{\lambda}^*_{TO}(t)$ with $\gamma_{EO} \geq 2\gamma_L$. In particular, γ_{EO} is shown to deviate from the analytical bound less than 1.5%, with $|\max S_{LEFO',1}(t) + 1|$ at worst ranging in $10^{-2\div 3}$.

As per the FO case, the experiment confirms the existence of lower bound $\gamma_{EO} \geq 2\gamma_L$, as anticipated in Section 4.8.2. It is also confirmed that all paths $\boldsymbol{\lambda}^*_{EO}(t) = \gamma_{EO}\boldsymbol{\lambda}^*_{TO}(t)$ for $\gamma_{EO} \geq 2\gamma_L$ are admissible and optimal.

Figure 6.37 is modified by adding $S_{EO} = S_{LEFO',1}([\mathbf{x}^*(t), \gamma_{EO}\boldsymbol{\lambda}^*_{TO}(t)])$, with γ_{EO} the lowest admissible with reference to Figure 6.42. In particular, Figure 6.43:

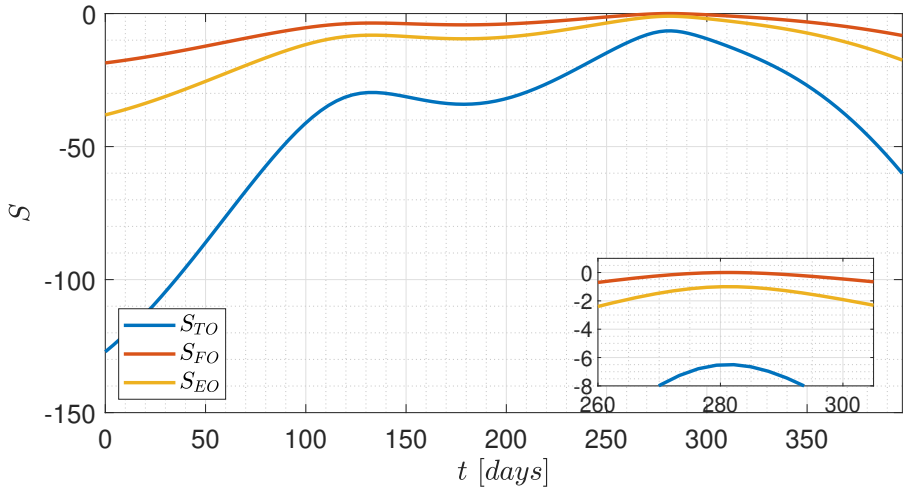


Figure 6.43: S_{TO}, S_{FO} and S_{EO} comparison employing $\boldsymbol{\lambda}^*_{FO}(t) = \gamma_{FO}\boldsymbol{\lambda}^*_{TO}(t)$ and $\boldsymbol{\lambda}^*_{EO}(t) = \gamma_{EO}\boldsymbol{\lambda}^*_{TO}(t)$.

Answer to Research Question:

It is concluded that, for an arbitrary formulation satisfying the criteria listed in Section 4.8, given the TO path $\mathbf{y}^*_{TO}(t) = [\mathbf{x}^*(t), \boldsymbol{\lambda}^*_{TO}(t)]^\top$ and the associated optimal time-of-flight $ToF^*(t_0)$ for departure date t_0, the FO optimal path $\mathbf{y}^*_{FO}(t)$ and the EO optimal path $\mathbf{y}^*_{EO}(t)$ for problems characterized by the same departure date t_0 and time-of-flight $ToF = ToF^*(t_0)$ are such that:

$$\begin{aligned} \exists \gamma_{FO} \in \mathbb{R}: \quad \mathbf{y}^*_{FO}(t) &= \begin{pmatrix} \mathbf{x}^*(t) \\ \gamma_{FO}\boldsymbol{\lambda}^*_{TO}(t) \end{pmatrix} \\ \exists \gamma_{FO} \in \mathbb{R}: \quad \mathbf{y}^*_{EO}(t) &= \begin{pmatrix} \mathbf{x}^*(t) \\ \gamma_{EO}\boldsymbol{\lambda}^*_{TO}(t) \end{pmatrix} \end{aligned} \tag{6.11}$$

In particular, regarding the formulations analyzed in this document, the structure of the switching functions allow to limit γ_{FO} and γ_{EO} so that:

$$\begin{aligned} \mathbf{y}^*_{FO}(t) &= \begin{pmatrix} \mathbf{x}^*(t) \\ \gamma_{FO}\boldsymbol{\lambda}^*_{TO}(t) \end{pmatrix} \qquad \forall \gamma_{FO} \in \mathbb{R}: \ \gamma_{FO} \geq \gamma_L \\ \mathbf{y}^*_{EO}(t) &= \begin{pmatrix} \mathbf{x}^*(t) \\ \gamma_{EO}\boldsymbol{\lambda}^*_{TO}(t) \end{pmatrix} \qquad \forall \gamma_{EO} \in \mathbb{R}: \ \gamma_{EO} \geq 2\gamma_L \end{aligned} \tag{6.12}$$

The experimental results presented confirm the assumptions advanced in Section 4.8. To further validate these hypotheses, particularly to demonstrate that the results are not tied to the problem at hand but apply broadly to problems satisfying the conditions outlined in Section 4.8, an additional experiment is presented in Appendix B.
Specifically, solution path scaling is tested for time, fuel and energy-optimal problems characterized by different system dynamics with respect to those of the M-ARGO mission. The corresponding problem formulations are found to satisfy the conditions set in Section 4.8, and, indeed, the paths to FO and EO problems that implicitly optimize time can be derived by scaling the costate path of the TO solution, therefore proving the validity of the set requirements.

6.3. Performance Comparison

This section presents a practical example that demonstrates the implementation of the results discussed in this chapter. Specifically, the convergence of FO and EO problems that implicitly optimize time, in the case of constant parameters, is evaluated using initial guesses of the form $\boldsymbol{\lambda}_{FO,g}(t_0) = \gamma_{FO}\boldsymbol{\lambda}^*_{TO}(t_0)$ and $\boldsymbol{\lambda}_{EO,g}(t_0) = \gamma_{EO}\boldsymbol{\lambda}^*_{TO}(t_0)$, respectively, with scaling parameters set to $\gamma_{FO} = 1.01 \cdot \gamma_L$ and $\gamma_{EO} = 1.01 \cdot 2\gamma_L$.
For each departure date t_0, the number of iteration steps needed to achieve convergence is recorded and compared to the iteration count of the final step in the ToF continuation scheme, described in Chapter 5.
The results, summarized in Table 6.7, are presented as average iteration counts per target over all considered departure dates t_0. It is worth noting that the iteration counts of the continuation scheme apply to both FO and EO implicitly time-optimal solutions, as these solutions are identical.

Target	Mean Number of Iterations		
	FO, $\boldsymbol{\lambda}_{FO,g}(t_0)$	EO, $\boldsymbol{\lambda}_{EO,g}(t_0)$	ToF Continuation
2000 $SG344$	1.0000	1.0000	21.4192
2010 $UE51$	1.0040	1.0040	31.0524
2011 MD	1.0000	1.0185	37.4630
2012 $UV136$	1.0000	1.0000	30.5480
2014 YD	1.0181	1.0120	29.4398

Table 6.7: Average iteration count for FO, EO convergence vs ToF continuation.

It is observed that, for both FO and EO problems, using a scaling parameter just 1% greater than the respective identified lower bounds ensures convergence at the given starting point in the vast majority of cases, significantly reducing the iterations count compared to the continuation scheme.
This comparison demonstrates how solutions obtained through parameter scaling can enhance the efficiency of continuation methods. When the scaling parameters satisfy the identified bounds, such solutions are optimal a priori and could be effectively used to either initiate or conclude the ToF continuation path for both FO and EO solutions.

7 | Conclusions And Future Development

7.1. Conclusions

This dissertation addresses a gap in the literature concerning the relationship between time-optimal and either fuel-optimal or energy-optimal problems, particularly those that implicitly optimize time.

It is argued that TO trajectories, characterized by time-of-flight $ToF^*(t_0)$, represent a limit case within the broader family of FO and EO solutions. In particular, FO and EO trajectories for which $ToF = ToF^*(t_0)$ coincide with those identified by the TO search. To establish the conditions under which this holds, a set dynamical requirements is explored, starting from the analysis of the M-ARGO mission case study. [30]

A plausible closed-form relationship linking the TO costate solutions to those of implicitly time-optimal FO and EO problems is formulated, specifically in the form of a linear scalar relationship between the initial costates. The validity and applicability of this formulation are examined, leading to the establishment of a set of analytical criteria that govern the Euler-Lagrange equations and shooting functions of the problem. It is shown that, regardless of the problem formulation, if these criteria are satisfied, the TO costate paths can be related to to those of the implicitly time-optimal FO and EO problems not only at initial time t_0, but throughout the entire trajectory.

The analytical propositions are successfully validated through numerical experimentation. In particular, tests have been conducted using:

- SEP variable propulsive parameters heliocentric 2BP dynamics, characteristic of M-ARGO mission
- Constant propulsive parameters heliocentric 2BP dynamics
- Constant propulsive parameters Earth-Moon CR3BP dynamics

The numerical results confirm the feasibility of applying scaling to the solution paths of the problems characterized by formulation compliant with the criteria devised. In particular, with reference the M-ARGO case study, it applies to the case characterized by constant propulsive parameters, but not to the variable ones.
Additionally, for the formulations analyzed characterized by constant propulsive parameters, it is anticipated and later verified that the path scaling parameter is not unique, and its lower bound can be determined using only information from the TO path.
Thus, it is proven that FO and EO trajectories and costate paths for implicitly time-optimal problems can be computed starting from the TO solution, eliminating the need of directly solving the associated OCPs.

These findings constitute an interesting advancement in optimal control theory. In fact, while established methods exist for relating fuel-optimal and energy-optimal solutions, time-optimal control problems are generally regarded as a separate problem. This work challenges that statement by presenting an analytical framework that bridges the gap between formulations.

7.2. Future Development

From a practical standpoint, the results presented in this dissertation can be leveraged to improve the performance of continuation methods. In fact, convergence of FO and EO solutions becomes increasingly difficult for time-of-flight values approaching the time-optimal limit. Given a set of TO solutions, a continuation scheme could constrain the ToF continuation process using solutions found through path scaling.

It is important to note that the indirect formulation of an OCP is specific to the problem statement in its dynamics, cost function, and constraints. Although the analytical criteria devised in this dissertation are general, the study of the bounds of the scaling parameter is tied to the formulations examined. Further exploration of additional problem statements could lead to the development of a more comprehensive methodology linking time-optimal and implicitly time-optimal solutions.

More specifically, further studies can be conducted on the SEP formulation of the M-ARGO mission. In particular, it is observed that, although solution scaling is feasible only in approximation, the bounds for the scaling parameter defined as per the constant propulsive parameters case appear to hold. The closed-form approach developed in this dissertation could potentially be modified, perhaps by adding biasing terms at initial time, between the solutions of interest. While this analysis falls beyond the scope the present work due to its case-specific nature, it represents a promising direction for future research.

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A | Appendix A

A.1. LTO jacobian

The analytical jacobian to the Lagrange TO formulation presented in Section 4.2, specifically Equation (4.12). To lighten the notation $\boldsymbol{\lambda}^*_{TO}(t_0) = \boldsymbol{\lambda}_0$ and $t^*_f = t_f$:

$$\mathbf{J}_{\boldsymbol{\Gamma}_{LTO}} = \frac{\partial \boldsymbol{\Gamma}_{LTO}\left([\boldsymbol{\lambda}_0, t_f]\right)}{\partial\left[\boldsymbol{\lambda}_0, t_f\right]} = \begin{bmatrix} \dfrac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\lambda}_0} & \dfrac{\partial \boldsymbol{\psi}}{\partial t_f} \\ \dfrac{\partial \lambda_m}{\partial \boldsymbol{\lambda}_0} & \dfrac{\partial \lambda_m}{\partial t_f} \\ \dfrac{\partial \Omega_f}{\partial \boldsymbol{\lambda}_0} & \dfrac{\partial \Omega_f}{\partial t_f} \end{bmatrix} \tag{A.1}$$

It is noted that t^*_f is interchangeable with $ToF^*(t_0)$. In particular:

$$\begin{aligned} t^*_f &= t_0 + ToF^*(t_0) \\ \partial t^*_f &= \partial ToF^*(t_0) \end{aligned} \tag{A.2}$$

The STM submatrices are identified as $\partial \mathbf{v}/\partial \mathbf{w} = \boldsymbol{\Phi}_{\mathbf{vw}}$. Recalling that $\mathbf{a}_t = -\mu \dfrac{\mathbf{r}_t}{r_t^3}$, the aforementioned quantities:

$$\frac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\lambda}_0} = \begin{bmatrix} \boldsymbol{\Phi}_{\mathbf{r}\boldsymbol{\lambda}}(t_f, t_0) \\ \boldsymbol{\Phi}_{\mathbf{v}\boldsymbol{\lambda}}(t_f, t_0) \end{bmatrix} \tag{A.3a}$$

$$\frac{\partial \boldsymbol{\psi}}{\partial t_f} = \begin{bmatrix} \dot{\mathbf{r}}(t_f) - \mathbf{v}_t(t_f) \\ \dot{\mathbf{v}}(t_f) - \mathbf{a}_t(t_f) \end{bmatrix} \tag{A.3b}$$

$$\frac{\partial \lambda_m}{\partial \boldsymbol{\lambda}_0} = \boldsymbol{\Phi}_{\lambda_m \boldsymbol{\lambda}}(t_f, t_0) \tag{A.3c}$$

$$\frac{\partial \lambda_m}{\partial t_f} = \dot{\lambda}_m\left(t_f\right) \tag{A.3d}$$

$$\frac{\partial \Omega_f}{\partial \boldsymbol{\lambda}_0} = \boldsymbol{\lambda}\left(t_f\right)^\top \left.\frac{\partial \mathbf{f}}{\partial \boldsymbol{\lambda}_0}\right|_{t_f} + \left[\mathbf{f} - \begin{pmatrix}\mathbf{v}_t \\ \mathbf{a}_t \\ 0\end{pmatrix}\right]_{t_f}^\top \boldsymbol{\Phi}_{\boldsymbol{\lambda}\boldsymbol{\lambda}}\left(t_f, t_0\right) \tag{A.3e}$$

$$\frac{\partial \Omega_f}{\partial t_f} = \boldsymbol{\lambda}\left(t_f\right)^\top \left[\frac{\partial \mathbf{f}}{\partial t_f} - \begin{pmatrix}\mathbf{a}_t \\ \dot{\mathbf{a}}_t \\ 0\end{pmatrix}\right]_{t_f} + \left[\mathbf{f} - \begin{pmatrix}\mathbf{v}_t \\ \mathbf{a}_t \\ 0\end{pmatrix}\right]_{t_f}^\top \dot{\boldsymbol{\lambda}}\left(t_f\right) \tag{A.3f}$$

where

$$\left.\frac{\partial \mathbf{f}}{\partial \boldsymbol{\lambda}_0}\right|_{t_f} = \left[\begin{array}{c} \boldsymbol{\Phi}_{\mathbf{v}\boldsymbol{\lambda}} \\ \left(3\mu\frac{\mathbf{r}\mathbf{r}^\top}{r^5} - \mu\frac{\mathbf{I}_{3\times 3}}{r^3}\right)\boldsymbol{\Phi}_{\mathbf{r}\boldsymbol{\lambda}} - u\frac{\boldsymbol{\lambda}_\mathbf{v}}{\lambda_\mathbf{v} m}\left(\mathbf{t}_r^\top \boldsymbol{\Phi}_{\mathbf{r}\boldsymbol{\lambda}} - \frac{T_{max}}{m}\boldsymbol{\Phi}_{m\boldsymbol{\lambda}}\right) + \dots \\ \cdots - u\frac{T_{max}}{\lambda_\mathbf{v} m}\left(\mathbf{I}_{3\times 3} - \frac{\boldsymbol{\lambda}_\mathbf{v}\boldsymbol{\lambda}_\mathbf{v}^\top}{\lambda_\mathbf{v}^2}\right)\boldsymbol{\Phi}_{\boldsymbol{\lambda}_\mathbf{v}\boldsymbol{\lambda}} \\ -\frac{u}{c^2}\left(c\mathbf{t}_r^\top - g_0 T_{max}\mathbf{i}_r^\top\right)\boldsymbol{\Phi}_{\mathbf{r}\boldsymbol{\lambda}} \end{array}\right]_{t_f} \tag{A.4a}$$

$$\left.\frac{\partial \mathbf{f}}{\partial t_f}\right|_{t_f} = \left[\begin{array}{c} \dot{\mathbf{v}} \\ 3\mu\frac{\mathbf{r}^\top\mathbf{v}}{r^5}\mathbf{r} - \mu\frac{\mathbf{v}}{r^3} - u\frac{\boldsymbol{\lambda}_\mathbf{v}}{\lambda_\mathbf{v} m}\left(\mathbf{t}_r^\top\mathbf{v} - T_{max}\frac{\dot{m}}{m}\right) + \dots \\ \cdots - u\frac{T_{max}}{\lambda_\mathbf{v} m}\left(\dot{\boldsymbol{\lambda}}_\mathbf{v} - \frac{\boldsymbol{\lambda}_\mathbf{v}^\top\dot{\boldsymbol{\lambda}}_\mathbf{v}}{\lambda_\mathbf{v}^2}\boldsymbol{\lambda}_\mathbf{v}\right) \\ -\frac{u}{c^2}\left(c\mathbf{t}_r^\top - g_0 T_{max}\mathbf{i}_r^\top\right)\mathbf{v} \end{array}\right] \tag{A.4b}$$

$$\dot{\mathbf{a}}_t = 3\mu\frac{\mathbf{r}_t^\top\mathbf{r}_t}{r_t^5}\mathbf{r}_t - \mu\frac{\mathbf{v}_t}{r_t^3} \tag{A.4c}$$

A.2. LEFO jacobian

The analytical jacobian to the Lagrange energy-to-fuel-optimal formulation presented in Section 4.3, specifically Equation (4.21):

$$\mathbf{J}_{\mathbf{\Gamma}_{LEFO}} = \frac{\partial \mathbf{\Gamma}_{LEFO}\left(\boldsymbol{\lambda}_0\right)}{\partial \boldsymbol{\lambda}_0} = \begin{bmatrix} \dfrac{\partial \boldsymbol{\psi}}{\partial \boldsymbol{\lambda}_0} & \dfrac{\partial \boldsymbol{\psi}}{\partial t_f} \\ \dfrac{\partial \lambda_m}{\partial \boldsymbol{\lambda}_0} & \dfrac{\partial \lambda_m}{\partial t_f} \end{bmatrix} \tag{A.5}$$

with elements defined as per Equation (A.3)

B | Appendix B

In order to present additional results supporting the assumptions presented in Section 4.8, an additional analysis is conducted on a problem characterized by different formulation. In particular, time, fuel and energy-optimal trajectories describing a transfer from a geostationary transfer orbit (GTO) to an Earth-Moon L_1 halo orbit are sought. [39]
The case study employs low-thrust propulsion, with propulsive parameters $T_{max} = 10\,N$ and $I_{sp} = 3000\,s$, and the satellite initial wet mass $m_0 = 1500\,kg$.
The circular restricted three-body problem (CR3BP) equations of motions in the Earth-Moon rotating frame read as [39]:

$$\dot{\mathbf{x}} = \mathbf{f}\left(\mathbf{x},\, u,\, \hat{\boldsymbol{\alpha}}\right) = \begin{Bmatrix} \dot{\mathbf{r}} \\ \dot{\mathbf{v}} \\ \dot{m} \end{Bmatrix} = \begin{Bmatrix} \mathbf{v} \\ \mathbf{g}\left(\mathbf{r}\right) + \mathbf{h}\left(\mathbf{v}\right) + u\dfrac{T_{max}}{m}\hat{\boldsymbol{\alpha}} \\ -u\dfrac{T_{max}}{c} \end{Bmatrix} \tag{B.1}$$

Where $c = I_{sp}g_0$ is the engine exhaust velocity, while the quantities $\mathbf{g}\left(\mathbf{r}\right)$ and $\mathbf{h}\left(\mathbf{v}\right)$ are defined as [39]:

$$\mathbf{g}\left(\mathbf{r}\right) = \begin{Bmatrix} x - \dfrac{\left(1-\mu\right)\left(x+\mu\right)}{r_1^3} - \mu\dfrac{x+\mu-1}{r_2^3} \\ y - \dfrac{\left(1-\mu\right)y}{r_1^3} - \mu\dfrac{y}{r_2^3} \\ -\dfrac{\left(1-\mu\right)z}{r_1^3} - \mu\dfrac{z}{r_2^3} \end{Bmatrix} \tag{B.2a}$$

$$\mathbf{h}\left(\mathbf{v}\right) = \begin{Bmatrix} 2v_x \\ -2v_y \\ 0 \end{Bmatrix} \tag{B.2b}$$

With $r_1 = \sqrt{\left(x+\mu\right)^2 + y^2 + z^2}$ and $r_2 = \sqrt{\left(x+\mu-1\right)^2 + y^2 + z^2}$.

The mass parameter $\mu = \dfrac{m_{Moon}}{m_{Earth} + m_{Moon}}$ is reported in Table B.1, together with physical constants of importance.

Physical Constant	Value
Mass parameter, μ	$1.21506683 \cdot 10^{-2}$
Earth gravitational acceleration, g_0	$9.80665\, m/s^2$
Length Unit, LU	$3.84405000 \cdot 10^5\, km$
Velocity Unit, VU	$1.02323281\, km/s$
Mass Unit, MU	$1500\, kg$

Table B.1: Physical constants for Earth-Moon CR3BP problem. [39]

The energy and fuel-optimal solutions are sought by employing the smoothing technique described in [3]. Therefore, the formulations are characterized by cost functions for the time-optimal and energy-to-fuel-optimal problems defined as per Equation (4.7) and Equation (4.15), respectively.
The trajectories are constrained by the initial GTO state $[\mathbf{r}_0, \mathbf{v}_0]$, the initial mass m_0 and the final state belonging to the L_1 halo orbit $[\mathbf{r}_f, \mathbf{v}_f]$. Those values of the boundary conditions are reported in Tables B.2 and B.3. They read as:

$$\boldsymbol{\psi}_0\left(\mathbf{x}\left(t_0\right)\right) = \begin{Bmatrix} \mathbf{r}\left(t_0\right) - \mathbf{r}_0 \\ \mathbf{v}\left(t_0\right) - \mathbf{v}_0 \\ m\left(t_0\right) - m_0 \end{Bmatrix} \tag{B.3a}$$

$$\boldsymbol{\psi}\left(\mathbf{x}\left(t_0\right)\right) = \begin{Bmatrix} \mathbf{r}\left(t_f\right) - \mathbf{r}_f \\ \mathbf{v}\left(t_f\right) - \mathbf{v}_f \end{Bmatrix} \tag{B.3b}$$

Physical Constant	Value
$\mathbf{r}_0$, $[LU]$	$\begin{bmatrix} -0.019488511458668 \\ -0.016033479812051 \\ 0 \end{bmatrix}$
$\mathbf{v}_0$, $[VU]$	$\begin{bmatrix} 8.918881923678198 \\ -4.081793688818725 \\ 0 \end{bmatrix}$

Table B.2: Trajectory initial boundary conditions. [39]

Physical Constant	Value
$\mathbf{r}_f$, $[LU]$	$\begin{bmatrix} 0.823385182067467 \\ 0 \\ -0.022277556273235 \end{bmatrix}$
$\mathbf{v}_f$, $[VU]$	$\begin{bmatrix} 0 \\ 0.134184170262437 \\ 0 \end{bmatrix}$

Table B.3: Trajectory final boundary conditions. [39]

The Hamiltonians to the TO, H_{TO}, and EFO problems, $H_{EFO,\varepsilon}$, and are defined according to Equation (4.8) and Equation (4.17), respectively. Therefore, the primer vector is defined as per Equation (4.9a), and the throttle control logics are equivalent to those of the constant parameter case presented in Section 4.8.2. The switching functions S_{TO} and $S_{EFO,\varepsilon}$ are defined as per Equation (4.10) and Equation (4.19), respectively.
Time-optimal and energy-to-fuel-optimal Euler-Lagrange equations satisfy the conditions imposed by Equation (4.54) and Equation (4.55): they are equivalent and, despite being highly non-linear, they are homogeneous with degree one with respect to the costate magnitude. In particular, given $\mathbf{y} = [\mathbf{x}, \boldsymbol{\lambda}]^\top$ the Euler-Lagrange equations read as [39]:

$$\mathbf{F}(\mathbf{y}) = \begin{Bmatrix} \mathbf{v} \\ \mathbf{g}(\mathbf{r}) + \mathbf{h}(\mathbf{v}) - u\dfrac{\boldsymbol{\lambda}_\mathbf{v}}{\lambda_\mathbf{v}}\dfrac{T_{max}}{m} \\ -u\dfrac{T_{max}}{c} \\ -\mathbf{G}^\top\boldsymbol{\lambda}_\mathbf{v} \\ -\boldsymbol{\lambda}_\mathbf{r} - \mathbf{H}^\top\boldsymbol{\lambda}_\mathbf{v} \\ -u\lambda_\mathbf{v}\dfrac{T_{max}}{m^2} \end{Bmatrix} \tag{B.4}$$

Where $\mathbf{G} = \dfrac{\partial \mathbf{g}}{\partial \mathbf{r}}$ and $\mathbf{H} = \dfrac{\partial \mathbf{h}}{\partial \mathbf{v}}$. [39]

The shooting functions $\mathbf{\Gamma}_{TO}$ and $\mathbf{\Gamma}_{EFO,\varepsilon}$ are defined as [39]:

$$\mathbf{\Gamma}_{TO}\left(\left[\boldsymbol{\lambda}_{TO}^{*}\left(t_0\right),\, ToF^{*}\left(t_0\right)\right]\right) = \begin{Bmatrix} \boldsymbol{\psi}\left(\mathbf{x}\left(t_f\right)\right) \\ \lambda_m\left(t_f\right) \\ H_{TO}\left(t_f\right) \end{Bmatrix} \tag{B.5a}$$

$$\mathbf{\Gamma}_{EFO,\varepsilon}\left(\boldsymbol{\lambda}_{TO}^{*}\left(t_0\right)\right) = \begin{Bmatrix} \boldsymbol{\psi}\left(\mathbf{x}\left(t_f\right)\right) \\ \lambda_m\left(t_f\right) \end{Bmatrix} \tag{B.5b}$$

They satisfy the conditions described in Section 4.8. Therefore, the problem is expected to be suitable for solution path scaling in the case of implicitly time-optimal trajectories.

The problems are solved with the same modalities described in Chapter 5, namely single shooting gradient search. The identified time-optimal trajectory, presented in Figure B.1, is consistent with the results presented in [39], and characterized by time-of-flight $ToF^{*}\left(t_0\right) = 7.8549\, days$ and mass ratio $m\left(t_f\right)/m_0 = 0.8462$. The errors on the final constraints are well within acceptable tolerance, in particular $\|\Delta\mathbf{r}\left(t_f\right)\| = 9.9776 \cdot 10^{-6}\, km$ and $\|\Delta\mathbf{v}\left(t_f\right)\| = 5.8846 \cdot 10^{-11}\, km/s$.

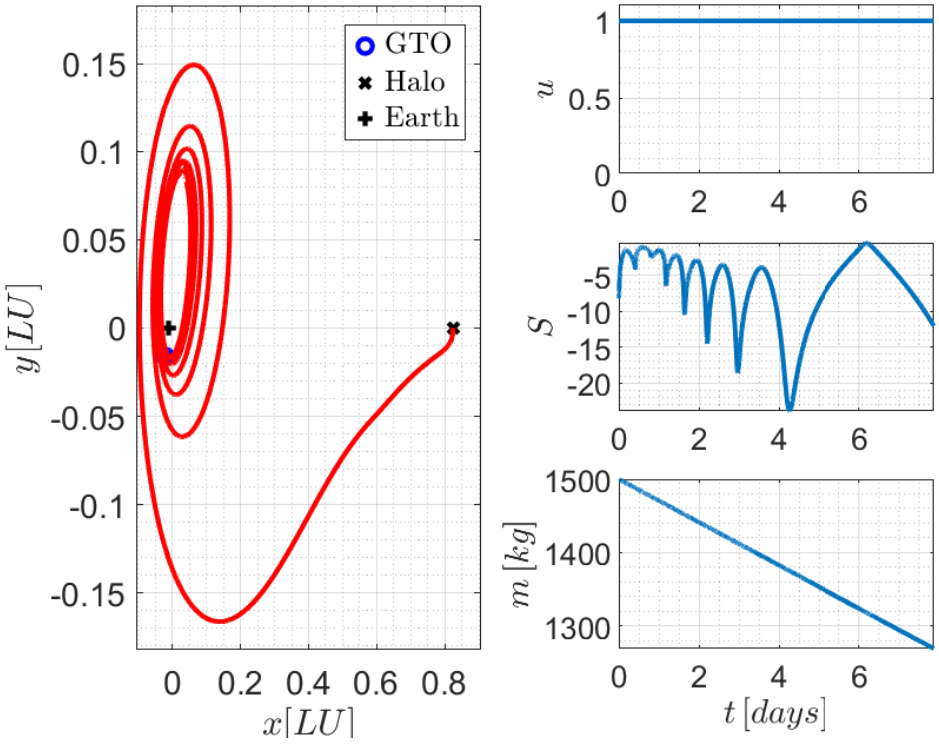


Figure B.1: GTO to Halo time-optimal transfer.

In accordance with the criteria devised in Section 4.8, it is observed that, given the optimal path for the TO problem $\boldsymbol{\lambda}^*_{TO}(t)$, the paths $\boldsymbol{\lambda}^*_{FO}(t) = \gamma_L \boldsymbol{\lambda}^*_{TO}(t)$ and $\boldsymbol{\lambda}^*_{EO}(t) = 2\gamma_L \boldsymbol{\lambda}^*_{TO}(t)$ are optimal for the FO and EO problems, respectively, which implicitly optimize time. In particular, the trajectories are not altered by the scaling, the shooting functions are satisfied within specified tolerance, and the throttle is constant and maximum ($u = 1$) for both the FO and EO solutions: respectively, $S_{EFO,0}(t) < 0$ and $S_{EFO,1}(t) < -1$ for all $t \in [t_0, ToF^*(t_0)]$.
Figure B.2 presents the comparison between the switching functions of the different problems, using the paths $\boldsymbol{\lambda}^*_{TO}(t)$, $\boldsymbol{\lambda}^*_{FO}(t) = \gamma_L \boldsymbol{\lambda}^*_{TO}(t)$ and $\boldsymbol{\lambda}^*_{EO}(t) = 2\gamma_L \boldsymbol{\lambda}^*_{TO}(t)$ for the TO, FO and EO problems, respectively.

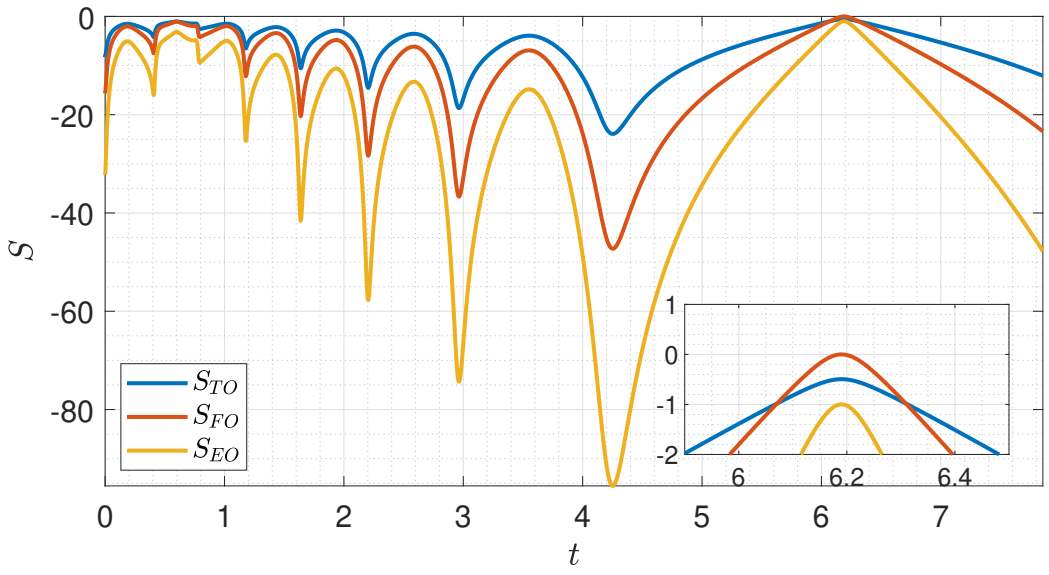


Figure B.2: S_{TO}, $S_{FO} = S_{EFO,0}$ and $S_{EO} = S_{EFO,1}$ comparison employing $\boldsymbol{\lambda}^*_{FO}(t) = \gamma_L \boldsymbol{\lambda}^*_{TO}(t)$ and $\boldsymbol{\lambda}^*_{EO}(t) = 2\gamma_L \boldsymbol{\lambda}^*_{TO}(t)$.

The experimental results obtained are in agreement with the analysis advanced in Section 4.8, therefore supporting the hypotheses advanced on the criteria for suitability of solution path scaling.

List of Figures

1.1 Relative fraction of launched EP-based GEO satellites per year between 1981-2018. [14] . . . 1
1.2 EP mission types per year between 1981-2018. [14] . . . 2

3.1 Homotopy curve types and critical points. . . . 21
3.2 Geometric interpretation of 0NPCM. [22] . . . 22
3.3 Geometric interpretation of PAM. [22] . . . 24
3.4 RAMSES mission porkchop plot solution space exploration strategy. [19] . 27
3.5 M-ARGO departure date t_0 continuation scheme for TO solutions. [30] . . 28
3.6 M-ARGO time-of-flight $\tau = ToF$ continuation scheme for FO solutions. [30] 28

4.1 Thruster operation logic representation. [31][32] . . . 30
4.2 Qualitative representation of FO or EO solution curves in the $(ToF, m(t_f))$ space. . . . 40
4.3 Qualitative representation of FO or EO solution curves in the (t_0, ToF) space. . . . 40
4.4 Qualitative thrust parameters profile for the proof by contradiction. . . . 43

5.1 Time-optimal departure date t_0 continuation scheme. . . . 60
5.2 Energy-optimal departure date t_0 continuation scheme. . . . 61
5.3 Energy-optimal time-of-flight ToF continuation scheme. . . . 62
5.4 Solution space exploration strategy. . . . 62
5.5 Energy-to-fuel-optimal ε continuation scheme. . . . 63

6.1 $2014\,YD$ porkchop plot. . . . 66
6.2 SEL2 to $2014\,YD$ (AST) transfer, sample trajectory A. . . . 66
6.3 SEL2 to $2014\,YD$ (AST) transfer, sample trajectory B. . . . 67
6.4 SEL2 to $2014\,YD$ (AST) transfer, sample trajectory C. . . . 67
6.5 SEL2 to $2014\,YD$ (AST) transfer, sample trajectory D. . . . 68
6.6 $2000\,SG344$ porkshop plot. . . . 69
6.7 $2010\,UE51$ porkchop plot. . . . 69

6.8 2011 MD porkchop plot. 69
6.9 2012 $UV136$ porkshop plot. 69
6.10 Study of 2000 $SG344$ solutions that implicitly optimize time. 71
6.11 Study of 2010 $UE51$ solutions that implicitly optimize time. 71
6.12 Study of 2011 MD solutions that implicitly optimize time. 71
6.13 Study of 2012 $UV136$ solutions that implicitly optimize time. 72
6.14 Study of 2014 YD solutions that implicitly optimize time. 72
6.15 Table 1.3 targets final constraints violations using $\boldsymbol{\lambda}_{FO}(t_0) = \overline{\gamma}\boldsymbol{\lambda}^*_{TO}(t_0)$. . . 73
6.16 2014 YD porkchop plot (CP). 75
6.17 SEL2 to 2014 YD (AST) transfer, sample trajectory A (CP). 76
6.18 SEL2 to 2014 YD (AST) transfer, sample trajectory B (CP). 76
6.19 SEL2 to 2014 YD (AST) transfer, sample trajectory C (CP). 77
6.20 SEL2 to 2014 YD (AST) transfer, sample trajectory D (CP). 77
6.21 2000 $SG344$ porkchop plot (CP). 79
6.22 2010 $UE51$ porkchop plot (CP). 79
6.23 2011 MD porkchop plot (CP). 79
6.24 2012 $UV136$ porkchop plot (CP). 79
6.25 Study of 2000 $SG344$ solutions that implicitly optimize time (CP). 80
6.26 Study of 2010 $UE51$ solutions that implicitly optimize time (CP). 80
6.27 Study of 2011 MD solutions that implicitly optimize time (CP). 80
6.28 Study of 2012 $UV136$ solutions that implicitly optimize time (CP). 81
6.29 Study of 2014 YD solutions that implicitly optimize time (CP). 81
6.30 Table 1.3 targets final constraints violations using $\boldsymbol{\lambda}^*_{FO}(t) = \overline{\gamma}\boldsymbol{\lambda}^*_{TO}(t)$ (CP). 82
6.31 TO and implicitly time-optimal FO costate paths for SEL2-2012 to $UV136$ transfer, $t_0 = 8662.3\,MJD2000$. 83
6.32 Scaling parameter γ_{FO} lower bound test, 2000 $SG344$. 85
6.33 Scaling parameter γ_{FO} lower bound test, 2010 $UE51$. 85
6.34 Scaling parameter γ_{FO} lower bound test, 2011 MD. 85
6.35 Scaling parameter γ_{FO} lower bound test, 2012 $UV136$. 86
6.36 Scaling parameter γ_{FO} lower bound test, 20014 YD. 86
6.37 S_{TO} and S_{FO} comparison employing $\boldsymbol{\lambda}^*_{FO}(t) = \gamma_{FO}\boldsymbol{\lambda}^*_{TO}(t)$. 87
6.38 Scaling parameter γ_{EO} lower bound test, 2000 $SG344$. 87
6.39 Scaling parameter γ_{EO} lower bound test, 2010 $UE51$. 88
6.40 Scaling parameter γ_{EO} lower bound test, 2011 MD. 88
6.41 Scaling parameter γ_{EO} lower bound test, 2012 $UV136$. 88
6.42 Scaling parameter γ_{EO} lower bound test, 2014 YD. 89

6.43 S_{TO}, S_{FO} and S_{EO} comparison employing $\boldsymbol{\lambda}^*_{FO}(t) = \gamma_{FO}\boldsymbol{\lambda}^*_{TO}(t)$ and $\boldsymbol{\lambda}^*_{EO}(t) = \gamma_{EO}\boldsymbol{\lambda}^*_{TO}(t)$. 89

B.1 GTO to Halo time-optimal transfer. 106

B.2 S_{TO}, $S_{FO} = S_{EFO,0}$ and $S_{EO} = S_{EFO,1}$ comparison employing $\boldsymbol{\lambda}^*_{FO}(t) = \gamma_L\boldsymbol{\lambda}^*_{TO}(t)$ and $\boldsymbol{\lambda}^*_{EO}(t) = 2\gamma_L\boldsymbol{\lambda}^*_{TO}(t)$. 107

List of Tables

1.1 M-ARGO mission objectives. [30] . . . 3
1.2 M-ARGO spacecraft data assumptions. [30] . . . 4
1.3 Sample downselected M-ARGO targets. [30] . . . 4

4.1 Thruster model coefficients. [30][31][32] . . . 31

5.1 Physical constants. [31][32] . . . 56
5.2 Preliminary departure window. [30] . . . 58
5.3 ACT parameters bounds. [31][32] . . . 59

6.1 Summary of the results of the sample SEL2-2014 YD trajectories. . . . 68
6.2 Porkchop plot continuation scheme performance. . . . 70
6.3 Final constraints violations using $\boldsymbol{\lambda}_{FO}(t_0) = \overline{\gamma}\boldsymbol{\lambda}^*_{TO}(t_0)$ with lowest σ_γ. . . . 74
6.4 Constant propulsive parameters. . . . 75
6.5 Summary of the results of the sample SEL2-2014 YD trajectories (CP). . . . 78
6.6 Porkchop plot continuation scheme performance (CP). . . . 78
6.7 Average iteration count for FO, EO convergence vs ToF continuation. . . . 91

B.1 Physical constants for Earth-Moon CR3BP problem. [39] . . . 104
B.2 Trajectory initial boundary conditions. [39] . . . 104
B.3 Trajectory final boundary conditions. [39] . . . 105

List of Symbols

Variable	Description	SI unit
$\mathcal{M}$	ACT mapping	–
$\mathcal{U}$	Admissible control space	–
r_a	Aphelion	km
s	Arc length	s
ω	Argument of periapsis	deg
t_f	Arrival date	$MJD2000$
$\overline{J}$	Augmented cost function	–
$\mathbf{y}$	Complete state	–
θ	Continuation parameter	–
$\mathbf{u}$	Control vector	–
J	Cost function	–
$\boldsymbol{\lambda}$	Costate vector	–
$\overline{\mathbf{f}}$	Costate dynamics	–
t_0	Departure date	$MJD2000$
r	Distance	km
g_0	Earth acceleration constant at sea level	m/s^2
e	Eccentricity	–
ε	Energy-to-fuel smoothing parameter	–
$\mathbf{F}$	Euler-Lagrange equations	–
c	Exhaust velocity	m/s
Ω_f	Final transversality condition	–
$\boldsymbol{\psi}_f$, $\boldsymbol{\psi}$	Final boundary constraints	–
$\boldsymbol{\nu}_f$	Final boundary constraints multipliers	–
H	Hamiltonian	–
$\mathbf{c}$	Homotopy curve	–

$\mathbf{\Gamma}$	Homotopy function	–
κ	Homotopy parameter	–
$\mathbf{i}_r$	Specific impulse gradient	s/km
α	In-plane thrust angle	deg
i	Inclination	deg
$\boldsymbol{\xi}$	Indirect formulation unknown	–
Ω_0	Initial transversality condition	–
$\boldsymbol{\psi}_0$	Initial boundary constraints	–
$\boldsymbol{\nu}_0$	Initial boundary constraints multipliers	–
P_{in}	Input power	W
$\mathbf{N}$	Interior point constraint	–
$\boldsymbol{\pi}$	Interior point constraint multiplier	–
L	Lagrange path cost	–
L_i	Lagrange point i	–
LU	Length Unit	km
m	Mass	kg
λ_m	Mass costate	–
MU	Mass unit	kg
P_{max}	Maximum engine power	W
T_{max}	Maximum thrust	N
ϕ	Mayer terminal cost	–
P_{min}	Minimum engine power	W
β	Out-of-plane thrust angle	deg
$\mathbf{J_a}$	PAM augmented jacobian	–
$\mathbf{C}$	Path constraints	–
$\boldsymbol{\mu}$	Path constraints multiplier	–
J	Cost function	–
r_p	Perihelion	km
$\mathbf{r}$	Position	km
$\boldsymbol{\lambda_r}$	Position costate	–
S_p	Power switching function	W
PU	Power unit	W
m_p	Propellant mass	kg

Ω	Right ascension of ascending node	deg
h	Runge-Kutta step size	–
γ	Scaling parameter	–
σ_γ	Scaling parameter covariance	–
a	Semimajor axis	km
$\mathbf{\Gamma}$	Shooting function	–
$\mathbf{J_\Gamma}$	Shooting function gradient	–
I_{sp}	Specific impulse	s
$\mathbf{f}$	State dynamics	km/s, km/s^2, kg/s
$\mathbf{\Phi}$	State transition matrix	–
$\mathbf{x}$	State vector	km, km/s, kg
μ	Sun gravitational constant	km^3/s^2
u	Throttle	–
τ	Throttle arc	s
$\mathcal{T}$	Throttle arcs set	s
S	Throttle switching function	–
$\hat{\boldsymbol{\alpha}}$	Thrust direction	–
$\mathbf{t}_r$	Thrust gradient	N/km
t	Time	s, day
TU	Time unit	s
ToF	Time-of-flight	s, $days$
$\mathbf{v}$	Velocity	km/s
$\boldsymbol{\lambda}_\mathbf{v}$	Velocity Costate	–
VU	Velocity Unit	km/s

List of Acronyms

Variable	Description
ACT	Adjoint Control Transformation
AU	Astronomical Unit
CoV	Calculus of Variations
DAE	Differential Algebraic Equation
DCM	Direct Continuation Method
EFO	Energy-to-Fuel-Optimal
EO	Energy-Optimal
EP	Electric Propulsion
ESA	European Space Agency
FO	Fuel-Optimal
GEO	Geostationary Orbit
GTO	Geostationary Transfer Orbit
HTS	Hyperbolic Tangent Smoothing
IPC	Interior Point Constraint
JAXA	Japanese Aerospace Exploration Agency
LEFO	Lagrange Energy-to-Fuel-Optimal
LEO	Low Earth Orbit
LTO	Lagrange Time-Optimal
LT	Low Thrust
MFO	Mayer Fuel-Optimal
MPBVP	Multiple Point Boundary Value Problem
MTO	Mayer Time-Optimal
NASA	National Aeronautics and Space Administration
NEA	Near Earth Asteroid
NEO	Near Earth Orbit

NLP	Non-Linear Programming
NPCM	Natural Parameter Continuation Method
OCP	Optimal Control Problem
PAM	Pseudo-Arclength Method
PL	Piecewise Linear
PMP	Pontryagin Minimum Principle
PC	Predictor Corrector
RK	Runge-Kutta
SEL2	Sun-Earth L_2
SEP	Solar Electric Propulsion
STM	State Transition Matrix
TFC	Theory of Functional Connections
TO	Time-Optimal
TPBVP	Two Point Boundary Value Problem
ZFP	Zero Finding Problem

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